

# Comparing Mutual Information in Finite Samples: Wald and Expanded Welch–Satterthwaite Inference

Daniel Zhao

Supervisor: Associate Professor Joseph T. Lizier

A thesis submitted in partial fulfilment of the requirements for the degree of

**Master of Complex Systems**

School of Computer Science

The University of Sydney

Australia

2026

# Abstract

Mutual information (MI) measures dependence between two variables. This thesis tests whether two independent categorical populations have equal MI, without requiring their joint distributions or margins to be the same.

Normal Wald inference standardises a bias-corrected MI difference using a plug-in standard error and a normal reference. Expanded Welch–Satterthwaite keeps that statistic but derives effective degrees of freedom from the sampling variability of the estimated MI variance. The derivation includes changes to the cell probabilities, margins and pointwise MI values. Its Student reference is a moment-matching approximation, not an exact finite-sample distribution.

The experiment covers fixed positive-support populations from  $2 \times 2$  to  $8 \times 8$ , with rectangular, unequal-sample, construction and extreme-sparsity checks. Each of 3,111 configurations uses 20,000 paired replicates at a two-sided nominal level of 0.05. False positives, detection, validity and runtime are reported by regime. The null cases distinguish identical populations from distinct populations with equal MI.

Expanded Welch improves some liberal Wald false-positive rates but worsens already conservative regimes. Its strict component-validity rule also produces fewer usable results in sparse samples. Wald is closer to nominal in all 24 main null regimes at sample sizes 500 or 1,000. At common MI 0.02 and sample size 50,000, both methods are near nominal in the tested populations; very small MI remains difficult. Expanded Welch has the same computational order and costs about 1.75 times as much per scalar call in the measured implementation.

Near independence, small effective degrees of freedom make Expanded Welch strongly conservative. A supplementary diagnostic shows why changing only the reference cannot correct errors in both the numerator and standard error.

The evidence supports Wald as the simpler general baseline, with Expanded Welch as a sensitivity analysis in selected liberal regimes.

# Acknowledgements

I am grateful to my supervisor, Associate Professor Joseph T. Lizier, for guidance throughout this research. I also thank the colleagues, friends, and family members whose support made this work possible.

# Contents

|                                                                    |           |
|--------------------------------------------------------------------|-----------|
| <b>Abstract</b>                                                    | <b>i</b>  |
| <b>Acknowledgements</b>                                            | <b>ii</b> |
| <b>1 Introduction</b>                                              | <b>1</b>  |
| 1.1 Why the comparison is difficult . . . . .                      | 1         |
| 1.2 Research questions . . . . .                                   | 2         |
| 1.3 Contributions to the field . . . . .                           | 2         |
| 1.4 Scope and organisation . . . . .                               | 3         |
| <b>2 Background and Literature Review</b>                          | <b>4</b>  |
| 2.1 Information-theoretic foundations . . . . .                    | 4         |
| 2.1.1 Entropy and information content . . . . .                    | 4         |
| 2.1.2 Pointwise and average mutual information . . . . .           | 5         |
| 2.2 Estimating discrete mutual information . . . . .               | 6         |
| 2.3 Sampling variance of plug-in MI . . . . .                      | 7         |
| 2.4 Related inferential methods . . . . .                          | 12        |
| 2.5 Literature synthesis and methodological gap . . . . .          | 16        |
| <b>3 Comparing Mutual Information with Normal Wald</b>             | <b>17</b> |
| 3.1 Problem setup and bias-corrected estimator . . . . .           | 17        |
| 3.2 Shared standard error and test statistic . . . . .             | 19        |
| 3.3 Normal Wald reference and motivation for expansion . . . . .   | 19        |
| <b>4 Expanded Welch–Satterthwaite Method</b>                       | <b>21</b> |
| 4.1 From Welch–Satterthwaite to the required MI quantity . . . . . | 21        |
| 4.2 Deriving the sensitivity of the MI variance . . . . .          | 22        |

|          |                                                                     |           |
|----------|---------------------------------------------------------------------|-----------|
| 4.3      | Component degrees of freedom and final test . . . . .               | 25        |
| 4.4      | Worked example and implementation . . . . .                         | 28        |
| 4.5      | Behaviour and limitations . . . . .                                 | 30        |
| <b>5</b> | <b>Experimental Design</b>                                          | <b>32</b> |
| 5.1      | A population pair before the design grid . . . . .                  | 32        |
| 5.2      | Constructing fixed joint distributions . . . . .                    | 32        |
| 5.3      | Sampling, tests and measured outcomes . . . . .                     | 34        |
| 5.4      | Experiment families and computation . . . . .                       | 35        |
| <b>6</b> | <b>Experimental Results</b>                                         | <b>37</b> |
| 6.1      | Main landscape: shape, margins and sample size . . . . .            | 37        |
| 6.2      | Baseline MI and the location of changed cells . . . . .             | 51        |
| 6.3      | Unequal samples and the direction of the difference . . . . .       | 54        |
| 6.4      | Larger differences and rectangular tables . . . . .                 | 57        |
| 6.5      | Population construction sensitivity . . . . .                       | 59        |
| 6.6      | Extreme skew, independence and larger samples . . . . .             | 61        |
| 6.7      | Overall calibration and a mechanism check . . . . .                 | 64        |
| 6.8      | Runtime and summary of evidence . . . . .                           | 65        |
| <b>7</b> | <b>Discussion</b>                                                   | <b>67</b> |
| 7.1      | Answers to the research questions . . . . .                         | 67        |
| 7.2      | What the correction can and cannot change . . . . .                 | 68        |
| 7.3      | Interpretation of invalid samples . . . . .                         | 69        |
| 7.4      | Practical conclusion and limits of the evidence . . . . .           | 69        |
| <b>8</b> | <b>Conclusion</b>                                                   | <b>71</b> |
| <b>A</b> | <b>Additional Derivation Details</b>                                | <b>73</b> |
| A.1      | Gradient of mutual information on the probability simplex . . . . . | 73        |
| A.2      | Why the variance of pointwise MI appears . . . . .                  | 74        |
| A.3      | Mean of the variance influence function . . . . .                   | 74        |
| A.4      | Direct table substitutions . . . . .                                | 75        |
| A.5      | Numerical derivative checks . . . . .                               | 76        |
| <b>B</b> | <b>Population Construction and Exact Design Settings</b>            | <b>77</b> |

---

|          |                                                  |           |
|----------|--------------------------------------------------|-----------|
| B.1      | Additive construction . . . . .                  | 77        |
| B.2      | Frozen statistical grids . . . . .               | 78        |
| <b>C</b> | <b>Evidence Index and Additional Diagnostics</b> | <b>81</b> |
| <b>D</b> | <b>The Independence Boundary</b>                 | <b>84</b> |
| D.1      | Constructing the independent reference . . . . . | 84        |
| D.2      | Why the first-order term vanishes . . . . .      | 85        |
| D.3      | Second-order behaviour . . . . .                 | 85        |
| D.4      | What alternative constructions produce . . . . . | 87        |
| <b>E</b> | <b>Reproducibility and Protocol Record</b>       | <b>88</b> |

# List of Figures

|      |                                                                       |    |
|------|-----------------------------------------------------------------------|----|
| 6.1  | Main landscape: $2 \times 2$ tables, uniform margins . . . . .        | 38 |
| 6.2  | Main landscape: $2 \times 2$ tables, same skew margins . . . . .      | 39 |
| 6.3  | Main landscape: $2 \times 2$ tables, different skew margins . . . . . | 40 |
| 6.4  | Main landscape: $3 \times 3$ tables, uniform margins . . . . .        | 41 |
| 6.5  | Main landscape: $3 \times 3$ tables, same skew margins . . . . .      | 42 |
| 6.6  | Main landscape: $3 \times 3$ tables, different skew margins . . . . . | 43 |
| 6.7  | Main landscape: $5 \times 5$ tables, uniform margins . . . . .        | 44 |
| 6.8  | Main landscape: $5 \times 5$ tables, same skew margins . . . . .      | 45 |
| 6.9  | Main landscape: $5 \times 5$ tables, different skew margins . . . . . | 46 |
| 6.10 | Main landscape: $8 \times 8$ tables, uniform margins . . . . .        | 47 |
| 6.11 | Main landscape: $8 \times 8$ tables, same skew margins . . . . .      | 48 |
| 6.12 | Main landscape: $8 \times 8$ tables, different skew margins . . . . . | 49 |
| 6.13 | Baseline-MI comparison, $3 \times 3$ tables . . . . .                 | 52 |
| 6.14 | Changed-cell comparison, $5 \times 5$ tables . . . . .                | 53 |
| 6.15 | Unequal samples, $Q$ higher . . . . .                                 | 55 |
| 6.16 | Unequal samples, $P$ higher . . . . .                                 | 56 |
| 6.17 | Wider MI differences, uniform $2 \times 2$ . . . . .                  | 57 |
| 6.18 | Rectangular tables . . . . .                                          | 58 |
| 6.19 | Additive versus log-linear construction . . . . .                     | 60 |
| 6.20 | Extreme-skew $3 \times 3$ tables . . . . .                            | 61 |
| 6.21 | Exact-independence boundary . . . . .                                 | 62 |
| 6.22 | Large-sample null behaviour . . . . .                                 | 63 |

# List of Tables

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |    |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 2.1 | Selected prior work, grouped by the question it addresses. . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 15 |
| 4.1 | Operational Expanded Welch–Satterthwaite algorithm. . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 29 |
| 5.1 | Row and column margins used for each population in the principal design. “Uniform” means the vector with every entry equal to $1/k$ . . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 34 |
| 5.2 | The fixed experiment families. All statistical points use 20,000 paired replicates and the 5% threshold. . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 36 |
| 6.1 | Two fully valid main-grid null regimes. Brackets beside each rate are pointwise Wilson 95% intervals; the final bracket is the approximate paired 95% interval for Wald minus Expanded Welch on the same samples. Source configurations are <code>config_313a5e5c05280f9e</code> and <code>config_700945d218841f2e</code> , respectively. . . . .                                                                                                                                                                                                                                               | 50 |
| 6.2 | Location, separately for each method, of the 108 main-grid null configurations relative to the descriptive rejection-rate interval 0.025–0.075. Each profile contains 36 configurations. The final column separately counts configurations with validity below 90%; it overlaps the three rate-location columns. In particular, all four $n = 2$ configurations per profile have zero validity for both methods. Uniform and same-skew are strong nulls; different-skew is the equal-MI weak null. Exact rates and validity are available in the saved results described in Appendix C. . . . . | 64 |
| 6.3 | Scalar-call times in microseconds for the different-skew null at $n_P = n_Q = 100$ and $I(P) = I(Q) = 0.02$ . Entries give median [interquartile range] over five passes on 200 saved inputs per regime. Population construction and multinomial sampling are excluded. . . . .                                                                                                                                                                                                                                                                                                                 | 65 |
| B.1 | Exact statistical experiment grids; MI values are in nats. . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 79 |
| C.1 | Saved files and the question each answers. . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 82 |



# Chapter 1

## Introduction

Suppose two independent groups are observed on the same categorical variables  $X$  and  $Y$ . Mutual information (MI) describes the strength of association between those variables in each group. If the estimated MI differs, the difference alone does not tell us whether the populations differ: both estimates vary when new observations are drawn. The question in this thesis is whether the two *population* MI values can plausibly be equal.

This is not the usual test of whether  $X$  and  $Y$  are independent within one group. The null hypothesis here permits dependence in both groups and permits their complete joint distributions to differ. It specifies only

$$H_0 : I(P) = I(Q), \quad H_1 : I(P) \neq I(Q), \quad (1.1)$$

where  $P$  and  $Q$  are the two joint distributions. The samples from  $P$  and  $Q$  are independent of each other; that sampling assumption does not imply independence of  $X$  and  $Y$  within either population.

### 1.1 Why the comparison is difficult

MI is a nonlinear function of all cell probabilities and their row and column margins. With a small sample or rare population cells, an observed table can contain many zeros even though every population probability is positive. The same sampled tables determine both the MI difference and its standard error, so sparsity affects both estimates. A first-order Normal Wald test is simple, but its finite-sample rejection rate need not be close to the nominal level.

Welch–Satterthwaite inference offers a possible adjustment. Instead of comparing the standardised difference with a normal reference, it compares the same statistic

with a Student reference whose effective degrees of freedom describe uncertainty in the estimated variance. For MI, that variance is itself a nonlinear function of the full table. Hutcheson's earlier information-theoretic test assigns each variance component its sample size as degrees of freedom; the familiar ordinary-Welch analogue instead uses  $n - 1$ . Neither assignment accounts for the way a changed cell also changes the margins and pointwise MI values. The method examined here estimates those degrees of freedom from the sensitivity of the full variance functional.

The resulting reference is an approximation, not a guarantee of better inference. A heavier-tailed reference can reduce excessive false positives, but it can also suppress useful detection when Wald already rejects too rarely. Moreover, one method can fail to produce a valid p-value on samples where the other succeeds. Calibration, detection and valid-result frequency must therefore be considered together.

## 1.2 Research questions

The thesis asks four related questions. First, how can the sampling uncertainty of the estimated MI variance be turned into an MI-specific Welch–Satterthwaite reference? Second, how does that reference affect false positives, detection and numerical validity relative to Normal Wald? Third, how do table size, margins, baseline dependence, the location of association and unequal sample sizes change the comparison? Fourth, do the observed calibration errors diminish with larger samples, and what additional computation does the correction require?

## 1.3 Contributions to the field

This thesis contributes to two-sample inference for discrete information measures in four connected ways. The established ingredients are first-order MI inference and Satterthwaite moment matching; the contribution lies in connecting, implementing and testing them for the equal-MI problem.

1. **Mathematical contribution.** The thesis derives the sensitivity of the plug-in MI variance to every cell probability. The derivative retains the induced changes in both margins and in the pointwise MI values, providing the quantities needed to estimate MI-specific component degrees of freedom.
2. **Methodological contribution.** The resulting Expanded Welch procedure turns that variance sensitivity into a usable two-sample test. Its relationship

to the Hutcheson sample-size assignment, the ordinary  $n - 1$  assignment and Normal Wald is made explicit, so the effect of the full variance-sensitivity expansion can be distinguished from generic Welch correction; the detailed ablation is retained in the experiment supplement.

3. **Empirical contribution.** A frozen, reproducible experiment evaluates false positives, detection, valid-result frequency and runtime on fixed populations with known MI. It covers  $2 \times 2$  through  $8 \times 8$  square tables, two rectangular shapes, unequal samples and stress regimes, while preserving every configuration-level result in a companion atlas instead of replacing the regimes by an overall average.
4. **Practical contribution.** The evidence establishes the limits of a heavier-tailed reference correction. Expanded Welch can improve a liberal Wald test, but it can also worsen an already conservative test and return fewer valid results in sparse samples under the strict implemented component-validity rule. Some of those failures come from the rule rather than the combined formula. Expanded Welch therefore should not replace Wald generally. This negative result identifies when reference-tail adjustment is useful and when bias, sparsity or failure of the first-order approximation requires a different solution.

## 1.4 Scope and organisation

The analysis concerns independent multinomial samples on fixed, aligned categorical alphabets. The principal theory assumes strictly positive cell probabilities and positive first-order MI variances. Exact independence is studied separately as a boundary case, not treated as an ordinary positive-MI regime. The final experiment compares Normal Wald with Expanded Welch at a two-sided nominal level of 0.05. It does not establish performance for paired data, continuous MI estimators or alphabets that grow with sample size.

Chapter 2 supplies the statistical background. Chapters 3 and 4 define the shared statistic and derive the correction. Chapter 5 explains how the populations and samples are generated and how the results are measured. Chapter 6 shows the complete landscape before the focused comparisons. Chapters 7 and 8 discuss the consequences and the resulting recommendation. The appendices provide the detailed algebra, construction and source-to-result record.

# Chapter 2

## Background and Literature Review

This chapter develops the concepts needed to compare MI between two categorical populations. It begins with entropy, pointwise MI, and population MI. It then reviews plug-in estimation, first-order inference, the Welch–Satterthwaite approximation, and prior comparisons of information quantities. The final section identifies the gap addressed by the expanded method.

### 2.1 Information-theoretic foundations

#### 2.1.1 Entropy and information content

For a discrete random variable  $X$  with probability mass function  $p_X(x)$ , the information content of outcome  $x$  is  $-\log p_X(x)$ . Rare outcomes have larger information content because they are less expected. Using natural logarithms gives information in nats. The Shannon entropy is the expected information content,

$$H(X) = - \sum_x p_X(x) \log p_X(x). \quad (2.1)$$

Entropy summarises the uncertainty of one random variable before its value is observed ([Shannon, 1948](#)).

For a pair  $(X, Y)$ , the joint entropy  $H(X, Y)$  is defined from the joint probabilities in the same way. The conditional entropy  $H(X | Y) = H(X, Y) - H(Y)$  measures the uncertainty remaining in  $X$  after observing  $Y$ .

### 2.1.2 Pointwise and average mutual information

Let  $P = (p_{ij})$  be the joint distribution of categorical variables  $X \in \{1, \dots, r\}$  and  $Y \in \{1, \dots, c\}$ . Write

$$p_{i+} = \sum_{j=1}^c p_{ij}, \quad p_{+j} = \sum_{i=1}^r p_{ij} \quad (2.2)$$

for the row and column margins. The pointwise mutual information associated with cell  $(i, j)$  is

$$\ell_P(i, j) = \log \left( \frac{p_{ij}}{p_{i+}p_{+j}} \right). \quad (2.3)$$

The denominator is the cell probability under independence with the same margins. A positive value means that the cell occurs more often than independence predicts; a negative value means that it occurs less often.

Mutual information is the probability-weighted average of these pointwise values (Shannon, 1948; Bossomaier et al., 2016):

$$I(P) = \sum_{i=1}^r \sum_{j=1}^c p_{ij} \ell_P(i, j). \quad (2.4)$$

Here the summations are nested: for each row  $i$ , the inner sum runs over all columns  $j$ . Equivalently, the expression sums once over every cell in the table and may be written with the shorthand  $\sum_{i,j}$ . Because  $P$  is the joint distribution of  $(X, Y)$ , the same quantity is conventionally written as  $I(X; Y)$ . Equivalently,

$$I(X; Y) = H(X) + H(Y) - H(X, Y). \quad (2.5)$$

The quantity is nonnegative and equals zero exactly when  $p_{ij} = p_{i+}p_{+j}$  for every cell, including zero-probability cells.

Equation (2.4) is also the Kullback–Leibler divergence between the joint distribution and the product of its margins. MI therefore measures the overall departure from independence, not one particular shape of association.

This scalar summary deliberately discards much of the information in the joint distribution. Two populations can have the same MI while assigning probability to different cells, having different margins, or exhibiting different association patterns. Equality of MI is therefore weaker than equality of distributions. This distinction is central to the present test: the null fixes one scalar functional while leaving the remaining features of each population unrestricted.

The logarithm base determines the reporting unit. This thesis uses natural logarithms, so MI is measured in nats; dividing by  $\log 2$  converts to bits. This

conversion rescales MI and its standard error together but leaves the degrees of freedom unchanged, so the test result is identical in either unit.

## 2.2 Estimating discrete mutual information

Suppose a sample of size  $n$  produces counts  $N_{ij}$ . The empirical cell probabilities are

$$\hat{p}_{ij} = \frac{N_{ij}}{n}. \quad (2.6)$$

Substitution into Equation (2.4) gives the plug-in estimator

$$\hat{I}(P) = \sum_{i,j:\hat{p}_{ij}>0} \hat{p}_{ij} \log \left( \frac{\hat{p}_{ij}}{\hat{p}_{i+}\hat{p}_{+j}} \right). \quad (2.7)$$

The zero-cell convention follows from  $\lim_{x \downarrow 0} x \log x = 0$ . Empty cells contribute no direct term, but they still affect the empirical margins and the finite-sample behaviour of the estimator.

Under multinomial sampling, the cell counts are random but constrained to sum to  $n$ , so they are dependent. The plug-in estimator remains consistent for a fixed alphabet, but its finite-sample behaviour depends on the complete probability table. In particular, rare cells can change the sampling behaviour even when the alphabet and sample sizes are unchanged (Paninski, 2003).

The plug-in estimator is upward biased: random variation in the cell counts tends to make the empirical table appear more dependent than the population table. For a fixed positive  $r \times c$  support, the leading bias is

$$\text{Bias}\{\hat{I}(P)\} = \frac{(r-1)(c-1)}{2n} + O(n^{-2}). \quad (2.8)$$

The factor  $(r-1)(c-1)$  increases with table size, while division by  $n$  shows that the bias decreases as the sample grows. The bias is therefore largest for larger tables with small samples. The result is closely related to Miller's entropy correction (Miller, 1955), and the same histogram-MI term appears in Moddemeijer (1989). Sparse sampling can produce further error beyond this leading approximation (Paninski, 2003).

Bias matters directly when comparing two populations. If  $I(P) = I(Q)$  but  $n_P \neq n_Q$ , the uncorrected difference has leading bias

$$\frac{(r-1)(c-1)}{2} \left( \frac{1}{n_P} - \frac{1}{n_Q} \right). \quad (2.9)$$

For a  $2 \times 2$  table with  $n_P = 50$  and  $n_Q = 100$ , Equation (2.9) gives 0.005 nats despite equal population MIs. Chapter 3 corrects this sample-size effect by subtracting  $(r-1)(c-1)/(2n)$  from each estimate.

## 2.3 Sampling variance of plug-in MI

To compare two estimated MI values, it is necessary to know how much each estimate would vary across repeated samples. The goal of this section is to derive the large-sample approximation

$$\text{Var}\{\hat{I}(P)\} \approx \frac{V(P)}{n}. \quad (2.10)$$

Here  $V(P)$ , defined below, measures how much the contribution of one observation varies across the population. The derivation assumes that the population cell probabilities are positive, so the logarithms and their derivatives are well defined. This first-order variance is established for histogram and discrete MI (Moddemeijer, 1989; Brillinger, 2004). The argument proceeds in three steps. First, sampling changes the empirical cell probabilities. Second, a first-order Taylor expansion translates those probability errors into an MI error. Third, that error can be written as a sample average, whose variance is straightforward to calculate.

### Step 1: From probability errors to MI error.

Using the entropy identity in Equation (2.5), MI can be written directly in terms of the joint and marginal probabilities:

$$I(P) = \sum_{i,j} p_{ij} \log p_{ij} - \sum_i p_{i+} \log p_{i+} - \sum_j p_{+j} \log p_{+j}. \quad (2.11)$$

Let  $f(u) = u \log u$ . The derivative needed for its first-order Taylor expansion follows directly from the product rule:

$$f'(u) = \frac{d}{du} \{u \log u\} = \log u + u \frac{1}{u} = \log u + 1. \quad (2.12)$$

The general first-order Taylor formula is

$$f(x) \approx f(a) + f'(a)(x - a), \quad (2.13)$$

where  $a$  is the value around which the function is expanded and  $x - a$  is the distance from that value. Here the new value is the estimated probability  $x = \hat{u}$ , the expansion point is the population probability  $a = u$ , and  $f(t) = t \log t$ . Therefore,

$x - a = \hat{u} - u$ , which is the sampling error in the estimated probability. Substituting these quantities into the general formula gives

$$\hat{u} \log \hat{u} \approx \underbrace{u \log u}_{\text{population value}} + \underbrace{(\log u + 1)(\hat{u} - u)}_{\text{first-order change}}. \quad (2.14)$$

Subtracting the population value from both sides makes the change explicit:

$$\hat{u} \log \hat{u} - u \log u \approx (\log u + 1)(\hat{u} - u). \quad (2.15)$$

Applying this expansion to the three sums in Equation (2.11) gives

$$\begin{aligned} \hat{I}(P) - I(P) &\approx \sum_{i,j} (\log p_{ij} + 1)(\hat{p}_{ij} - p_{ij}) \\ &\quad - \sum_i (\log p_{i+} + 1)(\hat{p}_{i+} - p_{i+}) \\ &\quad - \sum_j (\log p_{+j} + 1)(\hat{p}_{+j} - p_{+j}). \end{aligned} \quad (2.16)$$

The margin errors are sums of the corresponding cell errors:

$$\hat{p}_{i+} - p_{i+} = \sum_j (\hat{p}_{ij} - p_{ij}), \quad \hat{p}_{+j} - p_{+j} = \sum_i (\hat{p}_{ij} - p_{ij}). \quad (2.17)$$

For example, the row-margin sum can now be written as

$$\begin{aligned} \sum_i (\log p_{i+} + 1)(\hat{p}_{i+} - p_{i+}) &= \sum_i \left[ (\log p_{i+} + 1) \left\{ \sum_j (\hat{p}_{ij} - p_{ij}) \right\} \right] \\ &= \sum_{i,j} (\log p_{i+} + 1)(\hat{p}_{ij} - p_{ij}). \end{aligned} \quad (2.18)$$

The inner sum over  $j$  is the probability error for a fixed row  $i$ . It is multiplied by that row's coefficient  $\log p_{i+} + 1$ ; the outer sum then adds these products over all rows. The column-margin sum can be rearranged in the same way. Consequently, the same cell error  $\hat{p}_{ij} - p_{ij}$  appears in the joint term, its row-margin term and its column-margin term. Its combined coefficient is

$$\begin{aligned} &(\log p_{ij} + 1) - (\log p_{i+} + 1) - (\log p_{+j} + 1) \\ &= \log \left( \frac{p_{ij}}{p_{i+} p_{+j}} \right) - 1 \\ &= \ell_P(i, j) - 1. \end{aligned} \quad (2.19)$$



Therefore,

$$\hat{I}(P) - I(P) \approx \sum_{i,j} \{\ell_P(i,j) - 1\}(\hat{p}_{ij} - p_{ij}). \quad (2.20)$$

The empirical and population cell probabilities both sum to one, so

$$\sum_{i,j} (\hat{p}_{ij} - p_{ij}) = \sum_{i,j} \hat{p}_{ij} - \sum_{i,j} p_{ij} = 1 - 1 = 0. \quad (2.21)$$

The constant  $-1$  in Equation (2.19) therefore contributes zero. Explicitly,

$$\begin{aligned} \hat{I}(P) - I(P) &\approx \sum_{i,j} \{\ell_P(i,j) - 1\}(\hat{p}_{ij} - p_{ij}) \\ &= \sum_{i,j} \ell_P(i,j)(\hat{p}_{ij} - p_{ij}) - \sum_{i,j} (\hat{p}_{ij} - p_{ij}) \\ &= \sum_{i,j} \ell_P(i,j)(\hat{p}_{ij} - p_{ij}). \end{aligned} \quad (2.22)$$

The complicated changes in the joint table and its two margins have therefore reduced to a weighted sum of cell-probability errors. The weight for a cell is its population pointwise MI.

### Step 2: From cell errors to observation-level errors.

Let  $Z_a = (X_a, Y_a)$  be observation  $a$ . An empirical cell probability is the proportion of observations in that cell, so

$$\hat{p}_{ij} - p_{ij} = \frac{1}{n} \sum_{a=1}^n \{\mathbf{1}(Z_a = (i,j)) - p_{ij}\}. \quad (2.23)$$

Substituting this expression into Equation (2.22) first gives

$$\hat{I}(P) - I(P) \approx \sum_{i,j} \ell_P(i,j) \left[ \frac{1}{n} \sum_{a=1}^n \{\mathbf{1}(Z_a = (i,j)) - p_{ij}\} \right]. \quad (2.24)$$

The finite sums can be exchanged, so the contribution from observation  $a$  is gathered inside one pair of brackets:

$$\begin{aligned} \hat{I}(P) - I(P) &\approx \frac{1}{n} \sum_{a=1}^n \left[ \sum_{i,j} \ell_P(i,j) \mathbf{1}(Z_a = (i,j)) - \sum_{i,j} p_{ij} \ell_P(i,j) \right] \\ &= \frac{1}{n} \sum_{a=1}^n \{\ell_P(Z_a) - I(P)\}. \end{aligned} \quad (2.25)$$

The first sum inside the brackets selects the single cell containing  $Z_a$ , giving  $\ell_P(Z_a)$ . The second is the probability-weighted average of all pointwise-MI values, which is  $I(P)$ . This equation is the key step. To first order, the error in the plug-in

MI estimate behaves like the average of  $n$  observation-level contributions. The contribution of an observation in cell  $(i, j)$  is simply that cell's pointwise MI minus the population average,  $\ell_P(i, j) - I(P)$ . Its mean is zero because

$$\begin{aligned} E_P\{\ell_P(X, Y) - I(P)\} &= \sum_{i,j} p_{ij} \{\ell_P(i, j) - I(P)\} \\ &= \underbrace{\sum_{i,j} p_{ij} \ell_P(i, j)}_{I(P)} - I(P) \underbrace{\sum_{i,j} p_{ij}}_1 = 0. \end{aligned} \quad (2.26)$$

In formal asymptotic terminology, the centred function  $\ell_P(i, j) - I(P)$  is the influence function of MI; Equation (2.25) is its first-order linear representation (Hampel, 1974; van der Vaart, 1998; Kandasamy et al., 2015).

**Step 3: Calculate the sampling variance.**

Define the population variance of these centred contributions by

$$V(P) = \text{Var}_P\{\ell_P(X, Y) - I(P)\} = \sum_{i,j} p_{ij} \{\ell_P(i, j) - I(P)\}^2. \quad (2.27)$$

Under independent sampling, the  $n$  terms in Equation (2.25) are independent and each has variance  $V(P)$ . The covariance between any two different observations is therefore zero. Hence

$$\begin{aligned} \text{Var}\{\hat{I}(P)\} &\approx \text{Var}\left\{\frac{1}{n} \sum_{a=1}^n \{\ell_P(Z_a) - I(P)\}\right\} \\ &= \frac{1}{n^2} \sum_{a=1}^n \text{Var}\{\ell_P(Z_a) - I(P)\} \\ &= \frac{1}{n^2} \sum_{a=1}^n V(P) = \frac{nV(P)}{n^2} = \frac{V(P)}{n}. \end{aligned} \quad (2.28)$$

This proves the approximation stated in Equation (2.10). It is the familiar result that the variance of an average of  $n$  independent contributions is the variance of one contribution divided by  $n$ .

When  $V(P) > 0$ , the central limit theorem also gives

$$\frac{\hat{I}(P) - I(P)}{\sqrt{V(P)/n}} \xrightarrow{d} N(0, 1). \quad (2.29)$$

This does not claim that every finite-sample MI estimate is normally distributed. It is a local, large-sample approximation for a fixed table with positive population cell probabilities. The  $1/n$  leading bias from Section 2.2 decreases faster than the  $1/\sqrt{n}$  sampling error, so it does not change this limiting distribution.

**The independence boundary.**

At exact independence,  $p_{ij} = p_{i+}p_{+j}$ , so  $\ell_P(i, j) = \log 1 = 0$  in every cell. Hence  $I(P) = 0$ ,  $V(P) = 0$ , and the first-order average in Equation (2.25) vanishes.

A finite sample can still have positive MI. Its leading variation comes from the squared cell-probability errors in a second-order Taylor expansion, which produces the usual chi-squared limit (Brillinger, 2004; Marinescu and Balcau, 2025). The normal limit for comparing positive MI values and the chi-squared limit for testing independence therefore apply in different settings.

## 2.4 Related inferential methods

### Normal Wald inference.

A Wald test estimates a parameter difference, divides it by an estimated standard error, and compares the result with its asymptotic normal reference (Wald, 1943; van der Vaart, 1998). For two independent MI estimates, the two first-order variance contributions add:

$$\text{Var}\{\hat{I}(P) - \hat{I}(Q)\} \approx \frac{V(P)}{n_P} + \frac{V(Q)}{n_Q}. \quad (2.30)$$

Each term is the sampling variance of one MI estimate. Independence of the two samples makes their covariance zero. Replacing the population variances by estimates gives the standardisation

$$\frac{\hat{I}(P) - \hat{I}(Q)}{\sqrt{\hat{V}(P)/n_P + \hat{V}(Q)/n_Q}} \xrightarrow{d} N(0, 1) \quad \text{under } I(P) = I(Q). \quad (2.31)$$

Here  $\hat{V}(P)$  and  $\hat{V}(Q)$  estimate the pointwise-MI variances  $V(P)$  and  $V(Q)$ . This limit assumes the regular positive-support conditions of Section 2.3 and increasing sample sizes. Chapter 3 applies the leading bias correction to the numerator and specifies the full Normal Wald test.

With smaller samples, the MI difference may not follow the normal approximation closely, and the estimated standard error can vary between samples. The Wald test still uses a normal reference; it makes no separate small-sample adjustment for that standard-error variability.

### The Welch–Satterthwaite principle.

Welch’s test compares two means when their population variances may differ (Welch, 1947). For two independent samples with means  $\bar{X}_P, \bar{X}_Q$  and variances  $s_P^2, s_Q^2$ , the statistic for testing equal population means and its effective degrees of freedom are

$$T_{\text{Welch}} = \frac{\bar{X}_P - \bar{X}_Q}{\sqrt{s_P^2/n_P + s_Q^2/n_Q}}, \quad (2.32)$$

$$\nu = \frac{\left(s_P^2/n_P + s_Q^2/n_Q\right)^2}{\frac{(s_P^2/n_P)^2}{n_P - 1} + \frac{(s_Q^2/n_Q)^2}{n_Q - 1}}. \quad (2.33)$$

The denominator of the statistic contains one estimated variance contribution from each sample. Under normal sampling, their component degrees of freedom are  $n_P - 1$  and  $n_Q - 1$ . If  $s_P^2/n_P$  dominates, the formula makes the combined  $\nu$  close to  $n_P - 1$ ; when both contributions matter, both component values affect  $\nu$ . Satterthwaite obtains this form by matching the mean and variance of their sum to a scaled chi-squared variable (Satterthwaite, 1946).

Welch compares the statistic with a Student distribution using the calculated  $\nu$ . At the same significance level, this requires a larger absolute statistic to reject the null than a standard normal reference does. The difference between the Student and normal rejection thresholds becomes smaller as  $\nu$  increases.

For MI, the estimated variance contributions are  $\hat{V}(P)/n_P$  and  $\hat{V}(Q)/n_Q$ . Their component degrees of freedom need their own calculation because  $\hat{V}(P)$  and  $\hat{V}(Q)$  are nonlinear functions of the probability tables. Chapter 4 derives these MI-specific inputs to the same moment-matching structure.

Using the variability of a variance estimate to choose a reference distribution also has precedents in regression. Pan and Wall (2002) account for variability in sandwich variance estimates using approximate Student and  $F$  references. Bell and McCaffrey (2002) use Satterthwaite moment matching for regression variance estimates under multi-stage sampling after showing that traditional linearisation estimators can have substantial negative bias. Kauermann and Carroll (2001) examine the variability and efficiency of sandwich covariance estimation and propose an adjustment for the resulting confidence-interval undercoverage. Related adjustments and practical guidance are given by Fay and Graubard (2001) and Imbens and Kolesár (2016). These are close methodological precedents: estimating the variability of a variance estimator and using it in a finite-sample reference are established strategies. The present task is to derive the required inputs for the discrete MI functional and assess whether the resulting reference is useful.

The direction of the finite-sample problem is nevertheless different here. The cited regression corrections respond to underestimated uncertainty or anti-conservative coverage, so a heavier-tailed reference can move inference in the required direction. Near independence, the plug-in MI variance can instead be biased upward. A heavier-tailed Student reference may then reinforce, rather than offset, the resulting conservatism. This possibility is examined in the supplementary mechanism check in Section 6.7.

### Prior MI inference and related comparisons.

MI uncertainty also has a broader literature than its leading normal approximation. Roulston (1999) derives entropy and MI error estimates using error propagation

and examines them numerically. [Hutter \(2002\)](#) and [Hutter and Zaffalon \(2005\)](#) derive moments of the posterior distribution of MI under a Dirichlet model, including higher-order variance and shape information. Their posterior uncertainty has a different conditioning from the repeated-sampling distribution studied here. Their results establish that higher-order MI uncertainty is not a new topic, but do not directly validate the two-sample Student reference in this thesis.

Hutcheson’s test is a direct information-theoretic Welch predecessor. It compares two Shannon diversity estimates with an estimated variance and a Student reference, using

$$\nu_H = \frac{(s_{H,P}^2 + s_{H,Q}^2)^2}{(s_{H,P}^2)^2/n_P + (s_{H,Q}^2)^2/n_Q} \quad (2.34)$$

as its effective degrees of freedom ([Hutcheson, 1970](#)). Here each of  $s_{H,P}^2$  and  $s_{H,Q}^2$  is the estimated sampling variance of the corresponding entropy estimator  $\widehat{H}$ , and each variance component is assigned its sample size, not  $n_i - 1$ . Applying the same assignment to the two MI variance contributions gives a clear historical baseline. It does not, however, derive how the plug-in MI variance itself fluctuates: MI depends on a joint distribution and both margins, so moving mass into one cell changes pointwise values elsewhere in its row and column. The expanded calculation retains these changes.

The question of comparing MI values also has predecessors. In a segregation application, [Mora and Ruiz-Castillo \(2009\)](#) develop inference for an MI index and discuss pairwise comparisons across groups and periods; the later peer-reviewed account places that MI index within the wider segregation literature ([Mora and Ruiz-Castillo, 2011](#)). They do not derive the variance-sensitivity degrees of freedom used here, but the general aim of comparing MI values is not new. [Guha and Chothia \(2014\)](#) use MI for a different two-sample question: whether two continuous distributions are identical. Their MI is between a pooled observation and its sample label; their null is equality of distributions, not equality of two within-population MI values. For discrete data, [Drake and Guha \(2014\)](#) develop an MI-based  $k$ -sample test for the same distributional-equality question.

The exact-independence boundary has likewise been studied separately. [Marinescu and Balcau \(2025\)](#) use a second-order delta-method argument for MI under independence. More generally, their expansion combines a normal term with a linear combination of chi-squared terms; the normal term vanishes at exact independence. That setting explains why a first-order positive-MI comparison must not be presented as a replacement for a one-table independence test and suggests why second-order effects can remain relevant near that boundary.

Table 2.1 distinguishes the relevant established ideas by the problem each

addresses. This is not a claim that no other MI comparison has been proposed. It states which pieces are used here and why the final comparison still needs empirical validation.

**Table 2.1:** Selected prior work, grouped by the question it addresses.

| Topic              | Established result and relation to this thesis                                                                                                                                                                                                                                                                                               |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MI estimation      | <a href="#">Miller (1955)</a> and <a href="#">Paninski (2003)</a> establish bias and sparse-sample limitations of plug-in information estimates.                                                                                                                                                                                             |
| MI variance        | <a href="#">Moddemeijer (1989)</a> and <a href="#">Brillinger (2004)</a> give predecessors for the first-order MI variance, while <a href="#">Moddemeijer (1999)</a> treats dependent observation pairs. This thesis instead studies the sampling uncertainty of the plug-in variance estimate itself in the independent two-sample problem. |
| Welch principle    | <a href="#">Satterthwaite (1946)</a> and <a href="#">Welch (1947)</a> provide moment-matching ideas; <a href="#">Hutcheson (1970)</a> applies the same structure to Shannon diversity using component degrees of freedom $n_i$ .                                                                                                             |
| Other comparisons  | <a href="#">Mora and Ruiz-Castillo (2009, 2011)</a> study an MI-based segregation index. They do not derive the variance-sensitivity degrees of freedom used here.                                                                                                                                                                           |
| Different MI nulls | <a href="#">Guha and Chothia (2014)</a> and <a href="#">Drake and Guha (2014)</a> test distributional equality; <a href="#">Marinescu and Balcau (2025)</a> study one-table independence. These are not the two-population equal-MI null.                                                                                                    |
| Weak-null tests    | <a href="#">Chung and Romano (2013)</a> explain why a raw pooled-label permutation need not be valid when the compared parameters agree but $P \neq Q$ ; <a href="#">Janssen (1997)</a> gives an earlier studentized permutation result for heterogeneous two-sample nulls.                                                                  |

### Resampling and the weak null.

Bootstrap methods can approximate a statistic’s repeated-sampling behaviour by drawing from fitted empirical populations, at greater computational cost than a direct analytic table scan ([Efron, 1979](#)). Sparse empirical support can still be a problem. A label permutation instead pools the observations and reallocates their group labels; it targets a different symmetry assumption.

Permutation requires a separate distinction. Pooling two samples and permuting their group labels is exact under the strong null  $P = Q$ , because the observations are exchangeable. Under the weak null  $I(P) = I(Q)$ , the two complete distributions may differ, so raw label permutation generally targets the wrong null. Studentized permutation can recover asymptotic validity for some weak-null parameter comparisons ([Janssen, 1997](#); [Chung and Romano, 2013](#)), but it is not the standard one-table MI shuffle used to test independence.

The final comparison uses Normal Wald and the proposed Expanded Welch method, which share the same effect estimate and standard error. Resampling is relevant future work, but is not an exact finite-sample solution for the unrestricted weak null without additional assumptions.

## 2.5 Literature synthesis and methodological gap

The literature establishes all of the broad ingredients used in this thesis: plug-in MI estimation and bias correction, first-order MI variance, Wald comparison of smooth functionals, and Welch–Satterthwaite moment matching. It also contains prior comparisons of entropy and MI-related indices.

This thesis makes three contributions to two-sample inference for mutual information. First, it derives an explicit observation-level sensitivity for the estimated MI variance. Second, it uses this sensitivity to construct a plug-in Welch–Satterthwaite degrees-of-freedom correction. Third, it provides a controlled empirical evaluation across distinct equal-MI multinomial populations. Together, these contributions clarify when Welch–Satterthwaite correction is useful for MI comparison and, importantly, why it can become overly conservative.

The remaining task is specific to MI. Hutcheson’s direct information-theoretic predecessor assigns each component its sample size; an ordinary-Welch analogue uses  $n - 1$ . Both are fixed assignments. By contrast, established sandwich adjustments derive effective degrees of freedom from the estimated variability of the variance estimator itself. The plug-in MI variance requires its own version of that calculation because it is a nonlinear function of the same empirical joint table used to calculate MI. A change in one observation modifies the cell probabilities, margins, pointwise-MI values, and their weighted variance.

Chapters 3 and 4 develop the MI-specific variance sensitivity and its Welch–Satterthwaite reference. Chapter 6 evaluates when that adjustment helps.



## Chapter 3

# Comparing Mutual Information with Normal Wald

This chapter defines the estimand and statistic shared by the two methods in the final experiment. Normal Wald evaluates it against a standard normal reference; Chapter 4 develops a Student reference with estimated degrees of freedom.

### 3.1 Problem setup and bias-corrected estimator

Let

$$N^P = (N_{ij}^P), \quad N^Q = (N_{ij}^Q) \quad (3.1)$$

be independent  $r \times c$  count tables with totals  $n_P$  and  $n_Q$ . The row and column categories have the same definitions in both populations. The empirical probabilities are

$$\hat{p}_{ij} = \frac{N_{ij}^P}{n_P}, \quad \hat{q}_{ij} = \frac{N_{ij}^Q}{n_Q}. \quad (3.2)$$

The population parameter is the signed difference

$$\Delta = I(P) - I(Q). \quad (3.3)$$

The principal test is two-sided:

$$H_0 : \Delta = 0 \quad \text{against} \quad H_1 : \Delta \neq 0. \quad (3.4)$$

The first-order construction uses the following assumptions.

**Assumption 1** (Sampling). *Observations are independent and identically distributed within each population, and the two population samples are independent.*

**Assumption 2** (Aligned fixed alphabets). *The row and column categories are fixed in advance and aligned between the two tables.*

**Assumption 3** (Positive support). *Every population cell has positive probability. Empirical zero cells are allowed, but widespread observed support loss is recorded as a diagnostic.*

**Assumption 4** (Regular MI). *The first-order influence variances  $V(P)$  and  $V(Q)$  are positive. This excludes exact independence and other degenerate cases from the regular first-order approximation.*

All logarithms are natural. MI differences and standard errors are therefore reported in nats. Changing the logarithm base scales both by the same positive constant and leaves the standardised statistic unchanged.

#### Bias-corrected estimator.

For the first table, define empirical pointwise MI by

$$\hat{\ell}_P(i, j) = \log \left( \frac{\hat{p}_{ij}}{\hat{p}_{i+}\hat{p}_{+j}} \right) \quad \text{when } \hat{p}_{ij} > 0. \quad (3.5)$$

The plug-in MI estimate is

$$\hat{I}(P) = \sum_{i,j:\hat{p}_{ij}>0} \hat{p}_{ij} \hat{\ell}_P(i, j). \quad (3.6)$$

The same calculation gives  $\hat{I}(Q)$ .

The leading fixed-support MI bias appears in Equation (2.8) (Moddemeijer, 1989). Subtracting it from each plug-in estimate gives the estimated effect:

$$\hat{\Delta}_{\text{BC}} = \left\{ \hat{I}(P) - \frac{(r-1)(c-1)}{2n_P} \right\} - \left\{ \hat{I}(Q) - \frac{(r-1)(c-1)}{2n_Q} \right\}. \quad (3.7)$$

The two terms in braces are the bias-corrected MI estimates for the separate populations.

The correction uses the declared alphabet dimensions, not the observed nonempty dimensions. A corrected estimate can be negative in a sparse sample and is not clipped at zero, since clipping would define a different nonlinear estimator. When  $n_P = n_Q$ , the two corrections cancel in the difference; otherwise, their difference removes the leading unequal-sample bias described in Section 2.2.

## 3.2 Shared standard error and test statistic

Section 2.3 showed that the first-order sampling variance of  $\hat{I}(P)$  is  $V(P)/n_P$ , where  $V(P)$  is the population variance of pointwise MI (Moddemeijer, 1989; Brillinger, 2004). Subtracting the fixed leading bias term in Equation (3.7) does not change this variance.

The observed table estimates  $V(P)$  by

$$\hat{V}(P) = \sum_{i,j:\hat{p}_{ij}>0} \hat{p}_{ij} \{\hat{\ell}_P(i,j) - \hat{I}(P)\}^2. \quad (3.8)$$

Define  $\hat{V}(Q)$  analogously.

Because the two samples are independent, their estimated variance contributions add:

$$\widehat{\text{SE}}^2 = \frac{\hat{V}(P)}{n_P} + \frac{\hat{V}(Q)}{n_Q}. \quad (3.9)$$

The common standardised statistic is

$$T = \frac{\hat{\Delta}_{\text{BC}}}{\widehat{\text{SE}}}. \quad (3.10)$$

The null hypothesis is not imposed when calculating  $\hat{\Delta}_{\text{BC}}$ ,  $\widehat{\text{SE}}$  or  $T$ : they use the observed tables. Equal population MI enters when interpreting the reference distribution, which is centred at zero under  $H_0$ . Equations (3.7)–(3.10) are shared by Normal Wald and Expanded Welch. Their difference lies in the reference distribution, not the effect estimate or plug-in standard error.

## 3.3 Normal Wald reference and motivation for expansion

Under the regular first-order asymptotics for a scalar Wald comparison (Wald, 1943; van der Vaart, 1998),

$$T \xrightarrow{d} N(0, 1) \quad \text{under } H_0. \quad (3.11)$$

Here both  $n_P$  and  $n_Q$  increase while the aligned alphabet and the two positive-support population tables remain fixed. The two-sided Normal Wald p-value is

$$p_{\text{Wald}} = 2\{1 - \Phi(|T|)\}, \quad (3.12)$$

where  $\Phi$  is the standard normal distribution function. The thesis evaluates this test through p-values and rejection rates. It does not claim confidence-interval coverage, which would require a separate coverage experiment.

The normal reference is a large-sample approximation. Wald estimates the standard error from the tables but applies no degrees-of-freedom adjustment.

### Why consider an expanded reference?

The shared statistic supplies the estimated effect and its first-order standard error. The direct Hucheson-style information-theoretic analogue assigns component degrees of freedom  $n_P$  and  $n_Q$  (Hutcheson, 1970); a familiar ordinary-Welch analogue uses  $n_P - 1$  and  $n_Q - 1$  (Welch, 1947). For positive variance contributions, their combined degree of freedom is at least  $\min(n_P, n_Q) - 1$ , so at sample sizes of 50 or more their Student reference is already close to normal. The supplementary ablation retains these baselines alongside Expanded Welch. However,  $\hat{V}(P)$  is a nonlinear functional of a table whose margins and pointwise MI values also change with the data. The expanded method instead estimates component degrees of freedom from that full functional.

Satterthwaite moment matching requires the first two moments of each positive variance estimate (Satterthwaite, 1946). To first order,

$$\mathbb{E}\{\hat{V}(P)\} \approx V(P). \quad (3.13)$$

Chapter 4 derives  $\text{Var}\{\hat{V}(P)\}$  by calculating how the full variance functional changes when one observation changes the joint table.

# Chapter 4

## Expanded Welch–Satterthwaite Method

The expanded method retains the bias-corrected difference, standard error, and test statistic from Chapter 3. It evaluates that statistic against an estimated Student reference rather than the standard normal. This chapter derives the required degrees of freedom from the sampling variability of the full MI variance estimator.

### 4.1 From Welch–Satterthwaite to the required MI quantity

For the two MI estimates, the Welch–Satterthwaite equation combines their estimated variance contributions as in the original moment-matching framework (Satterthwaite, 1946; Welch, 1947):

$$\nu = \frac{\left\{ \widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q \right\}^2}{\left\{ \widehat{V}(P)/n_P \right\}^2 / \nu_P + \left\{ \widehat{V}(Q)/n_Q \right\}^2 / \nu_Q}. \quad (4.1)$$

The numerator contains the two terms in the squared standard error. The remaining quantities are the component degrees of freedom  $\nu_P$  and  $\nu_Q$ , which should measure how reliably  $\widehat{V}(P)$  and  $\widehat{V}(Q)$  estimate their population counterparts.

The derivation proceeds in four stages. First, Satterthwaite moment matching expresses  $\nu_P$  in terms of the mean and variance of  $\widehat{V}(P)$ . Second, a one-cell perturbation gives the observation-level sensitivity of  $V(P)$ . Third, the variability of these sensitivities gives the sampling variance of  $\widehat{V}(P)$ . Finally, the two estimated component degrees of freedom are combined through Equation (4.1).

**Moment matching for one component.**

Satterthwaite represents a positive variance estimate by a scaled chi-squared variable. For the first population, the working approximation represents  $\widehat{V}(P)$  by

$$\mathbb{E}\{\widehat{V}(P)\} \frac{\chi_{\nu_P}^2}{\nu_P}. \quad (4.2)$$

The approximation matches moments; it does not assert that the finite-sample distribution of  $\widehat{V}(P)$  is exactly chi-squared.

For a chi-squared variable with  $\nu_P$  degrees of freedom,

$$\mathbb{E}(\chi_{\nu_P}^2) = \nu_P, \quad \text{Var}(\chi_{\nu_P}^2) = 2\nu_P. \quad (4.3)$$

The scaling in Equation (4.2) therefore matches the mean automatically:

$$\mathbb{E} \left[ \mathbb{E}\{\widehat{V}(P)\} \frac{\chi_{\nu_P}^2}{\nu_P} \right] = \mathbb{E}\{\widehat{V}(P)\}. \quad (4.4)$$

Its variance is

$$\text{Var} \left[ \mathbb{E}\{\widehat{V}(P)\} \frac{\chi_{\nu_P}^2}{\nu_P} \right] = \frac{[\mathbb{E}\{\widehat{V}(P)\}]^2}{\nu_P^2} \text{Var}(\chi_{\nu_P}^2) \quad (4.5)$$

$$= \frac{2[\mathbb{E}\{\widehat{V}(P)\}]^2}{\nu_P}. \quad (4.6)$$

Matching this quantity to  $\text{Var}\{\widehat{V}(P)\}$  gives

$$\boxed{\nu_P = \frac{2[\mathbb{E}\{\widehat{V}(P)\}]^2}{\text{Var}\{\widehat{V}(P)\}}}. \quad (4.7)$$

To first order,  $\mathbb{E}\{\widehat{V}(P)\} \approx V(P)$ . The required calculation is therefore the sampling variance of  $\widehat{V}(P)$ .

**4.2 Deriving the sensitivity of the MI variance**

The sampling behaviour of a smooth plug-in functional can be obtained by measuring its first-order response to one observation ([Hampel, 1974](#); [van der Vaart, 1998](#)). Fix a cell  $(x, y)$  and define the perturbed distribution

$$P_\varepsilon = (1 - \varepsilon)P + \varepsilon\delta_{(x,y)}, \quad (4.8)$$

where  $\delta_{(x,y)}$  places probability one on cell  $(x, y)$ . At  $\varepsilon = 0$ , the distribution is  $P$ . Increasing  $\varepsilon$  moves a small amount of probability toward the selected cell while preserving total probability one.

For an arbitrary cell  $(i, j)$ ,

$$p_{ij}(\varepsilon) = (1 - \varepsilon)p_{ij} + \varepsilon \mathbf{1}\{i = x, j = y\}. \quad (4.9)$$

The joint and marginal probability derivatives at the original distribution are

$$\left. \frac{dp_{ij}(\varepsilon)}{d\varepsilon} \right|_0 = \mathbf{1}\{i = x, j = y\} - p_{ij}, \quad (4.10)$$

$$\left. \frac{dp_{i+}(\varepsilon)}{d\varepsilon} \right|_0 = \mathbf{1}\{i = x\} - p_{i+}, \quad (4.11)$$

$$\left. \frac{dp_{+j}(\varepsilon)}{d\varepsilon} \right|_0 = \mathbf{1}\{j = y\} - p_{+j}. \quad (4.12)$$

The derivatives of the joint probabilities sum to zero because

$$\sum_{i,j} \{\mathbf{1}(i = x, j = y) - p_{ij}\} = 1 - 1 = 0. \quad (4.13)$$

### Effect on pointwise MI and MI.

Along the perturbation path, pointwise MI is

$$\ell_{P_\varepsilon}(i, j) = \log p_{ij}(\varepsilon) - \log p_{i+}(\varepsilon) - \log p_{+j}(\varepsilon). \quad (4.14)$$

Differentiating the three logarithms and substituting the probability derivatives gives

$$\left. \frac{d}{d\varepsilon} \ell_{P_\varepsilon}(i, j) \right|_0 = \frac{\mathbf{1}\{i = x, j = y\}}{p_{ij}} - \frac{\mathbf{1}\{i = x\}}{p_{i+}} - \frac{\mathbf{1}\{j = y\}}{p_{+j}} + 1. \quad (4.15)$$

MI along the path is

$$I(P_\varepsilon) = \sum_{i,j} p_{ij}(\varepsilon) \ell_{P_\varepsilon}(i, j). \quad (4.16)$$

The product rule separates its derivative into the change in probability weights and the change in pointwise-MI values:

$$\begin{aligned} \left. \frac{dI(P_\varepsilon)}{d\varepsilon} \right|_0 &= \sum_{i,j} \{ \mathbf{1}(i=x, j=y) - p_{ij} \} \ell_P(i, j) \\ &\quad + \sum_{i,j} p_{ij} \left. \frac{d\ell_{P_\varepsilon}(i, j)}{d\varepsilon} \right|_0. \end{aligned} \quad (4.17)$$

The first sum is  $\ell_P(x, y) - I(P)$ . In the second sum, the selected-cell term contributes  $p_{xy}/p_{xy} = 1$ . The row term contributes  $-\sum_j p_{xj}/p_{x+} = -1$ ; the column term likewise contributes  $-1$ ; and the constant contributes  $\sum_{i,j} p_{ij} = 1$ . These four contributions cancel. Hence

$$\boxed{\left. \frac{dI(P_\varepsilon)}{d\varepsilon} \right|_0 = \ell_P(x, y) - I(P)}. \quad (4.18)$$

This result recovers the centred pointwise-MI influence function used in Chapter 3.

#### Effect on the MI variance.

The target is the first-order change in

$$V(P) = \text{Var}_P\{\ell_P(X, Y)\}. \quad (4.19)$$

Write this variance in second-moment form:

$$V(P) = \mathbb{E}_P\{\ell_P(X, Y)^2\} - I(P)^2. \quad (4.20)$$

Define the observation-level variance sensitivity by

$$g_P(x, y) = \left. \frac{dV(P_\varepsilon)}{d\varepsilon} \right|_0. \quad (4.21)$$

Differentiating Equation (4.20) gives

$$g_P(x, y) = \left. \frac{d}{d\varepsilon} \mathbb{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X, Y)^2\} \right|_0 - 2I(P) \left. \frac{dI(P_\varepsilon)}{d\varepsilon} \right|_0. \quad (4.22)$$

It remains to differentiate

$$\mathbb{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X, Y)^2\} = \sum_{i,j} p_{ij}(\varepsilon) \ell_{P_\varepsilon}(i, j)^2. \quad (4.23)$$



The product rule gives

$$\begin{aligned} \left. \frac{d}{d\varepsilon} E_{P_\varepsilon} \{ \ell_{P_\varepsilon}(X, Y)^2 \} \right|_0 &= \sum_{i,j} \left. \frac{dp_{ij}(\varepsilon)}{d\varepsilon} \right|_0 \ell_P(i, j)^2 \\ &\quad + 2 \sum_{i,j} p_{ij} \ell_P(i, j) \left. \frac{d\ell_{P_\varepsilon}(i, j)}{d\varepsilon} \right|_0. \end{aligned} \quad (4.24)$$

The first sum is

$$\ell_P(x, y)^2 - E_P \{ \ell_P(X, Y)^2 \}. \quad (4.25)$$

For the second sum, define the probability-weighted row and column means

$$E_P \{ \ell_P(X, Y) \mid X = x \} = \frac{\sum_j p_{xj} \ell_P(x, j)}{p_{x+}}, \quad (4.26)$$

$$E_P \{ \ell_P(X, Y) \mid Y = y \} = \frac{\sum_i p_{iy} \ell_P(i, y)}{p_{+y}}. \quad (4.27)$$

Substitution of Equation (4.15) gives

$$\begin{aligned} &\sum_{i,j} p_{ij} \ell_P(i, j) \left. \frac{d\ell_{P_\varepsilon}(i, j)}{d\varepsilon} \right|_0 \\ &= \ell_P(x, y) - \frac{\sum_j p_{xj} \ell_P(x, j)}{p_{x+}} - \frac{\sum_i p_{iy} \ell_P(i, y)}{p_{+y}} + I(P) \\ &= \ell_P(x, y) - E_P \{ \ell_P(X, Y) \mid X = x \} - E_P \{ \ell_P(X, Y) \mid Y = y \} + I(P). \end{aligned} \quad (4.28)$$

Combining Equations (4.18), (4.22), (4.25), and (4.28) yields

$$\boxed{\begin{aligned} g_P(x, y) &= \{ \ell_P(x, y) - I(P) \}^2 - V(P) \\ &\quad + 2 \left[ \ell_P(x, y) - E_P \{ \ell_P(X, Y) \mid X = x \} \right. \\ &\quad \left. - E_P \{ \ell_P(X, Y) \mid Y = y \} + I(P) \right]. \end{aligned}} \quad (4.29)$$

The first line is the change if the pointwise-MI values were held fixed. The bracket accounts for changes in those values, including the effects of the selected joint probability and both margins.

## 4.3 Component degrees of freedom and final test

**Population component degrees of freedom.**

Define the population variability of the observation-level sensitivities by

$$\tau^2(P) = \text{Var}_P\{g_P(X, Y)\}. \quad (4.30)$$

Averaging the one-cell perturbation directions gives  $\sum_{x,y} p_{xy}\{\delta_{(x,y)} - P\} = 0$ . The derivative is linear in its direction, so  $E_P\{g_P(X, Y)\} = 0$ ; Appendix A shows the steps. Explicit centring protects the numerical calculation if roundoff produces a small nonzero empirical mean.

**Assumption 5** (Regular variance sensitivity). *For the first-order Expanded Welch construction,  $\tau^2(P) > 0$  and  $\tau^2(Q) > 0$ .*

The empirical distribution is

$$\hat{P} - P = \frac{1}{n_P} \sum_{a=1}^{n_P} \{\delta_{Z_a^{(P)}} - P\}. \quad (4.31)$$

Taylor's first-order term is linear in this difference (van der Vaart, 1998, Chapter 20). By Equation (4.21), each summand has sensitivity  $g_P(Z_a^{(P)})$ . Therefore,

$$\hat{V}(P) - V(P) \approx \frac{1}{n_P} \sum_{a=1}^{n_P} g_P(Z_a^{(P)}). \quad (4.32)$$

The observations are independent, so

$$\text{Var}\{\hat{V}(P)\} \approx \frac{1}{n_P^2} \sum_{a=1}^{n_P} \text{Var}_P\{g_P(Z_a^{(P)})\} \quad (4.33)$$

$$= \frac{\tau^2(P)}{n_P}. \quad (4.34)$$

Substitution into Equation (4.7) gives the population component degree of freedom

$$\boxed{\nu_P \approx \frac{2n_P V(P)^2}{\tau^2(P)}}. \quad (4.35)$$

Equation (4.35) depends on relative, not absolute, uncertainty. Since

$$\frac{\text{Var}\{\hat{V}(P)\}}{V(P)^2} \approx \frac{\tau^2(P)}{n_P V(P)^2}, \quad (4.36)$$

the component degrees of freedom are approximately twice the reciprocal of this relative variance. A stable estimate relative to the size of  $V(P)$  has large degrees of freedom; a relatively unstable estimate has smaller degrees of freedom and produces a heavier-tailed final reference. This first-order component formula assumes  $\tau^2(P) > 0$ .

Positive  $V(P)$  alone does not guarantee that condition: at a stationary point of the variance functional, its first-order sensitivity can vanish even though MI and  $V(P)$  are positive. There, a higher-order analysis of  $\hat{V}(P)$  would be needed to describe its finite-sample distribution.

### Table-based component degrees of freedom.

The observed table replaces every population quantity in Equation (4.29) by its empirical counterpart. The sums below are over occupied cells, because pointwise MI is undefined at an empirical zero. For each occupied cell,

$$\begin{aligned} \hat{g}_P(i, j) = & \{\hat{\ell}_P(i, j) - \hat{I}(P)\}^2 - \hat{V}(P) \\ & + 2 \left[ \hat{\ell}_P(i, j) - \frac{\sum_{j'} \hat{p}_{ij'} \hat{\ell}_P(i, j')}{\hat{p}_{i+}} - \frac{\sum_{i'} \hat{p}_{i'j} \hat{\ell}_P(i', j)}{\hat{p}_{+j}} + \hat{I}(P) \right]. \end{aligned} \quad (4.37)$$

In all weighted sums, an empty cell contributes zero; no logarithm or sensitivity needs to be evaluated there. The empirical variance is centred explicitly:

$$\hat{\tau}^2(P) = \sum_{i,j} \hat{p}_{ij} \left\{ \hat{g}_P(i, j) - \sum_{k,l} \hat{p}_{kl} \hat{g}_P(k, l) \right\}^2. \quad (4.38)$$

The estimated component degree of freedom is

$$\boxed{\hat{\nu}_P = \frac{2n_P \hat{V}(P)^2}{\hat{\tau}^2(P)}}. \quad (4.39)$$

Repeating the same calculation for the second table gives  $\hat{\nu}_Q$ .

### Combining the two components.

Substituting the two variance contributions and component degrees of freedom into Equation (4.1) gives

$$\boxed{\hat{\nu}_{\text{expanded}} = \frac{\{\hat{V}(P)/n_P + \hat{V}(Q)/n_Q\}^2}{\{\hat{V}(P)/n_P\}^2 / \hat{\nu}_P + \{\hat{V}(Q)/n_Q\}^2 / \hat{\nu}_Q}}. \quad (4.40)$$

Substituting Equation (4.39) and its  $Q$  analogue gives the algebraically equivalent form

$$\hat{\nu}_{\text{expanded}} = \frac{\{\hat{V}(P)/n_P + \hat{V}(Q)/n_Q\}^2}{\hat{\tau}^2(P)/(2n_P^3) + \hat{\tau}^2(Q)/(2n_Q^3)}. \quad (4.41)$$

Equation (4.41) remains defined when one table has  $\hat{V} = \hat{\tau}^2 = 0$  and the other supplies a positive denominator contribution. The zero table drops out, so the

combined degree of freedom reduces to the other table's component. The implementation evaluated in this thesis does not use the direct form in that case: it first requires both component degrees of freedom in Equation (4.39) to be positive and finite. Reported Expanded-only invalidity can therefore reflect this rule rather than a failure of the combined Satterthwaite formula.

The expanded two-sided p-value is

$$p_{\text{expanded}} = 2 \left\{ 1 - F_{t_{\hat{\nu}_{\text{expanded}}}}(|T|) \right\}, \quad (4.42)$$

where  $F_{t_{\nu}}$  denotes the cumulative distribution function of a Student  $t$  variable with  $\nu$  degrees of freedom. As with Wald, only test behaviour is evaluated; no confidence-interval coverage claim is made.

The expanded calculation does not change  $\hat{\Delta}_{\text{BC}}$ ,  $\widehat{\text{SE}}$ , or  $T$ . It changes only the quantile used to evaluate that statistic.

**Proposition 1** (Nested rejection under the implemented rules). *If the two calculations use the same finite statistic  $T$  and Expanded Welch has positive finite degrees of freedom, its two-sided p-value is at least the Normal Wald p-value. The implemented Expanded-valid set is a subset of the Wald-valid set. Thus, at a common fixed threshold, every Expanded Welch rejection is also a Wald rejection, both replicate by replicate and in the unconditional rates that count invalid outputs as non-rejections.*

*Proof.* Write a Student variable as  $Z/\sqrt{S}$ , with independent  $Z \sim N(0, 1)$  and  $S \sim \chi_{\nu}^2/\nu$ . For  $c > 0$ , the conditional two-sided tail  $2\{1 - \Phi(c\sqrt{s})\}$  is strictly convex in  $s$ , because its second derivative is  $c\phi(c\sqrt{s})(s^{-3/2} + c^2s^{-1/2})/2 > 0$ . Since  $\mathbb{E} S = 1$ , Jensen's inequality gives  $\Pr(|Z/\sqrt{S}| > c) > \Pr(|Z| > c)$ . At  $c = 0$  both tails equal one.  $\square$

Invalid Expanded outputs count as non-rejections in the reported unconditional rates. Because the correction changes only the reference, it cannot raise rejection where Wald is already too conservative.

## 4.4 Worked example and implementation

Consider the two observed  $3 \times 3$  tables

$$N^P = \begin{pmatrix} 35 & 10 & 5 \\ 10 & 20 & 5 \\ 5 & 5 & 10 \end{pmatrix}, \quad N^Q = \begin{pmatrix} 30 & 5 & 5 \\ 5 & 30 & 5 \\ 5 & 5 & 10 \end{pmatrix}. \quad (4.43)$$

Their totals are  $n_P = 105$  and  $n_Q = 100$ . The plug-in MI estimates are 0.13693 and 0.25848 nats. Since  $(3 - 1)(3 - 1) = 4$ , the corrected difference is

$$\hat{\Delta}_{\text{BC}} = \left(0.13693 - \frac{4}{2(105)}\right) - \left(0.25848 - \frac{4}{2(100)}\right) = -0.12059. \quad (4.44)$$

The empirical influence variances are  $\hat{V}(P) = 0.27189$  and  $\hat{V}(Q) = 0.43371$ . Hence

$$\widehat{\text{SE}} = \left(\frac{0.27189}{105} + \frac{0.43371}{100}\right)^{1/2} = 0.08323, \quad T = -1.44898. \quad (4.45)$$

Normal Wald gives  $p = 0.14734$ .

For Expanded Welch, the table-based sensitivity calculation first gives  $\hat{\tau}^2(P) = 0.93212$  and  $\hat{\tau}^2(Q) = 0.82035$ . Substitution into Equation (4.39) gives component degrees of freedom

$$\hat{\nu}_P = 16.65, \quad \hat{\nu}_Q = 45.86. \quad (4.46)$$

Equation (4.40) combines them into  $\hat{\nu}_{\text{expanded}} = 59.03$ , giving  $p = 0.15264$ . Both methods fail to reject at the 5% level for these tables. The example shows that Expanded Welch changes the reference, not the observed statistic.

### Operational algorithm and complexity.

Table 4.1 summarises one comparison.

**Table 4.1:** Operational Expanded Welch–Satterthwaite algorithm.

| Step | Calculation                                                                                                                   |
|------|-------------------------------------------------------------------------------------------------------------------------------|
| 1    | Validate the two aligned count tables and their declared alphabet dimensions; independence of samples is a design assumption. |
| 2    | Compute empirical probabilities, pointwise MI, plug-in MI, and the leading bias-corrected difference.                         |
| 3    | Compute $\hat{V}(P)$ , $\hat{V}(Q)$ , the shared standard error, and $T$ .                                                    |
| 4    | Compute the row and column pointwise-MI means needed in $\hat{g}_P$ and $\hat{g}_Q$ .                                         |
| 5    | Compute $\hat{\tau}^2(P)$ , $\hat{\tau}^2(Q)$ , and the two component degrees of freedom.                                     |
| 6    | Combine the components using Equation (4.40).                                                                                 |
| 7    | Return the Student p-value, effective degrees of freedom, and validity diagnostics.                                           |

Each empirical probability, pointwise-MI value, row mean, column mean, and variance contribution is calculated by a fixed number of table scans. The time complexity is therefore  $O(rc)$ , and the working memory is also  $O(rc)$ . The method is deterministic once the two count tables are given.

## 4.5 Behaviour and limitations

### Near independence.

The adjustment has a local explanation. Let  $P_t = P_0 + tH$ , where  $P_0 = (q_{ij})$  is independent with positive cells  $q_{ij} = a_i b_j$ , and  $H$  has zero row and column sums. Thus  $t = 0$  is the expansion point and the margins stay fixed. For  $f(t) = I(P_t)$ , the second-order Taylor expansion has  $f(0) = 0$ ,  $f'(0) = \sum_{i,j} H_{ij} = 0$ , and  $f''(0) = \sum_{i,j} H_{ij}^2 / q_{ij}$ .

$$I(P_t) = \underbrace{0}_{\text{population value}} + \underbrace{0t}_{\text{first order}} + \underbrace{\frac{t^2}{2} \sum_{i,j} \frac{H_{ij}^2}{q_{ij}}}_{\text{second order}} + O(t^3), \quad (4.47)$$

$$\ell_{P_t}(i, j) = \log \left( 1 + \frac{tH_{ij}}{q_{ij}} \right) = \frac{tH_{ij}}{q_{ij}} + O(t^2), \quad (4.48)$$

$$V(P_t) = t^2 \sum_{i,j} \frac{H_{ij}^2}{q_{ij}} + O(t^3). \quad (4.49)$$

Appendix D gives the second-order MI result when the margins can also change. The row and column conditional means of pointwise MI have no first-order term because the corresponding sums of  $H$  are zero. Equation (4.29) then gives  $g_{P_t}(i, j) = 2tH_{ij}/q_{ij} + O(t^2)$ , and hence

$$\tau^2(P_t) = 4t^2 \sum_{i,j} \frac{H_{ij}^2}{q_{ij}} + O(t^3). \quad (4.50)$$

For a distribution close to independence on this path, these expansions give

$$V(P) \approx 2I(P), \quad \tau^2(P) \approx 4V(P). \quad (4.51)$$

Substitution into the component formula then gives

$$\nu_P = \frac{2n_P V(P)^2}{\tau^2(P)} \approx n_P I(P). \quad (4.52)$$

Thus low MI can produce a small effective degree of freedom and a much heavier-tailed reference. The approximation is local: it assumes fixed table dimension, positive probabilities and sufficiently regular cell counts. Very near independence, quadratic sampling variation can still be important, so a first-order Student adjustment need not improve finite-sample calibration.

### Theoretical status and validity conditions.

The influence-function calculations in Equations (4.18) and (4.29) are directional

derivatives of the corresponding population functionals. Under fixed positive support, they provide the usual first-order asymptotic linear representations.

The scaled chi-squared model in Equation (4.2) and the final Student reference are working approximations. They match the first two moments of the estimated variance and do not provide an exact finite-sample law. Moreover,  $\hat{I}(P)$  and  $\hat{V}(P)$  use the same observations; their first-order covariance is generally  $E_P[\{\ell_P(X, Y) - I(P)\}g_P(X, Y)]/n_P$ , not zero. Moment matching the denominator does not establish independence from the numerator or an exact Student law. Under regular large-sample conditions with positive variance sensitivities,  $\hat{\nu}_P$ ,  $\hat{\nu}_Q$ , and  $\hat{\nu}_{\text{expanded}}$  increase, so the Student reference approaches the same normal limit as Wald inference.

The software first requires matching table shapes, at least two rows and two columns, finite nonnegative integer counts, and more than one observation in each table. Wald is then valid only when the corrected difference and standard error are finite and  $\hat{V}(P) + \hat{V}(Q) > 10^{-14}$ . Expanded Welch additionally requires positive finite component and combined degrees of freedom. This is the strict component-validity convention stated after Equation (4.41); the software does not evaluate that direct combined form when a component degenerates. Failed conditions are recorded as invalid calculations. The validation in Chapters 5 and 6 evaluates how often these conditions occur and whether the finite-sample correction improves calibration.

# Chapter 5

## Experimental Design

The purpose of the experiment is to compare the calibration, detection and validity of two tests across controlled populations. Each design point fixes two population distributions in advance. Repeated independent samples from those populations show how often each procedure returns a valid result and how often it rejects. Because the population MI values are known, a rejection at equal MI is a false positive; a rejection at unequal MI is detection.

### 5.1 A population pair before the design grid

Consider two  $2 \times 2$  populations whose row and column margins are  $(0.7, 0.3)$  for  $P$  and  $(0.8, 0.2)$  for  $Q$ . The construction produces, to the displayed precision,

$$P \simeq \begin{pmatrix} 0.53273 & 0.16727 \\ 0.16727 & 0.13273 \end{pmatrix}, \quad Q \simeq \begin{pmatrix} 0.68175 & 0.11825 \\ 0.11825 & 0.08175 \end{pmatrix}. \quad (5.1)$$

These are *probability* tables, not observed counts. Their target MI values are 0.02 and 0.03 nats, so the true magnitude of the difference is 0.01 nats. Sampling new count tables changes the estimated MI values, but does not move this fixed population pair to a new horizontal-axis location. Calculations use the stored unrounded probabilities; the rounded display in Equation (5.1) does not determine the samples.

### 5.2 Constructing fixed joint distributions

For one population, let  $a = (a_1, \dots, a_r)$  and  $b = (b_1, \dots, b_c)$  be prescribed row and column probabilities. The independence table has cell probabilities  $a_i b_j$ . Association



is introduced by

$$R(t) = ab^T + tH, \quad H\mathbf{1}_c = 0, \quad \mathbf{1}_r^T H = 0. \quad (5.2)$$

The zero row and column sums mean that changing  $t$  leaves the margins fixed. For the main construction,  $H$  is zero except in the first two rows and columns, where it is

$$H_{1:2,1:2} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}. \quad (5.3)$$

Thus  $t$  is added to the two diagonal cells and removed from the other two cells. The changes cancel within each affected row and column. The same matrix specifies a  $2 \times 2$  association and a localised change in a larger table; only its embedding differs. This follows the minimal four-violation sparse alternative used by [Berrett and Samworth \(2021, Section 3\(b\)\)](#). Here it is adapted to compare two fixed populations: their margins are prescribed separately and the perturbation strength is chosen to attain a specified MI and is not used directly as the horizontal-axis effect.

For the pair in Equation (5.1), the starting independence tables are

$$P_0 = \begin{pmatrix} 0.49 & 0.21 \\ 0.21 & 0.09 \end{pmatrix}, \quad Q_0 = \begin{pmatrix} 0.64 & 0.16 \\ 0.16 & 0.04 \end{pmatrix}. \quad (5.4)$$

Adding the block in Equation (5.3) with strengths  $t_P \simeq 0.04273$  and  $t_Q \simeq 0.04175$ , respectively, gives the displayed  $P$  and  $Q$ . The strengths are chosen to reach their separate MI targets; the 0.01-nat difference is not a cell-probability increment.

The largest nonnegative strength before a cell with  $H_{ij} < 0$  reaches zero is

$$t_{\max} = \min_{H_{ij} < 0} \frac{a_i b_j}{-H_{ij}}. \quad (5.5)$$

The construction solves for  $0 \leq t < t_{\max}$  such that  $I\{R(t)\}$  equals the requested target. Targets at or beyond the endpoint are not admitted. This endpoint is specific to the chosen path, not the largest possible MI over every table with those margins. Both constructed tables are checked for strictly positive cells, correct margins and target MI before any sampling. Appendix B gives the numerical details.

For a dimension of size  $k$ , the skewed margin used in the design is

$$m_k(\rho) = \left( \rho, \frac{1-\rho}{k-1}, \dots, \frac{1-\rho}{k-1} \right). \quad (5.6)$$

The three principal profiles are specified separately for the two populations in

Table 5.1. For a rectangular  $r \times c$  table, the formula is applied with  $k = r$  to row margins and  $k = c$  to column margins. At a zero MI difference, the same-margin profiles produce  $P = Q$  under this construction; the different-skew profile has  $I(P) = I(Q)$  but  $P \neq Q$ .

**Table 5.1:** Row and column margins used for each population in the principal design. “Uniform” means the vector with every entry equal to  $1/k$ .

| Profile        | $P$ : rows and columns | $Q$ : rows and columns |
|----------------|------------------------|------------------------|
| Uniform        | uniform                | uniform                |
| Same skew      | $m_k(0.8)$             | $m_k(0.8)$             |
| Different skew | $m_k(0.7)$             | $m_k(0.8)$             |

In the usual effect direction,  $I(P) = I_0$  and  $I(Q) = I_0 + |\Delta|$ . The paired solve chooses separate strengths for  $P$  and  $Q$ ; it does not add random noise to their true MI values. The reverse-direction imbalance experiment assigns the higher target to  $P$ . The statistical convention remains  $\Delta = I(P) - I(Q)$ , which is negative in the usual direction. Figures use the nonnegative magnitude  $|\Delta|$ .

Two focused checks change where or how association is placed while keeping the same interpretation of target MI. A “rare block” moves the four-cell change in Equation (5.3) to the final two rows and columns, whose margins are smaller in skewed profiles. A “spread” pattern changes cells throughout the table using increasing row and column scores; its positive and negative changes cancel within each margin. The ordinal log-linear construction uses the product of those scores inside an exponential cell weight, then rescales the rows and columns to preserve the prescribed margins. Its strength is chosen to reach the target MI. Matching margins and MI across constructors does not imply matching all cell probabilities or expected counts.

### 5.3 Sampling, tests and measured outcomes

For each unique configuration, 20,000 independent pairs of multinomial count tables are drawn from its fixed  $P$  and  $Q$ . Normal Wald and Expanded Welch are applied to the *same* count-table pair. This pairing makes method differences more precisely measurable than unrelated simulations, though it does not remove Monte Carlo uncertainty. The two-sided nominal threshold is  $\alpha = 0.05$  throughout the statistical study.

For each method, the primary plotted rejection rate is

$$\frac{\#\{\text{sample pairs giving a valid p-value no larger than } 0.05\}}{20,000}. \quad (5.7)$$

At  $|\Delta| = 0$  this estimates the false-positive rate. At  $|\Delta| > 0$  it estimates the probability of detecting the specified difference at the fixed nominal threshold, often called power. A rate near 0.05 is desirable under the null; a smaller rate is not automatically better. Under an alternative, higher detection is useful only alongside acceptable null calibration and a usable test result.

An invalid calculation contributes no rejection to Equation (5.7). This convention reports the behaviour of the implemented procedure, but an invalid result is not a valid acceptance of the null. The companion valid-result rate is the fraction of replicates with a usable p-value. The saved data also give rejection conditional on each method’s valid outputs and on the subset where both are valid. Those denominators answer different questions: conditioning each method on its own valid samples can change which tables are compared. For a single rate, saved Wilson 95% intervals (Wilson, 1927) represent Monte Carlo uncertainty at the fixed population pair. Paired method differences use the same replicates and their saved paired standard errors. These intervals do not describe uncertainty over all possible population distributions.

The “minimum expected count” diagnostic is  $\min_{ij} n_P p_{ij}$  for  $P$ , or  $\min_{ij} n_Q q_{ij}$  for  $Q$ . It is not the expectation of the smallest observed count. No lower bound on that diagnostic was used to admit statistical configurations. Empty observed cells and rows remain possible. In the plots a hollow marker warns that fewer than 90% of outputs of that method were valid; 90% is a display convention, not a validated safety threshold.

## 5.4 Experiment families and computation

Table 5.2 records the question answered by each family. The common population construction above applies unless the named factor changes. Exact parameter sets are in Appendix B, and the machine-readable protocol is identified in Appendix E. There are 534 distinct population pairs  $(P, Q)$ , each specifying two fixed joint probability tables. Reusing a pair at different sample sizes creates distinct sampling configurations. There are 3,111 unique statistical configurations but 4,001 display points: some configurations are deliberately shown in more than one comparison. Each unique configuration is sampled once, giving 62,220,000 table pairs in the final statistical run. The within-family counts in Table 5.2 do not sum to 3,111

because the same configuration can occur in several families. Display reuse is not an independent replication.

**Table 5.2:** The fixed experiment families. All statistical points use 20,000 paired replicates and the 5% threshold.

| Family           | Displays | Unique | Factor examined                                                 |
|------------------|----------|--------|-----------------------------------------------------------------|
| Main             | 648      | 648    | Square table size, three margin profiles and equal sample size. |
| Baseline         | 648      | 648    | Starting positive MI of 0.0001, 0.001 or 0.02 nats.             |
| Changed cells    | 144      | 144    | First, rare or spread association patterns.                     |
| Imbalance        | 1,152    | 1,056  | Sample allocation and which population has higher MI.           |
| Larger effects   | 162      | 162    | MI differences up to 0.2 nats where feasible.                   |
| Rectangular      | 108      | 108    | $2 \times 3$ and $3 \times 5$ tables.                           |
| Construction     | 432      | 432    | Additive versus ordinal log-linear population tables.           |
| Extreme skew     | 315      | 315    | Dominant margins up to 0.9995, including $n = 1$ stress.        |
| Rare-cell stress | 140      | 140    | First- versus rare-block changes, including $n = 1$ stress.     |
| Independence     | 144      | 144    | Exact $I(P) = 0$ boundary.                                      |
| Convergence      | 108      | 108    | Fixed equal-MI populations with sample sizes up to 50,000.      |
| Runtime          | 60       | 60     | Full-call timing regimes, separate from simulation.             |

The runtime experiment measures a full method call after count tables are already available. It varies table shape, profile, sample size and null or alternative status. Each regime uses 200 saved inputs, 20 warm-up calls and five timed passes. Thus it measures method computation, not the cost of constructing populations or generating random tables. Median call times and interquartile ranges are reported in Chapter 6.

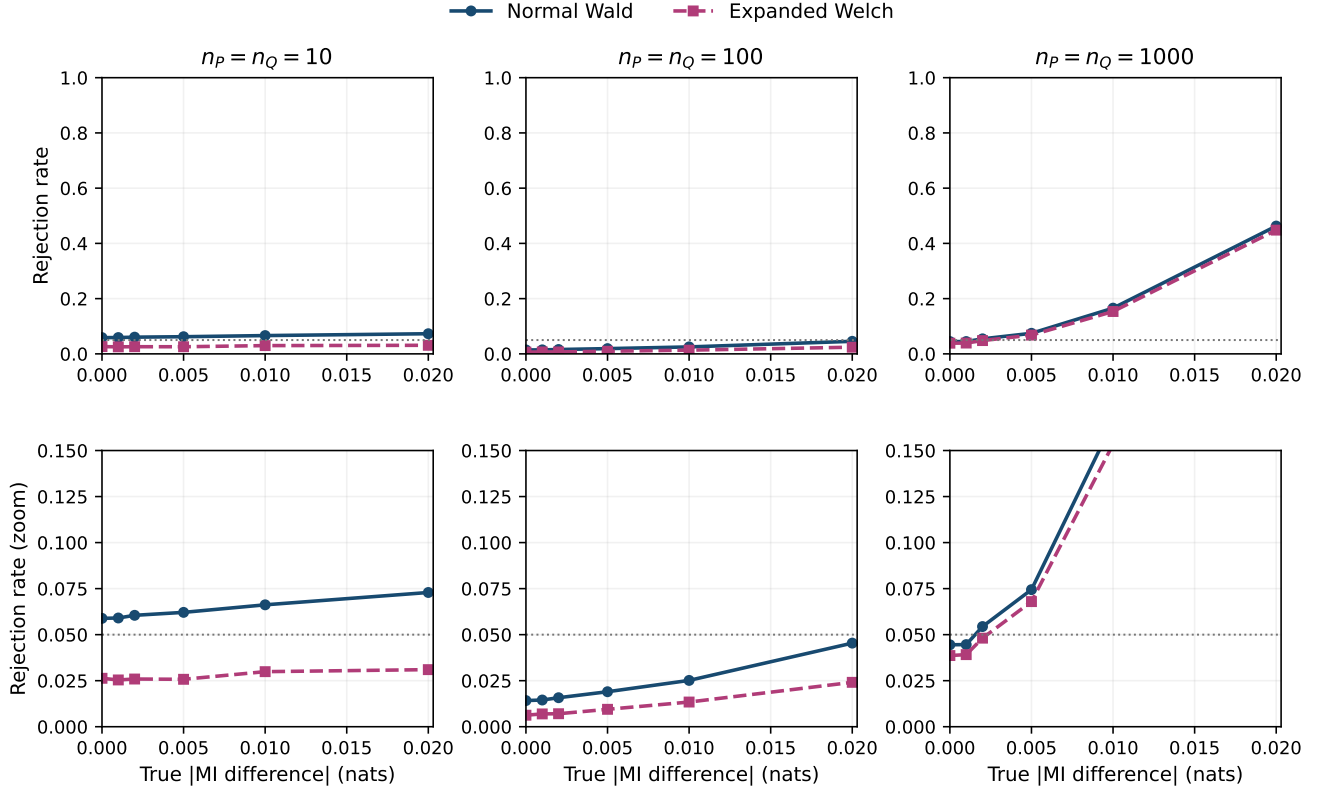
# Chapter 6

## Experimental Results

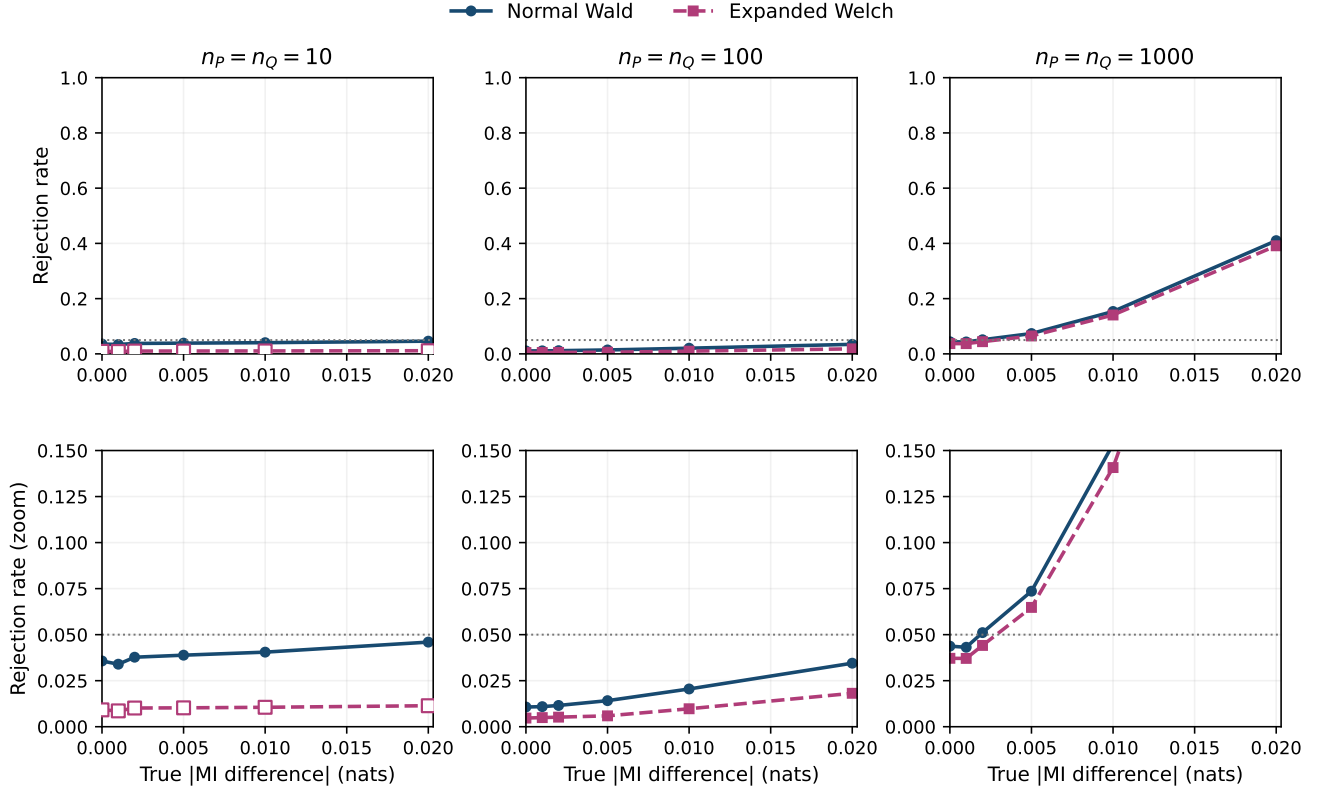
Every ordinary panel in this chapter begins at a true MI difference of zero. That point measures false-positive rejection under  $H_0$ . Points to its right measure detection under  $H_1$ . The horizontal coordinate is a *constructed population* difference in nats. The vertical coordinate is the rejection rate over 20,000 random table pairs. The blue solid circles are Normal Wald and the magenta dashed squares are Expanded Welch. The dotted horizontal line marks the nominal 0.05 rejection rate. Each plotted rate divides the number of valid rejections by all 20,000 replicates. A hollow marker indicates a valid-result rate below 0.90. The figure source manifest gives the exact saved configuration identifiers for every panel.

### 6.1 Main landscape: shape, margins and sample size

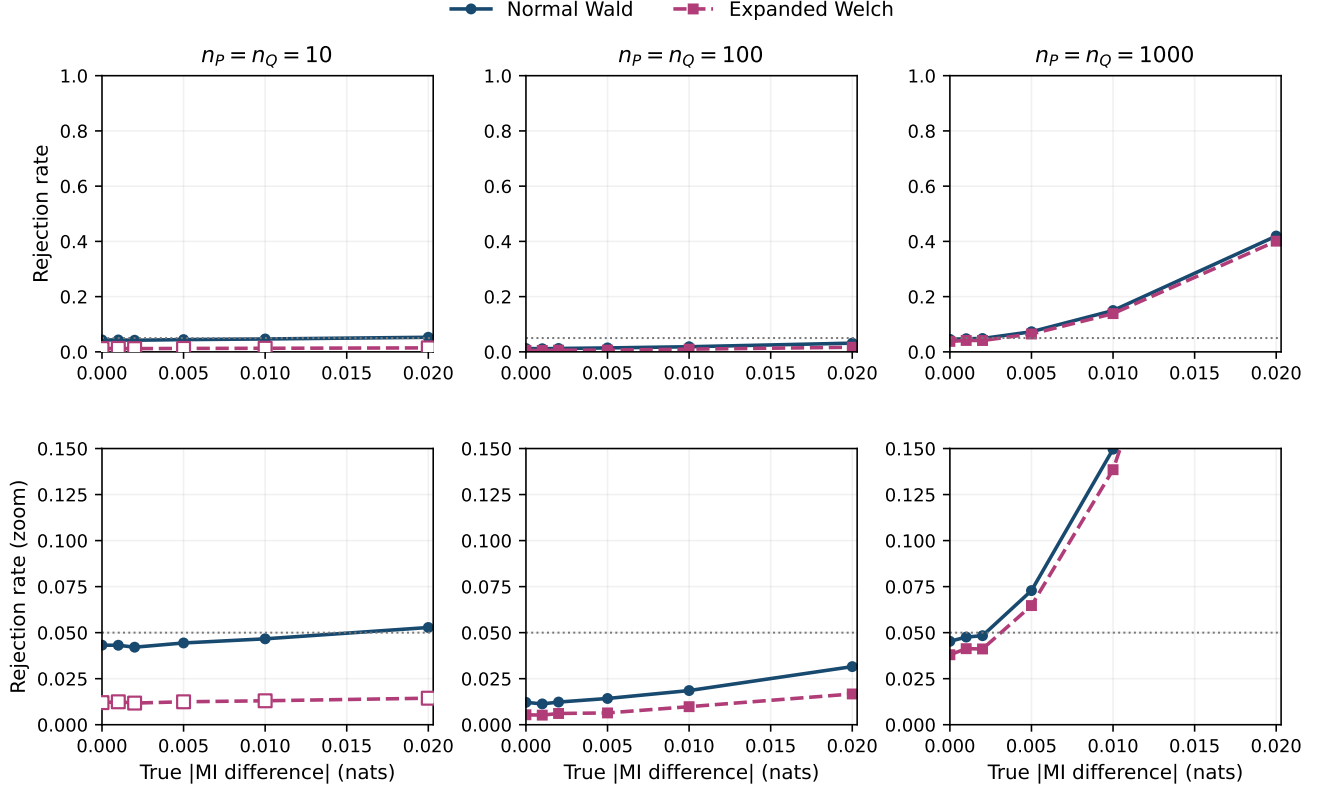
The first question is broad: do the methods behave differently as the table grows, margins become uneven and samples become larger? Figures 6.1–6.12 provide a systematic view, not a selection of favourable examples. Each of the four shapes is shown under all three margin profiles. Within every figure, the columns are the same three equal sample sizes. The full vertical scale preserves comparability of detection; the lower row enlarges the region in which calibration differences near 0.05 can be seen. The full nine-size ladder, including  $n = 2, 5, 20, 50, 250, 500$ , remains in the companion atlas.



**Figure 6.1:**  $2 \times 2$  tables, both  $P$  and  $Q$  have uniform row and column margins  $(1/2, 1/2)$ . Columns:  $n_P = n_Q = 10, 100, 1000$ , respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3);  $I(P) = 0.02$  and  $I(Q) = 0.02 + |\Delta|$ . The horizontal grid is  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.

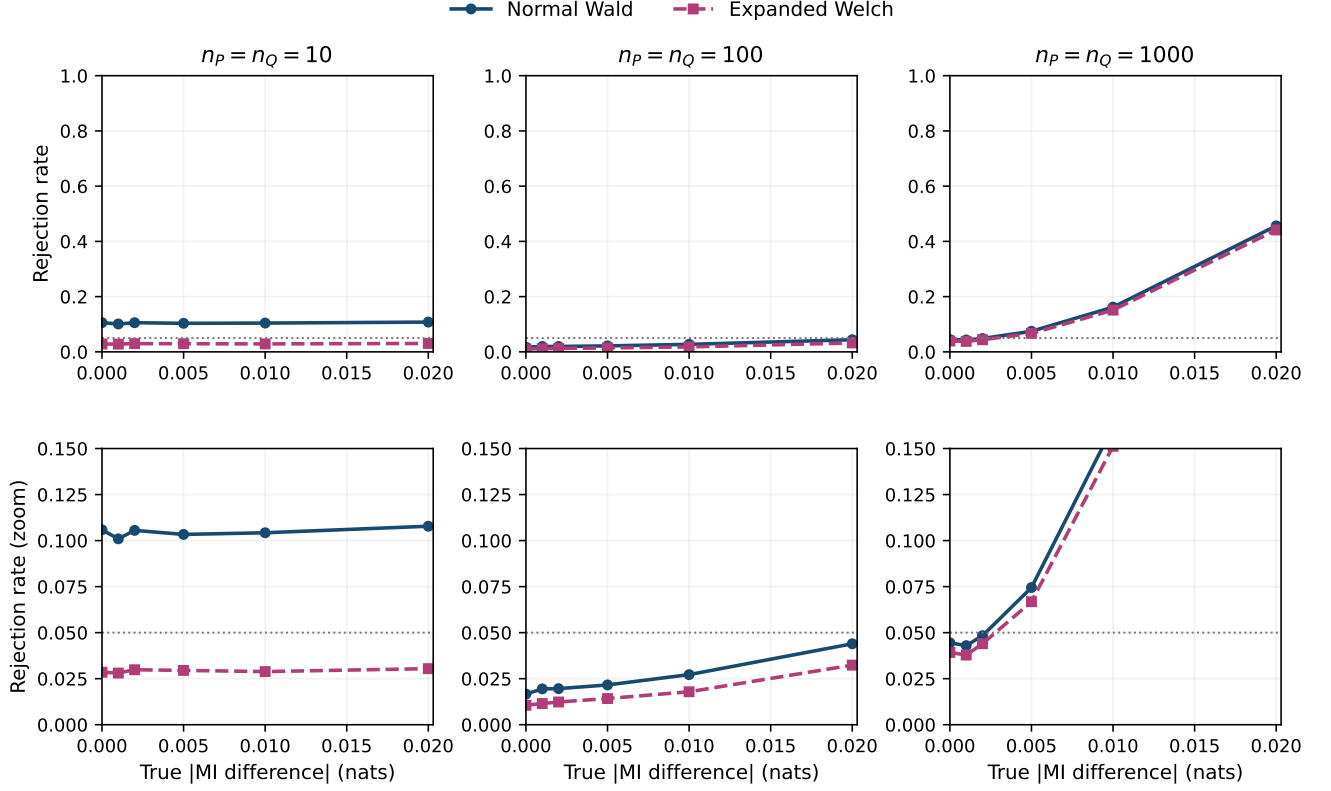


**Figure 6.2:**  $2 \times 2$  tables, both  $P$  and  $Q$  have row and column margins  $m_2(0.8) = (0.8, 0.2)$ . Columns:  $n_P = n_Q = 10, 100, 1000$ , respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3);  $I(P) = 0.02$  and  $I(Q) = 0.02 + |\Delta|$ . The horizontal grid is  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.

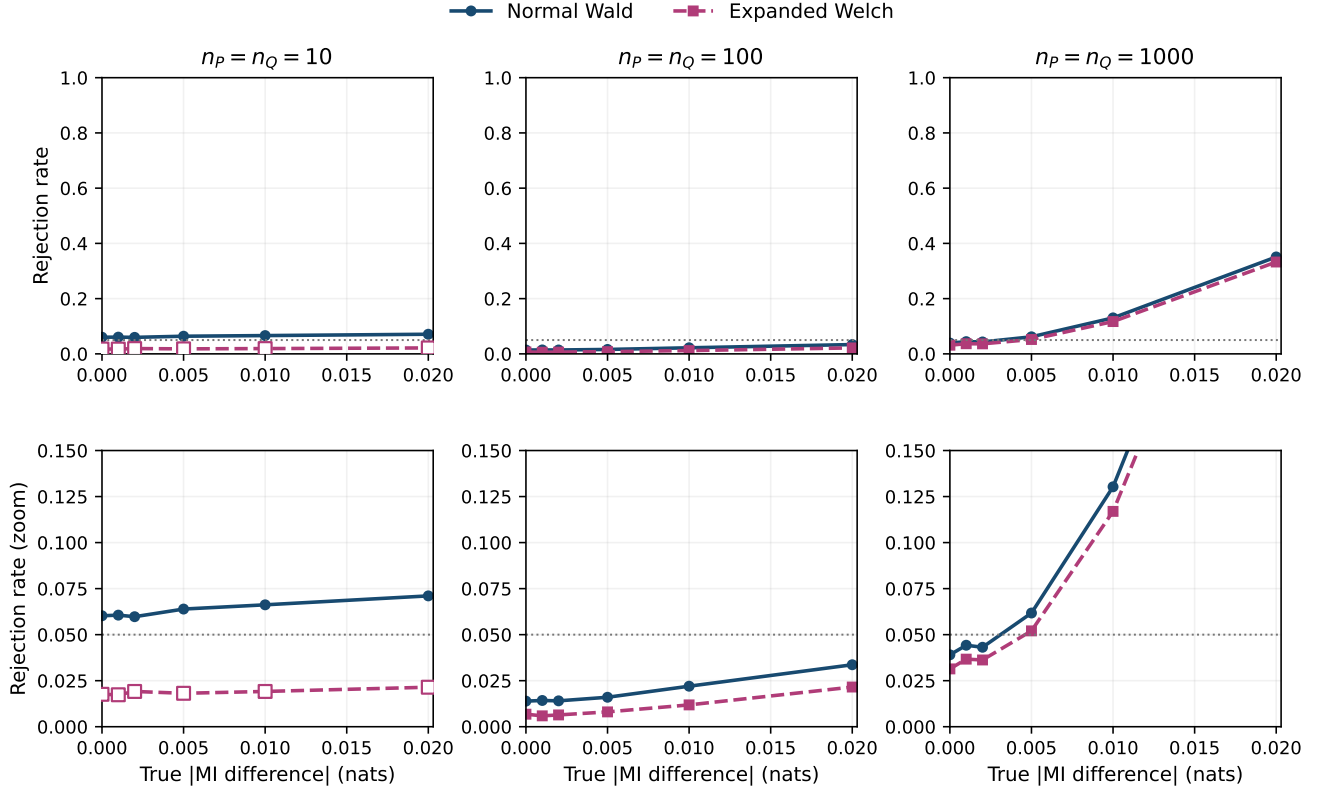


**Figure 6.3:**  $2 \times 2$  tables,  $P$  has row and column margins  $m_2(0.7) = (0.7, 0.3)$ , while  $Q$  has  $m_2(0.8) = (0.8, 0.2)$ . Columns:  $n_P = n_Q = 10, 100, 1000$ , respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3);  $I(P) = 0.02$  and  $I(Q) = 0.02 + |\Delta|$ . The horizontal grid is  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.

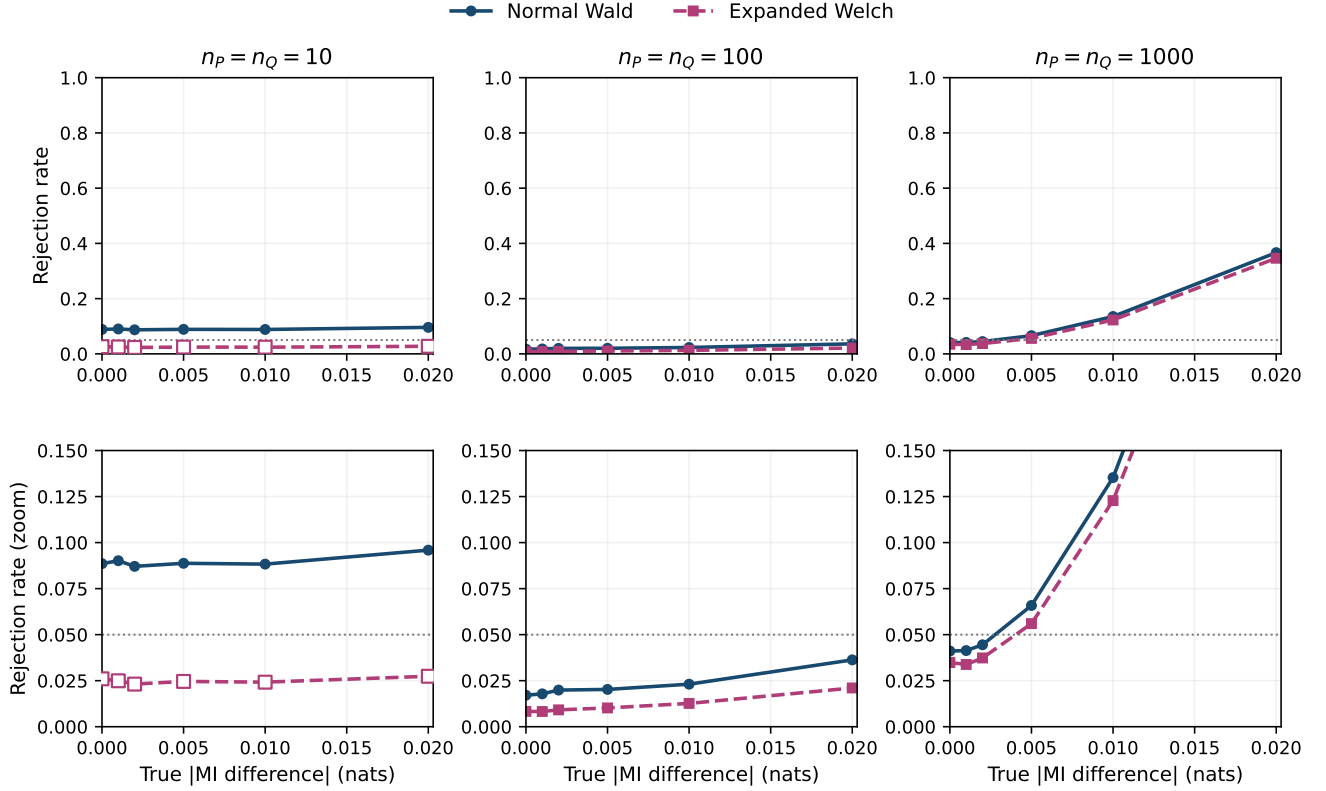




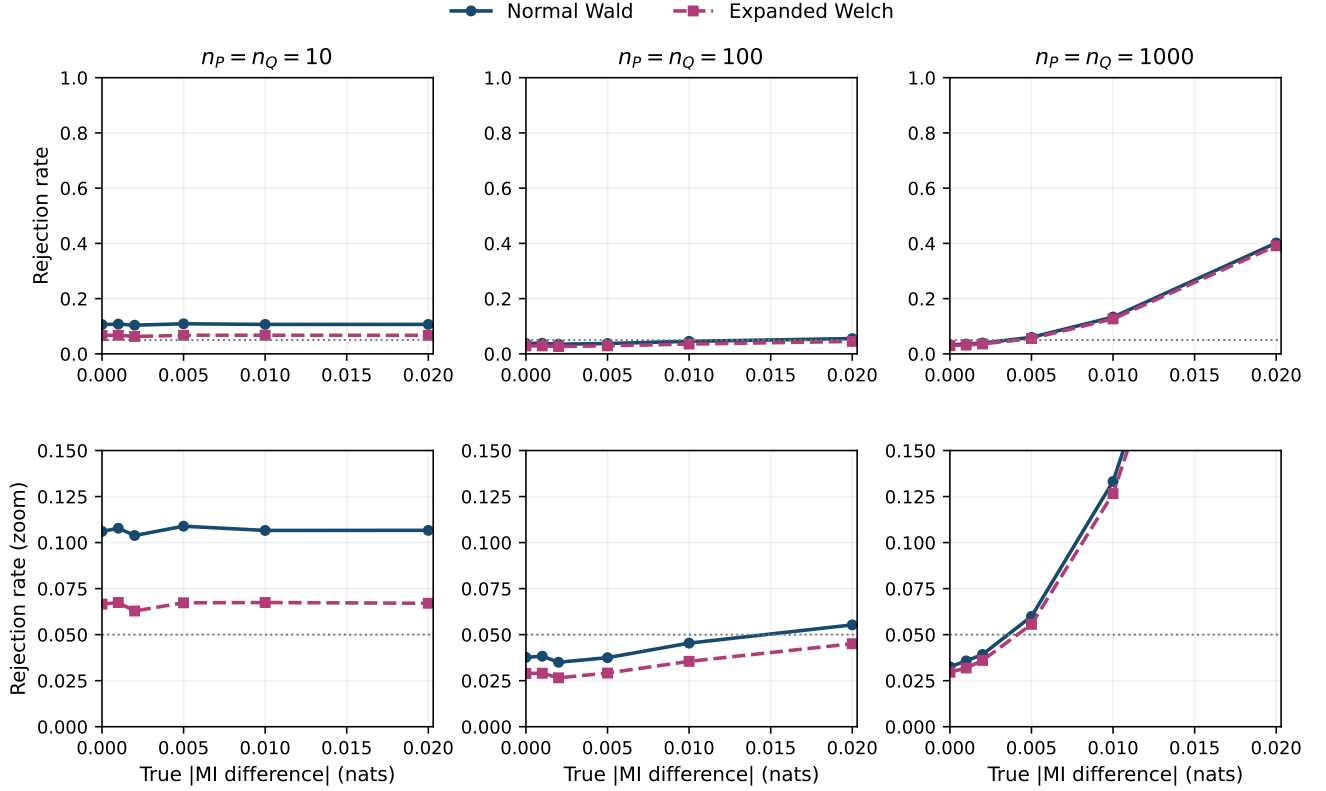
**Figure 6.4:**  $3 \times 3$  tables, both  $P$  and  $Q$  have uniform row and column margins  $(1/3, 1/3, 1/3)$ . Columns:  $n_P = n_Q = 10, 100, 1000$ , respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3);  $I(P) = 0.02$  and  $I(Q) = 0.02 + |\Delta|$ . The horizontal grid is  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.



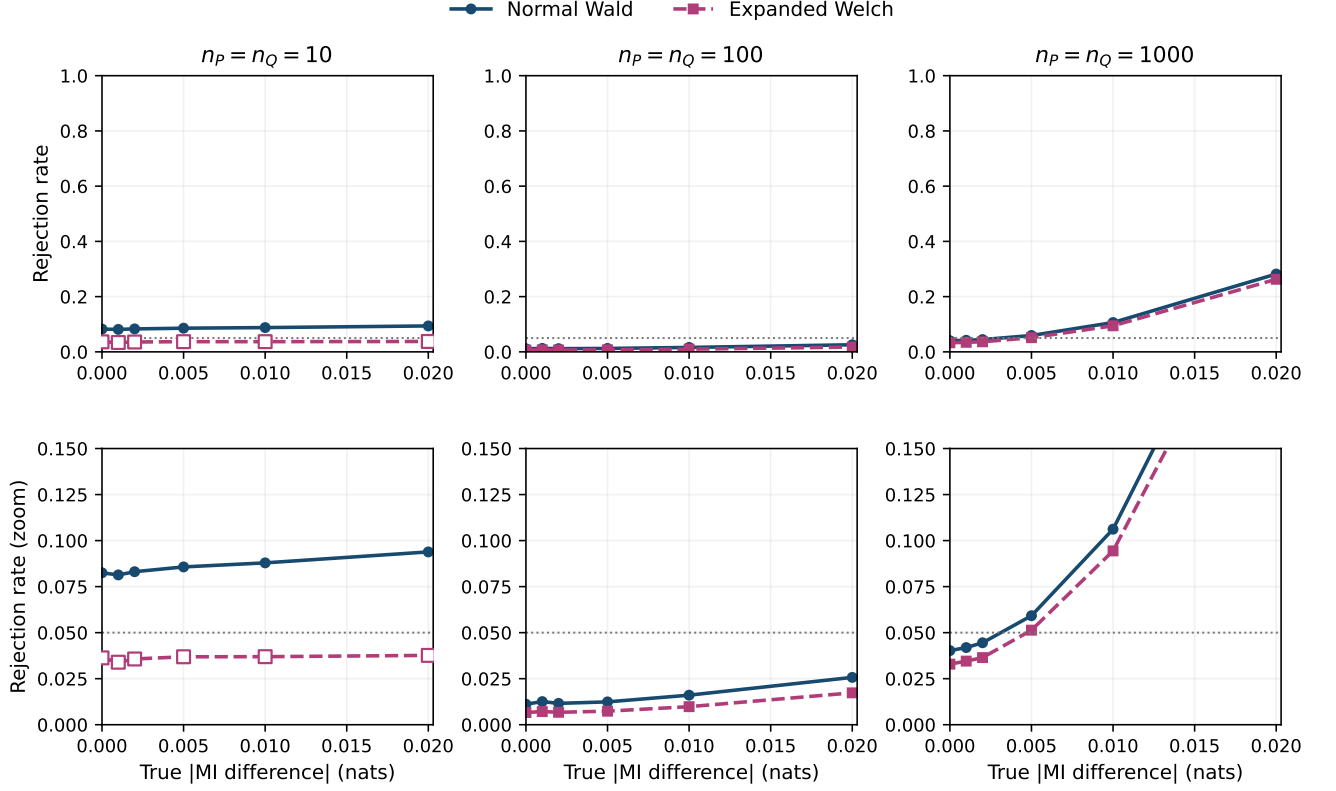
**Figure 6.5:**  $3 \times 3$  tables, both  $P$  and  $Q$  have row and column margins  $m_3(0.8) = (0.8, 0.1, 0.1)$ . Columns:  $n_P = n_Q = 10, 100, 1000$ , respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3);  $I(P) = 0.02$  and  $I(Q) = 0.02 + |\Delta|$ . The horizontal grid is  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.



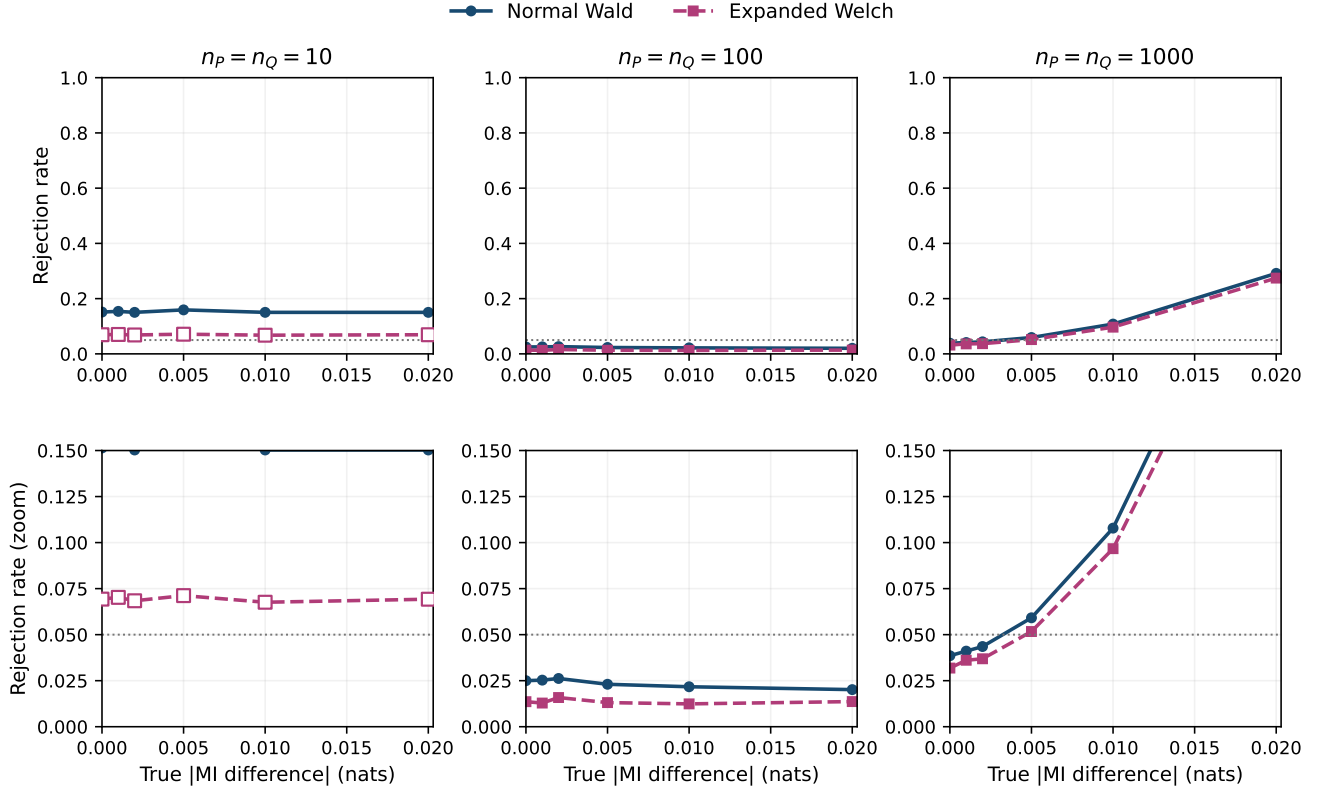
**Figure 6.6:**  $3 \times 3$  tables,  $P$  has row and column margins  $m_3(0.7) = (0.7, 0.15, 0.15)$ , while  $Q$  has  $m_3(0.8) = (0.8, 0.1, 0.1)$ . Columns:  $n_P = n_Q = 10, 100, 1000$ , respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3);  $I(P) = 0.02$  and  $I(Q) = 0.02 + |\Delta|$ . The horizontal grid is  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.



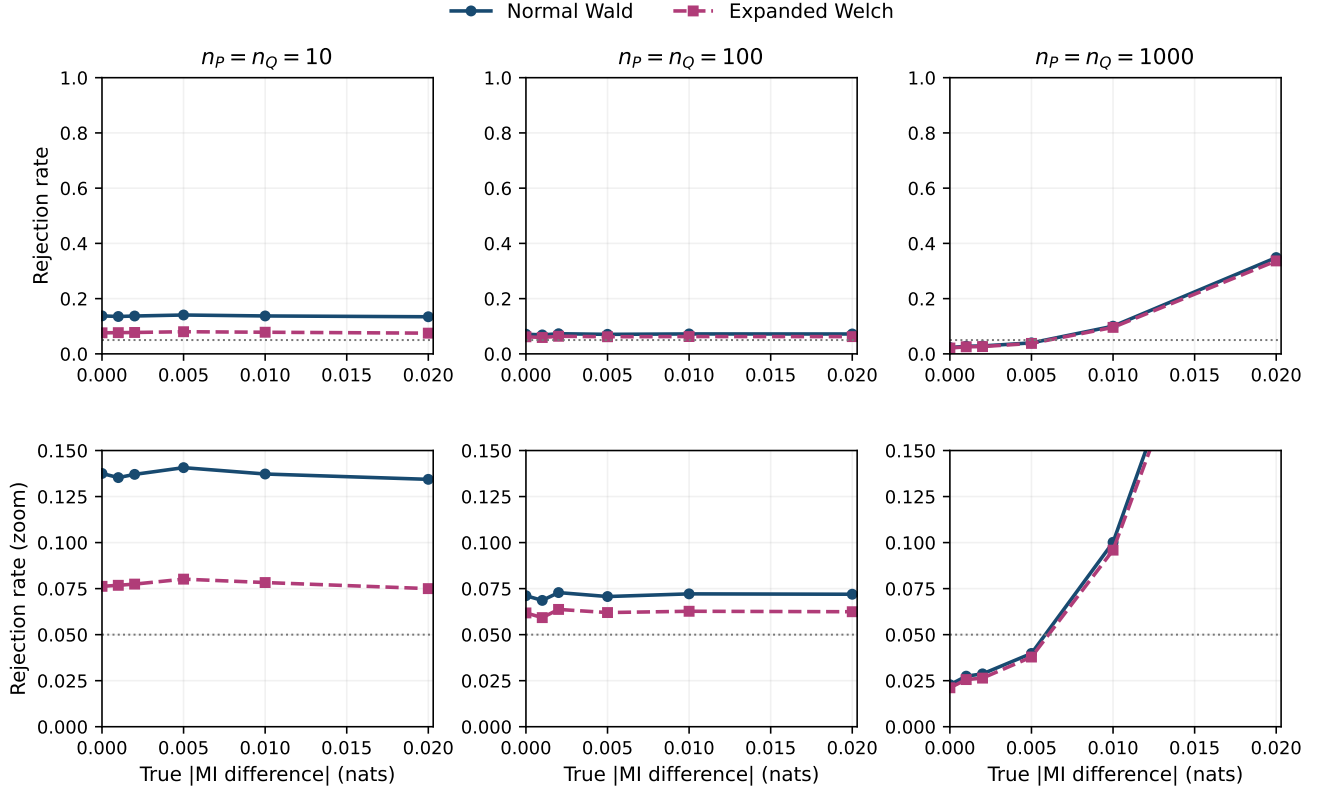
**Figure 6.7:**  $5 \times 5$  tables, both  $P$  and  $Q$  have uniform row and column margins  $(1/5, \dots, 1/5)$ . Columns:  $n_P = n_Q = 10, 100, 1000$ , respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3);  $I(P) = 0.02$  and  $I(Q) = 0.02 + |\Delta|$ . The horizontal grid is  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.



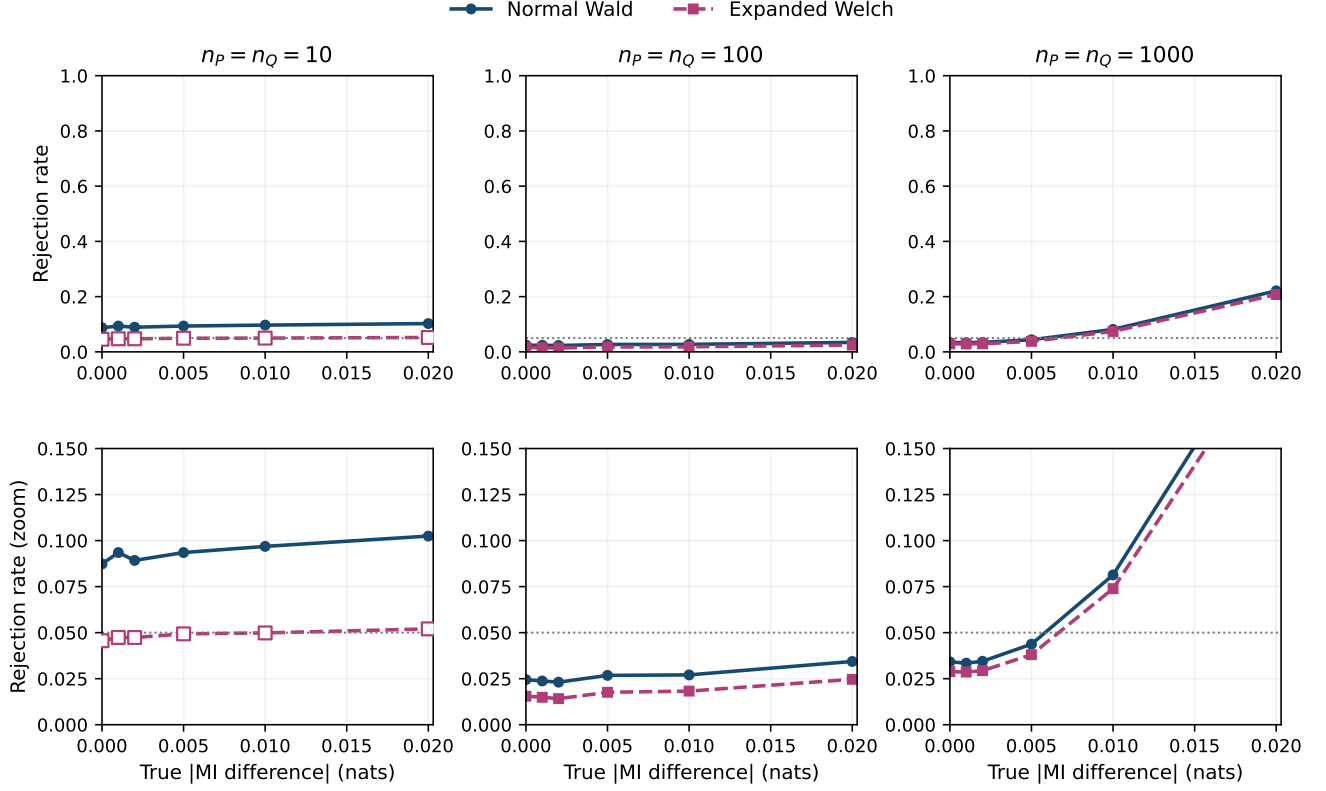
**Figure 6.8:**  $5 \times 5$  tables, both  $P$  and  $Q$  have row and column margins  $m_5(0.8) = (0.8, 0.05, 0.05, 0.05, 0.05)$ . Columns:  $n_P = n_Q = 10, 100, 1000$ , respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3);  $I(P) = 0.02$  and  $I(Q) = 0.02 + |\Delta|$ . The horizontal grid is  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.



**Figure 6.9:**  $5 \times 5$  tables,  $P$  has row and column margins  $m_5(0.7) = (0.7, 0.075, 0.075, 0.075, 0.075)$ , while  $Q$  has  $m_5(0.8) = (0.8, 0.05, 0.05, 0.05, 0.05)$ . Columns:  $n_P = n_Q = 10, 100, 1000$ , respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3);  $I(P) = 0.02$  and  $I(Q) = 0.02 + |\Delta|$ . The horizontal grid is  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.

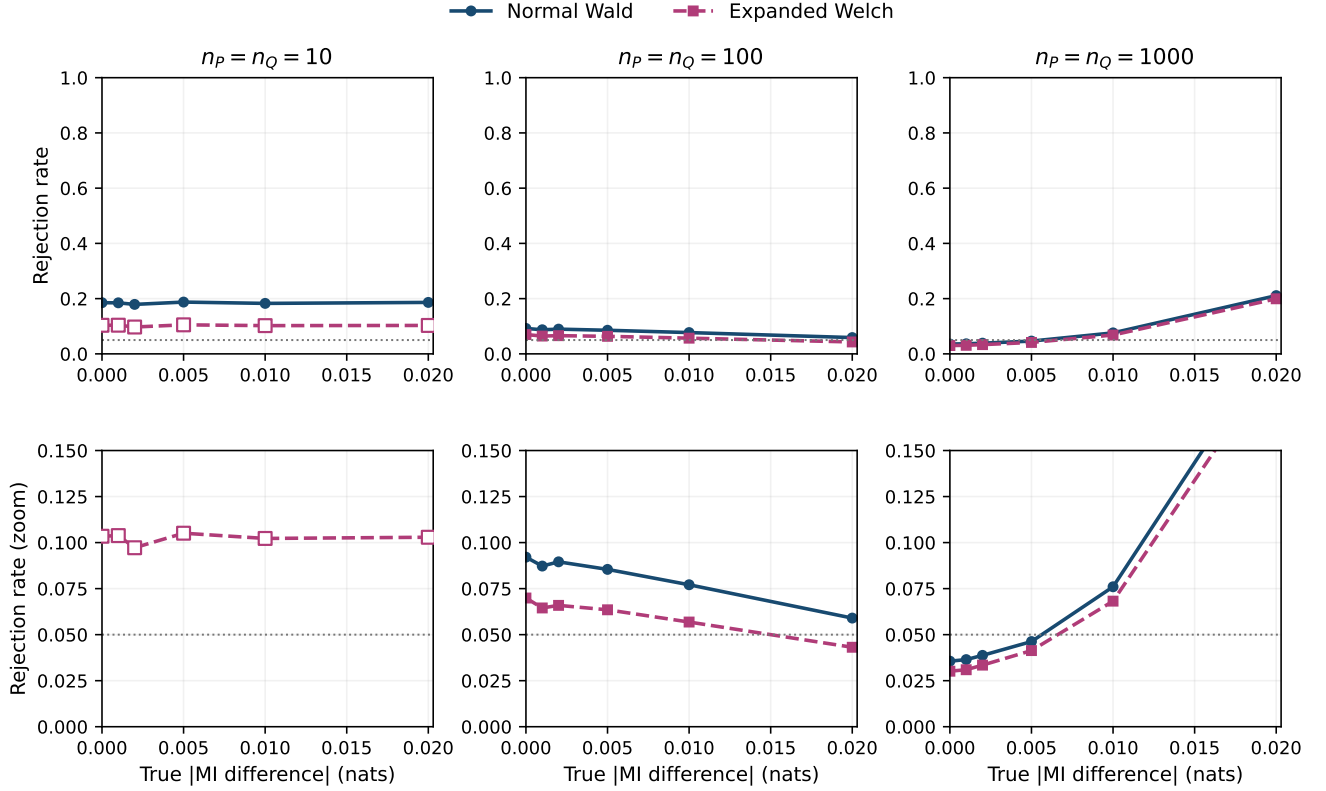


**Figure 6.10:**  $8 \times 8$  tables, both  $P$  and  $Q$  have uniform row and column margins ( $1/8, \dots, 1/8$ ). Columns:  $n_P = n_Q = 10, 100, 1000$ , respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3);  $I(P) = 0.02$  and  $I(Q) = 0.02 + |\Delta|$ . The horizontal grid is  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.



**Figure 6.11:**  $8 \times 8$  tables, both  $P$  and  $Q$  have row and column margins  $m_8(0.8) = (0.8, 0.2/7, \dots, 0.2/7)$ . Columns:  $n_P = n_Q = 10, 100, 1000$ , respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3);  $I(P) = 0.02$  and  $I(Q) = 0.02 + |\Delta|$ . The horizontal grid is  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.





**Figure 6.12:**  $8 \times 8$  tables,  $P$  has row and column margins  $m_8(0.7) = (0.7, 0.3/7, \dots, 0.3/7)$ , while  $Q$  has  $m_8(0.8) = (0.8, 0.2/7, \dots, 0.2/7)$ . Columns:  $n_P = n_Q = 10, 100, 1000$ , respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3);  $I(P) = 0.02$  and  $I(Q) = 0.02 + |\Delta|$ . The horizontal grid is  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.

The landscape does not yield a uniformly preferable method. At  $n = 5$ , Expanded Welch has a zero null rejection rate in all 12 main regimes. This includes the uniform  $3 \times 3$ ,  $5 \times 5$  and  $8 \times 8$  regimes, where more than 90% of calculations are valid, so the zero rates cannot be attributed only to invalid outputs. Among the 24 main null regimes with  $n = 10$  or 20, Expanded Welch is numerically closer to 0.05 in 15 and Wald in 9. The comparison reverses in the 24 main null settings at  $n = 500$  or 1,000: Wald is closer to 0.05 in all 24. At  $n = 1,000$ , Wald null rates across the main settings lie between 0.0228 and 0.0453; Expanded Welch lies between 0.0211 and 0.0393. Neither is uniformly at the nominal level.

The interpretation is directional. Expanded Welch uses a heavier-tailed reference and therefore lowers rejection; it does not diagnose whether Wald started above or below 0.05. The same adjustment can consequently improve a liberal regime and worsen a conservative one. Closeness to 0.05 must be read regime by regime and not treated as an inherent advantage of either method.

Two fully valid null points make the contrast concrete. With uniform  $8 \times 8$  populations,  $n_P = n_Q = 20$ , and equal true MI of 0.02, Wald rejects 0.06870 of samples and Expanded Welch 0.05695. The latter is closer to 0.05. With uniform  $2 \times 2$  populations at  $n_P = n_Q = 1,000$ , the respective rates are 0.04450 and 0.03865; the correction moves an already conservative procedure farther below the target. Both methods are valid on all replicates in these examples, so the change is attributable to their different references, not missing outputs.

These two examples isolate the central trade-off. In the  $8 \times 8$  case, the lower rejection rate corrects part of Wald's excess. In the  $2 \times 2$  case, the same downward movement is harmful because Wald was already conservative. Expanded Welch is therefore a one-direction adjustment, not an adaptive calibration procedure.

**Table 6.1:** Two fully valid main-grid null regimes. Brackets beside each rate are pointwise Wilson 95% intervals; the final bracket is the approximate paired 95% interval for Wald minus Expanded Welch on the same samples. Source configurations are `config_313a5e5c05280f9e` and `config_700945d218841f2e`, respectively.

| Regime                             | Wald                       | Expanded                   | Wald minus<br>Expanded     |
|------------------------------------|----------------------------|----------------------------|----------------------------|
|                                    | Rate [95%<br>interval]     | Rate [95%<br>interval]     | Rate [95%<br>interval]     |
| $8 \times 8$ , uniform, $n = 20$   | 0.0687 [0.0653,<br>0.0723] | 0.0570 [0.0538,<br>0.0602] | 0.0118 [0.0103,<br>0.0132] |
| $2 \times 2$ , uniform, $n = 1000$ | 0.0445 [0.0417,<br>0.0474] | 0.0387 [0.0361,<br>0.0414] | 0.0059 [0.0048,<br>0.0069] |

The  $n = 5$  records in the companion atlas also show an important limit to

interpreting small rejection rates. In the  $8 \times 8$  different-skew alternative with  $n_P = n_Q = 5$ ,  $I(P) = 0.02$  and  $I(Q) = 0.021$ , Wald rejects 0.30865 and Expanded Welch rejects 0.00000, but their valid-result rates are only 0.83840 and 0.31415. These are probabilities of a *valid rejection*, not a clean comparison of power at matched achieved false-positive rates. The corresponding null and validity records must be examined before calling the gap a power advantage.

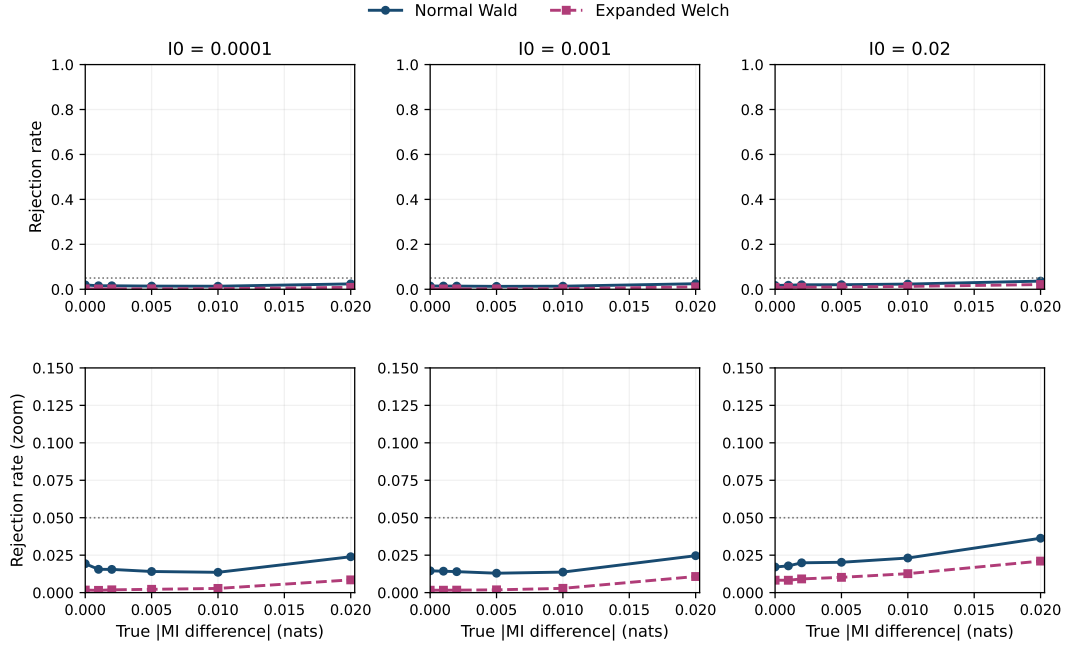
## 6.2 Baseline MI and the location of changed cells

The main grid fixes  $I(P) = 0.02$ . Does starting closer to independence change the conclusion? Figure 6.13 holds shape, margins, sample size and changed cells fixed while varying only the common starting MI. The zero-difference points are especially informative: for  $I_0 = 0.0001, 0.001, 0.02$ , Wald's rates are 0.0193, 0.0146 and 0.0171, while Expanded Welch's are 0.0017, 0.0016 and 0.0083. All the plotted null calculations in this focused set are valid. Replacing the normal reference with a heavier-tailed Student reference does not address the already-low null rejection in these settings.

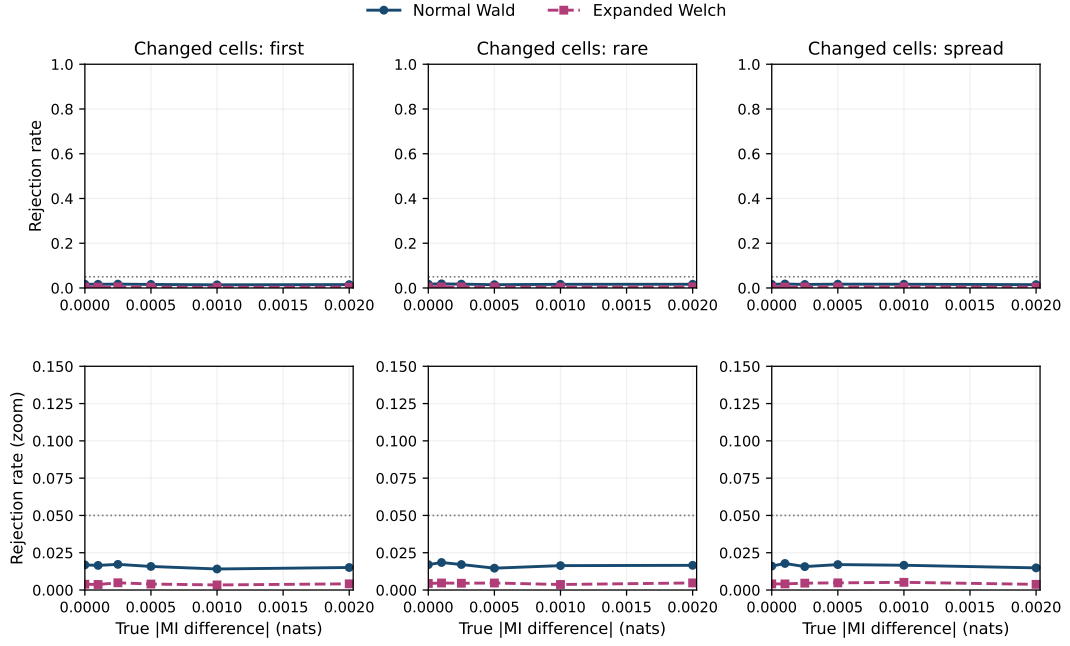
Starting closer to independence therefore does not create a general case for Expanded Welch. Here the main difficulty is already present in the first-order statistic and standard error; using a larger Student critical value only makes the conservative rejection rate smaller.

Figure 6.14 asks whether the association being placed in an early, rare or distributed set of cells changes performance when the margins and target MI are fixed. At the displayed null points, Wald rates are 0.0169, 0.0170 and 0.0160 and Expanded Welch rates are 0.0039, 0.0044 and 0.0041 for first, rare and spread patterns. The similarity of those three null rates in this selected setting is not evidence that the patterns are equivalent in all tables. The full  $3 \times 3$ ,  $5 \times 5$  and  $8 \times 8$  matched sets are in the atlas.

Within this controlled comparison, moving the association to different cells does not remove the conservative gap between the methods. The implication is limited but useful: the displayed result is not an artefact of only the first  $2 \times 2$  change block used in the main landscape.



**Figure 6.13:** Baseline-MI comparison for  $3 \times 3$  tables. Columns have  $I(P) = 0.0001, 0.001, 0.02$  nats, respectively;  $I(Q) = I(P) + |\Delta|$ .  $P$  has row and column margins  $(0.7, 0.15, 0.15)$  and  $Q$  has  $(0.8, 0.1, 0.1)$ . Both use the first-cell change block,  $n_P = n_Q = 100$ , and  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. Top: 0–1 rejection scale; bottom: 0–0.15 zoom. Each point has 20,000 replicates; hollow markers flag validity below 90%.

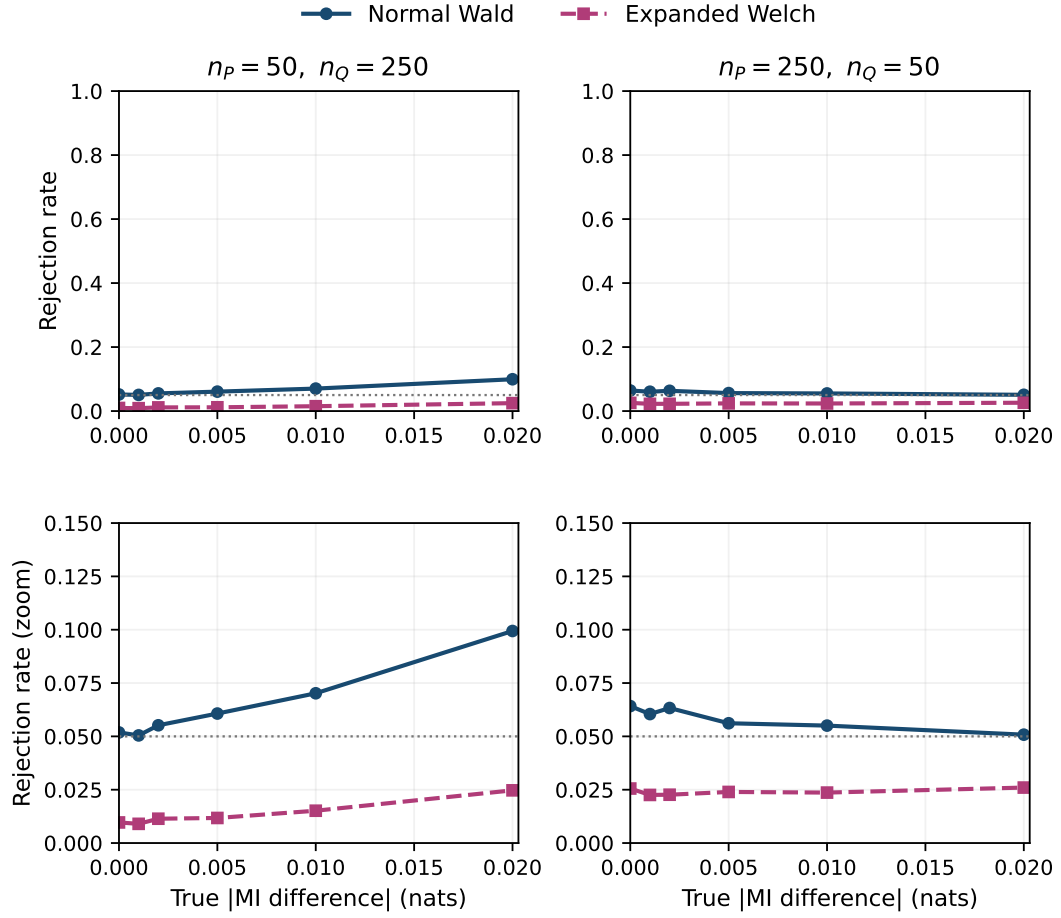


**Figure 6.14:** Changed-cell comparison for  $5 \times 5$  tables. Columns place association in the first  $2 \times 2$  block, last (rare)  $2 \times 2$  block, or a centred score outer product spread across all cells. For every column,  $P$ 's row and column margins are  $(0.7, 0.075, 0.075, 0.075, 0.075)$ , and  $Q$ 's are  $(0.8, 0.05, 0.05, 0.05, 0.05)$ . Here  $I(P) = 0.0001$ ,  $I(Q) = I(P) + |\Delta|$ ,  $n_P = n_Q = 100$ , and  $|\Delta| = 0, 0.0001, 0.00025, 0.0005, 0.001, 0.002$  nats. Top: 0–1; bottom: 0–0.15. Each point has 20,000 replicates.

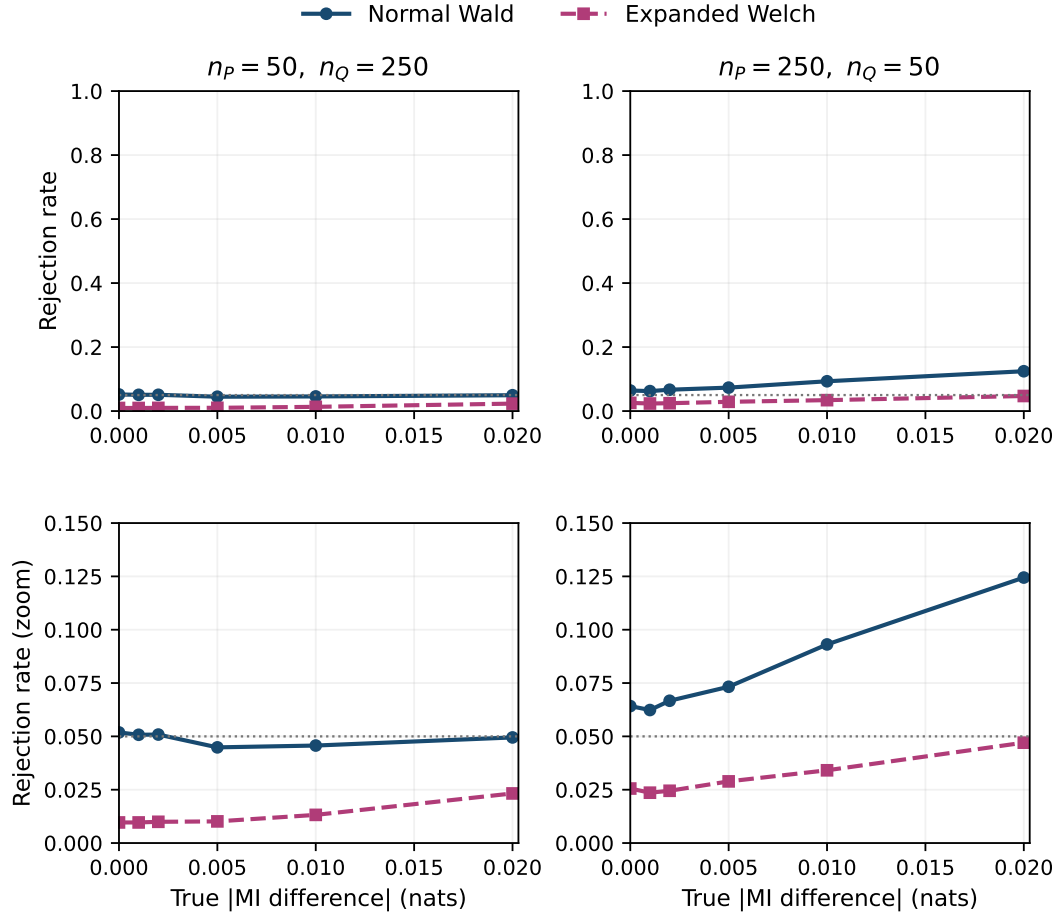
## 6.3 Unequal samples and the direction of the difference

Figures 6.15 and 6.16 hold the unordered sample sizes at 50 and 250 but reverse which population has the larger sample. They also show the two directions of the MI difference separately. The distinction matters because the finite bias corrections are  $(r-1)(c-1)/(2n_P)$  and  $(r-1)(c-1)/(2n_Q)$ , and the two populations have different margins. For example, at  $(n_P, n_Q) = (50, 250)$  and a 0.02-nat difference, Wald's rejection is 0.0994 when  $Q$  has higher MI and 0.0495 when  $P$  has higher MI. Expanded Welch gives 0.0247 and 0.0232. All four rates in that specific comparison have validity one. These alternative rates should still be read beside the matched null points; they are not size-adjusted power.

The practical implication is that an absolute MI difference does not fully specify the testing problem. Which population has the higher MI and which one has the larger sample changes the bias and variance contributions. Power should therefore not be averaged across the two directions of sample imbalance.



**Figure 6.15:** Unequal samples,  $Q$  higher. Columns have  $(n_P, n_Q) = (50, 250)$  and  $(250, 50)$ . For both columns,  $P$  has  $3 \times 3$  row and column margins  $(0.7, 0.15, 0.15)$  and  $Q$  has  $(0.8, 0.1, 0.1)$ ; the first-cell change block is used.  $I(P) = 0.02$  and  $I(Q) = 0.02 + |\Delta|$ , with  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. Top: 0–1; bottom: 0–0.15. Each point has 20,000 replicates.

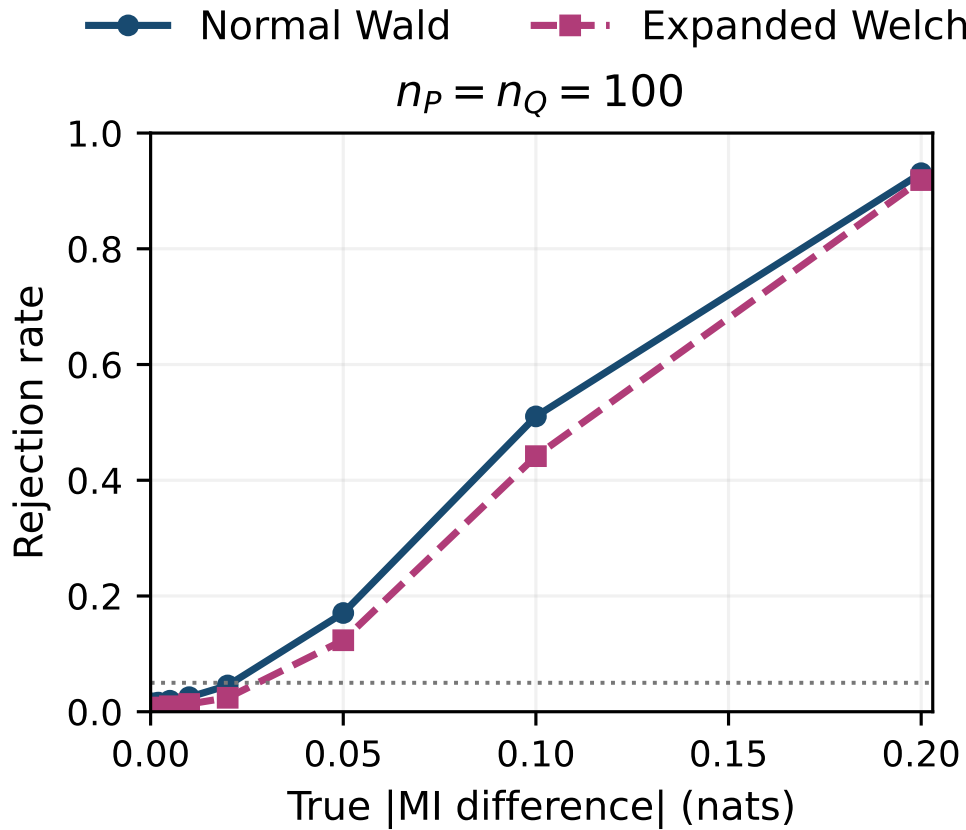


**Figure 6.16:** Unequal samples,  $P$  higher. Columns have  $(n_P, n_Q) = (50, 250)$  and  $(250, 50)$ .  $P$ 's  $3 \times 3$  row and column margins are  $(0.7, 0.15, 0.15)$ ;  $Q$ 's are  $(0.8, 0.1, 0.1)$ . Both use the first-cell change block.  $I(Q) = 0.02$  and  $I(P) = 0.02 + |\Delta|$ , with  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. Top: 0–1; bottom: 0–0.15. Each point has 20,000 replicates.



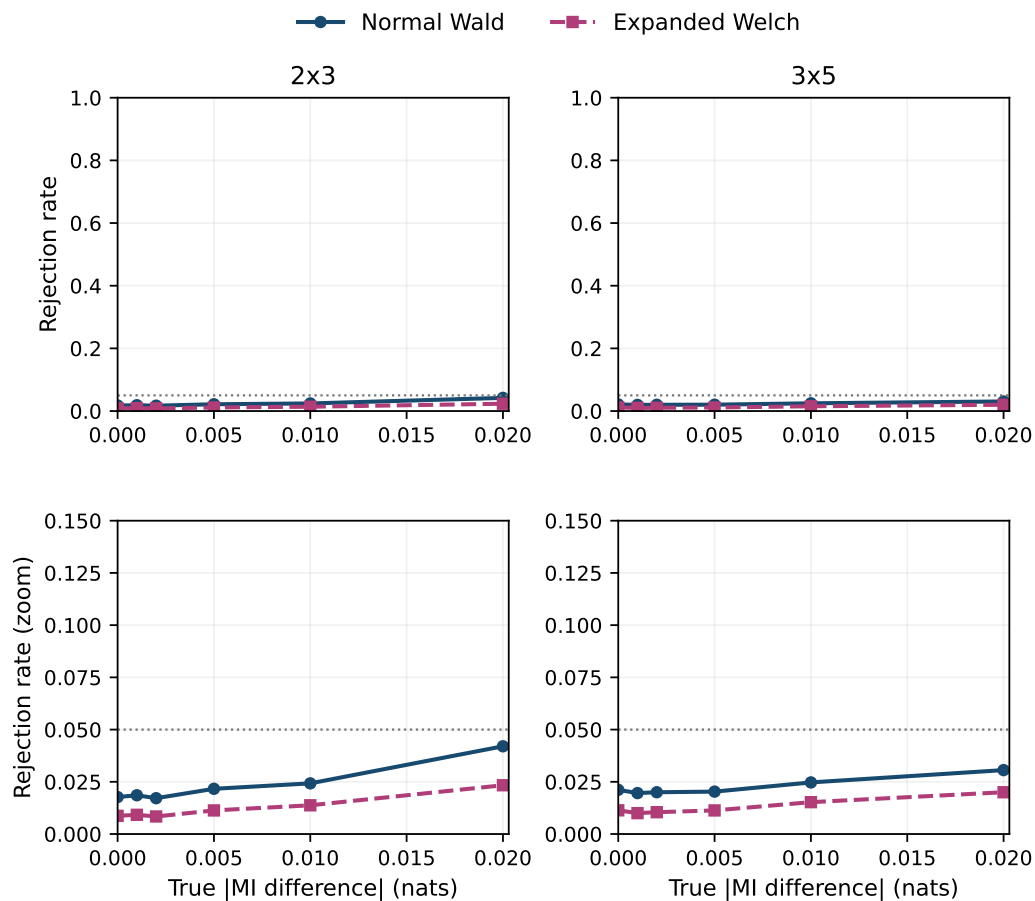
## 6.4 Larger differences and rectangular tables

The main grid deliberately concentrates on small MI differences. A separate  $2 \times 2$  family extends the feasible axis to 0.2 nats without changing the axis units, using only targets the construction can reach. This shows where detection eventually rises, while retaining the zero-difference point that defines null calibration. Rectangular  $2 \times 3$  and  $3 \times 5$  tables check that the discussion is not tied only to square alphabets. The panels do not average over shapes.



**Figure 6.17:** Wider MI differences in a uniform  $2 \times 2$  population pair. The single panel has  $P$  and  $Q$  row and column margins  $(0.5, 0.5)$ , first-cell changes,  $n_P = n_Q = 100$ ,  $I(P) = 0.02$ , and  $I(Q) = 0.02 + |\Delta|$ . The horizontal grid is  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02, 0.05, 0.1, 0.2$  nats; the rejection axis is 0–1. Each point has 20,000 replicates.

The wider  $2 \times 2$  curve shows that both methods eventually detect large MI differences, while Expanded Welch remains below Wald until the rejection rates approach one. This is the expected cost of its larger critical value: the power loss matters most at intermediate differences and becomes small when the signal is overwhelming.

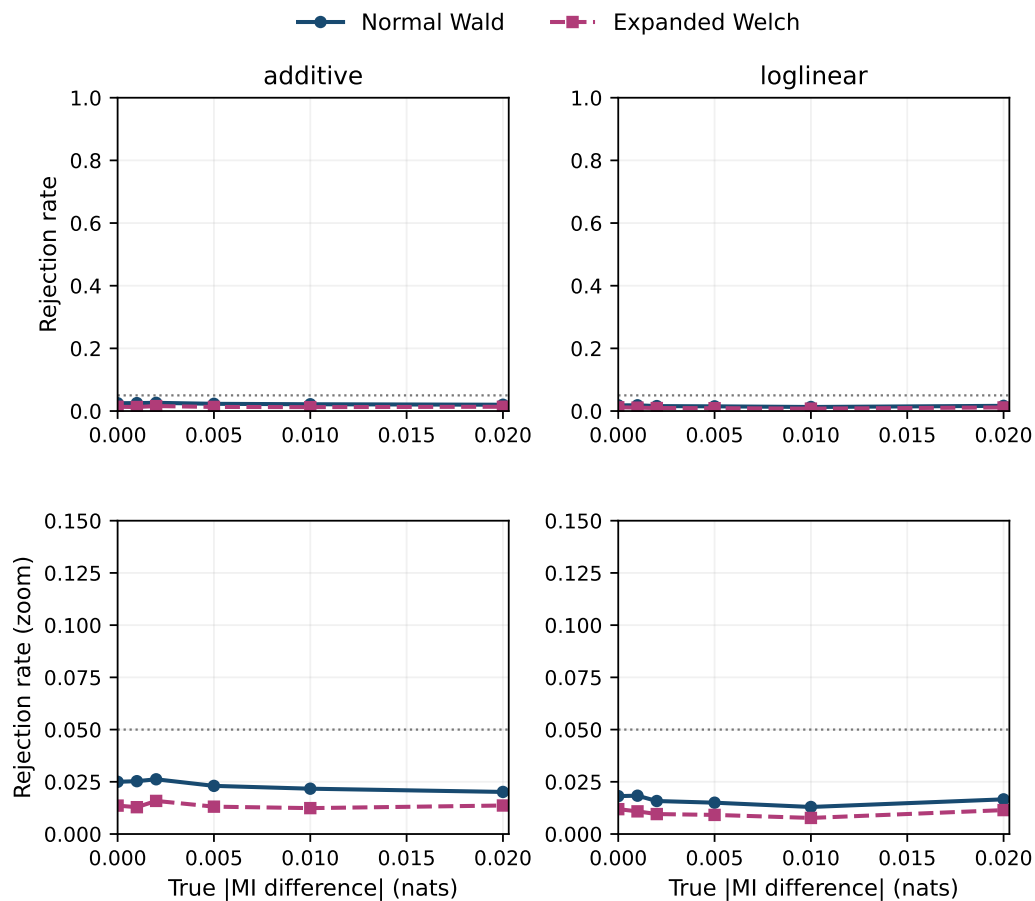


**Figure 6.18:** Rectangular tables. Columns are  $2 \times 3$  and  $3 \times 5$ . In each dimension  $k$ ,  $P$ 's corresponding margin is  $m_k(0.7)$  and  $Q$ 's is  $m_k(0.8)$ , for both rows and columns. Thus the first column uses  $P$  rows  $(0.7, 0.3)$ , columns  $(0.7, 0.15, 0.15)$ , and  $Q$  rows  $(0.8, 0.2)$ , columns  $(0.8, 0.1, 0.1)$ ; the second applies the same formula to 3 and 5 categories. Both use first-cell changes,  $n_P = n_Q = 100$ ,  $I(P) = 0.02$ , and  $I(Q) = 0.02 + |\Delta|$  for  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. Top: 0–1; bottom: 0–0.15. Each point has 20,000 replicates.

In the rectangular tables, both methods remain below 0.05 throughout the displayed range, with Expanded Welch lower still. Those low alternative rejection rates indicate weak detection. The rectangular tables therefore show the conservative half of the trade-off, while the wider  $2 \times 2$  curve shows where detection eventually becomes strong.

## 6.5 Population construction sensitivity

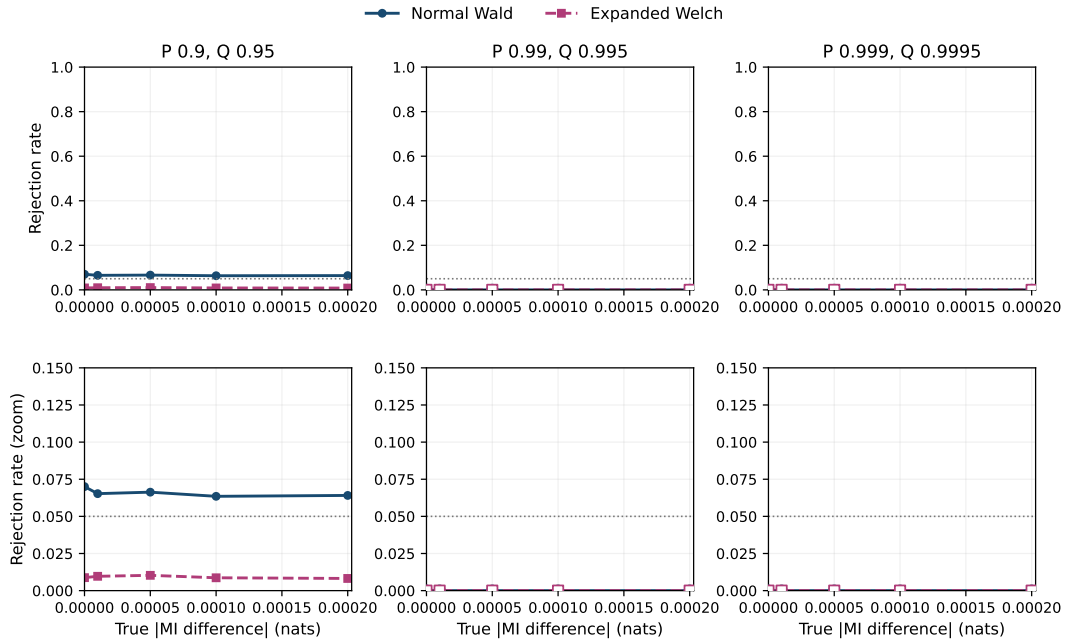
The additive block is interpretable, but it is only one way to make tables with fixed margins and target MI. Figure 6.19 compares it with an ordinal log-linear construction under matched targets. The null rates are not identical: for the displayed  $5 \times 5$  different-skew comparison at  $n = 100$ , Wald gives 0.0250 under additive construction and 0.0181 under log-linear construction; Expanded Welch gives 0.0136 and 0.0119. All are below nominal and all are valid. Thus matching the scalar MI and margins does not fix the full finite-sample behaviour. The constructor comparison provides a sensitivity check across two dependence structures.



**Figure 6.19:** Additive versus ordinal log-linear construction for  $5 \times 5$  tables. The left column uses the first-cell additive block; the right fits an ordinal interaction while preserving the margins. In both,  $P$ 's row and column margins are  $(0.7, 0.075, 0.075, 0.075, 0.075)$  and  $Q$ 's are  $(0.8, 0.05, 0.05, 0.05, 0.05)$ . Both have  $I(P) = 0.02$ ,  $I(Q) = 0.02 + |\Delta|$ ,  $n_P = n_Q = 100$ , and  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. Top: 0–1; bottom: 0–0.15. Each point has 20,000 replicates.

## 6.6 Extreme skew, independence and larger samples

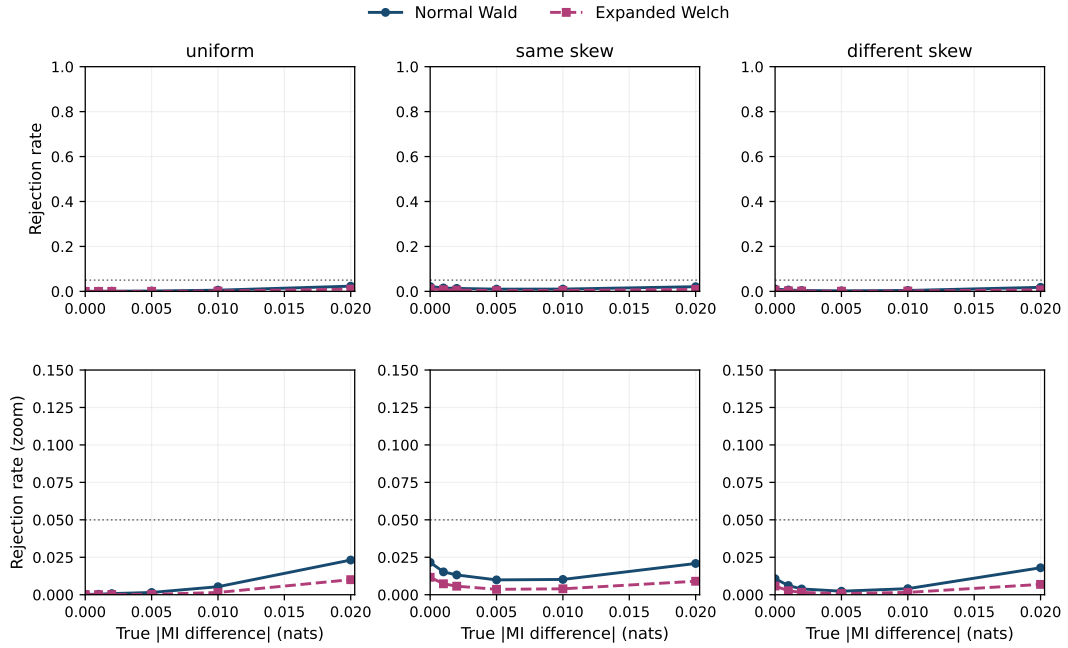
The extreme-skew check deliberately keeps configurations with very small expected cell counts. At  $n_P = n_Q = 100$ ,  $3 \times 3$  tables with dominant margins 0.999 for  $P$  and 0.9995 for  $Q$  yield valid null results in only 0.0118 of Wald replicates and 0.00005 of Expanded Welch replicates under the strict component-validity rule. Both unconditional false-positive rates are zero, but this is mostly an absence of valid tests, not evidence of excellent calibration. With dominant margins 0.9 and 0.95, the corresponding null rates are 0.0701 and 0.0088; Expanded Welch lowers an excessive Wald rate past the target. Figure 6.20 compares all three specified skew levels.



**Figure 6.20:** Extreme-skew  $3 \times 3$  tables. Across columns the dominant row and column probabilities for  $(P, Q)$  are  $(0.9, 0.95)$ ,  $(0.99, 0.995)$ , and  $(0.999, 0.9995)$ . Each remaining margin entry splits the leftover probability equally between the two other categories. Both use first-cell changes,  $n_P = n_Q = 100$ ,  $I(P) = 0.00001$ , and  $I(Q) = I(P) + |\Delta|$ , for  $|\Delta| = 0, 0.00001, 0.00005, 0.0001, 0.0002$  nats. Top: 0–1; bottom: 0–0.15. Each point has 20,000 replicates; hollow markers flag validity below 90%.

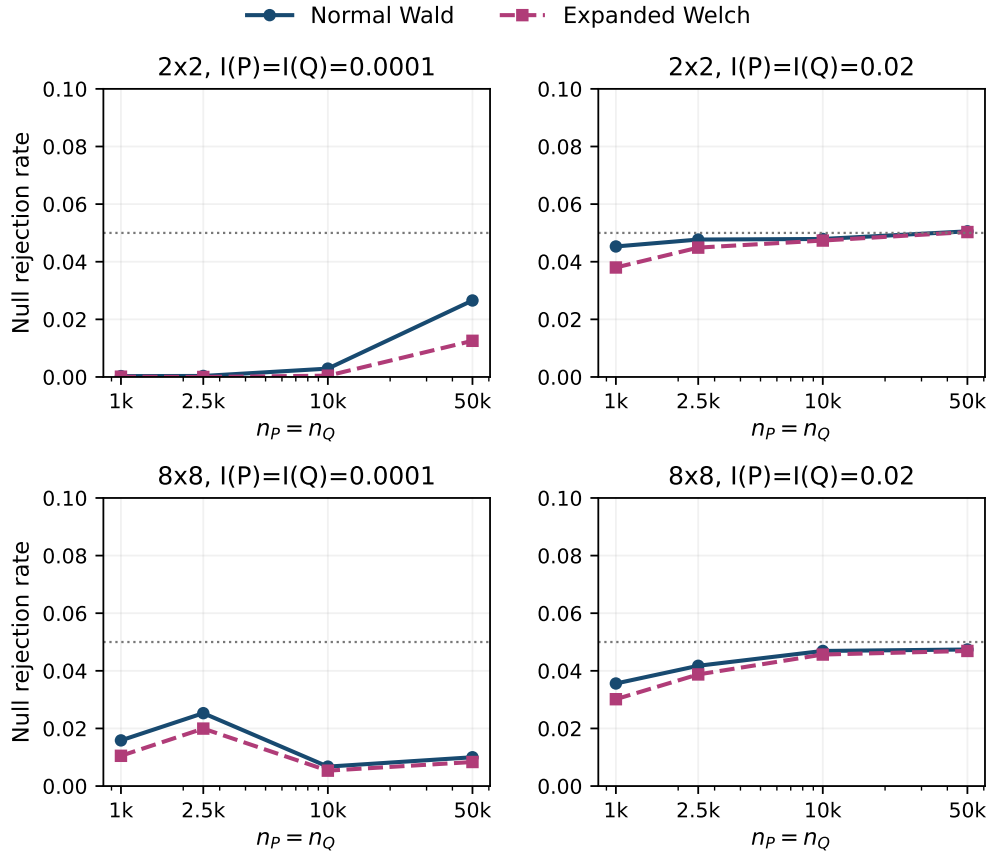
Exact independence is a separate theoretical boundary. Figure 6.21 starts  $P$  at MI zero and includes the equal-MI point where both populations are independent. This is not a routine positive-MI validation of Expanded Welch: the population first-order MI variance is zero at independence. At  $n = 100$ , for example, the uniform  $2 \times 2$  null rates are only 0.00025 and 0.00010 for Wald and Expanded Welch. Appendix D explains why a second-order independence test has a different

reference.



**Figure 6.21:** Exact-independence boundary for  $2 \times 2$  tables. Columns are uniform margins (0.5, 0.5) for both populations, same skew (0.8, 0.2) for both, and different skew with  $P = (0.7, 0.3)$  and  $Q = (0.8, 0.2)$ ; each vector is used for both rows and columns. The first-cell block is used,  $I(P) = 0$ ,  $I(Q) = |\Delta|$ ,  $n_P = n_Q = 100$ , and  $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$  nats. Top: 0–1; bottom: 0–0.15. Each point has 20,000 replicates.

Does increasing  $n$  resolve the null error when the populations remain fixed? Figure 6.22 keeps equal MI and changes only sample size. Across the 12 first-block shape–margin combinations with  $I(P) = I(Q) = 0.02$  and  $n = 50,000$ , Wald’s false-positive rates range from 0.0470 to 0.0527 and Expanded Welch’s from 0.0466 to 0.0526. At equal MI 0.0001, however, the ranges at the same  $n$  are 0.0089–0.0280 and 0.0076–0.0138. These fixed-population results show that convergence can remain slow when the true MI is very small.



**Figure 6.22:** Large-sample null behaviour. Rows compare  $2 \times 2$  and  $8 \times 8$  tables; columns have common true MI 0.0001 and 0.02 nats. For both shapes,  $P$ 's row and column margins are  $m_k(0.7)$  and  $Q$ 's are  $m_k(0.8)$ , with  $k = 2$  or  $8$ . Both populations use the first-cell block. The horizontal axis is the equal sample size  $n_P = n_Q = 1,000, 2,500, 10,000, 50,000$  on a labelled log scale; the vertical axis is null rejection from 0 to 0.10. Each point has 20,000 replicates.

## 6.7 Overall calibration and a mechanism check

The main-grid null contains two cases distinguished by whether the population tables themselves are equal. The 72 uniform and same-skew configurations have  $P = Q$ , whereas the 36 different-skew configurations have  $P \neq Q$  but  $I(P) = I(Q)$ . The latter are the weak-null cases that most directly match the thesis motivation. Table 6.2 reports these profiles separately. The broad interval 0.025–0.075 is used only to describe the landscape; it is not a formal acceptance region for calibration.

**Table 6.2:** Location, separately for each method, of the 108 main-grid null configurations relative to the descriptive rejection-rate interval 0.025–0.075. Each profile contains 36 configurations. The final column separately counts configurations with validity below 90%; it overlaps the three rate-location columns. In particular, all four  $n = 2$  configurations per profile have zero validity for both methods. Uniform and same-skew are strong nulls; different-skew is the equal-MI weak null. Exact rates and validity are available in the saved results described in Appendix C.

| Margins        | Method         | Below | Inside | Above | Validity below 90% |
|----------------|----------------|-------|--------|-------|--------------------|
| Uniform        | Normal Wald    | 10    | 18     | 8     | 4                  |
| Uniform        | Expanded Welch | 19    | 16     | 1     | 5                  |
| Same skew      | Normal Wald    | 14    | 15     | 7     | 8                  |
| Same skew      | Expanded Welch | 25    | 11     | 0     | 12                 |
| Different skew | Normal Wald    | 9     | 16     | 11    | 8                  |
| Different skew | Expanded Welch | 19    | 13     | 4     | 12                 |

The table makes the direction of the adjustment clear. Expanded Welch greatly reduces the number of rates above the descriptive band, but it also increases the number below it in every margin profile. Its apparent protection against liberal rejection is therefore purchased with additional conservatism rather than with calibration in both directions.

A separate supplementary study tested the mechanism without altering the confirmatory results. For the representative  $3 \times 3$  different-skew null with  $I(P) = I(Q) = 0.02$  and  $n = 100$ , ordinary Wald divided each observed bias-corrected MI difference by its own plug-in standard error and rejected 0.0166 of sample pairs. Keeping those same differences but dividing each by one fixed finite-sample standard deviation estimated from an independent pilot gave a rejection rate of 0.0551. Dividing instead by the fixed population first-order standard deviation gave 0.0976. The confirmatory Wald rate was 0.0171; its small difference from 0.0166 is Monte Carlo variation between independent runs.



Across the evaluated pairs, the average squared plug-in standard error was about 1.17 times the finite-sample variance of the MI difference estimated from the independent pilot. The population first-order variance was smaller than that finite-sample variance. These differences help explain why the three denominators produce different rejection rates. The two fixed denominators are diagnostics, not proposed tests: Expanded Welch changes neither the observed MI differences nor their plug-in standard errors. Full pilot and ablation results are retained in the experiment supplement.

## 6.8 Runtime and summary of evidence

The full method calls have the same  $O(rc)$  asymptotic complexity. The observed median time across the 60 timing regimes is about 85 microseconds for Wald and 150 microseconds for Expanded Welch. The median Expanded-to-Wald ratio is about 1.75, with regime ratios about 1.72–1.79. For the explicitly timed  $3 \times 3$ , different-skew,  $n_P = n_Q = 100$ , equal-MI input regime, the medians are 85.083 and 149.979 microseconds. These are implementation- and machine-specific timings, not universal complexity constants. Their near-flat values mainly measure fixed per-call overhead and do not empirically establish the  $O(rc)$  scaling, which follows instead from the operation count in Section 4.4. Exact per-regime medians and interquartile ranges are in the saved runtime file cited in Appendix C.

Computationally, the correction is feasible: it has the same cell-linear order as Wald and adds roughly 75% to a scalar call in this implementation. The overhead is modest compared with resampling, but it is still an extra cost. Because the statistical results do not show a general improvement, runtime does not provide a separate reason to prefer Expanded Welch.

**Table 6.3:** Scalar-call times in microseconds for the different-skew null at  $n_P = n_Q = 100$  and  $I(P) = I(Q) = 0.02$ . Entries give median [interquartile range] over five passes on 200 saved inputs per regime. Population construction and multinomial sampling are excluded.

| Shape        | Normal Wald       | Expanded Welch       | Expanded/Wald |
|--------------|-------------------|----------------------|---------------|
| $2 \times 2$ | 85.8 [85.5, 86.3] | 150.0 [149.4, 151.3] | 1.75          |
| $3 \times 3$ | 85.1 [84.6, 85.6] | 150.0 [148.9, 150.9] | 1.76          |
| $3 \times 5$ | 85.2 [84.9, 85.5] | 150.1 [149.5, 150.9] | 1.76          |
| $5 \times 5$ | 85.3 [85.1, 85.7] | 149.2 [148.5, 150.1] | 1.75          |
| $8 \times 8$ | 88.0 [87.1, 88.9] | 153.4 [152.4, 154.8] | 1.74          |

The evidence gives three conclusions. Expanded Welch can move a liberal Wald null rate closer to 0.05, but its heavier-tailed reference also reduces rejection where

---

Wald is already conservative. In sparse samples, the unconditional rate can fall because Expanded Welch is more often invalid under the strict component-validity rule. Some failures would not occur under the direct combined formula and are not needed for the first two findings. These behaviours cannot be summarised by one number; the matched curves and validity records underpin the discussion in Chapter 7.

# Chapter 7

## Discussion

The proposed correction models how the estimated MI variance changes across samples. The experiments show why this alone is insufficient: Wald and Expanded Welch use the same observed MI difference and plug-in standard error. Changing the reference distribution leaves any finite-sample error in those quantities untouched.

### 7.1 Answers to the research questions

The first research question asked how an MI-specific effective degree of freedom can be constructed. Chapter 4 does so by differentiating the variance of pointwise MI along one-cell probability perturbations. The derivative includes changes to row and column margins, not just to the selected cell. Its sampling variance supplies the component degrees of freedom used in Satterthwaite moment matching. These derivative and moment-matching steps define a reproducible reference distribution.

The second question concerned false positives, detection and validity. Table 6.1 contains both a fully valid small-sample null case where Expanded Welch moves an excessive Wald false-positive rate toward 0.05 and a larger-sample null case where the same adjustment moves a below-nominal Wald rate farther from 0.05. In the sparsest panels, a low unconditional rejection rate may largely reflect invalid outputs under the implemented rule. Thus improved calibration at selected null points does not imply that the expanded procedure performs better overall. Under alternatives, its rejection rate is generally lower; these comparisons are not power at a matched achieved size.

The third question asked whether population features alter this judgement. The main landscape varies alphabet size and margins explicitly. The focused comparisons show that baseline MI, changed-cell location, sample allocation and population constructor can change the finite-sample behaviour even when the target

MI difference is held fixed. The construction comparison is particularly instructive: equal margins and MI do not determine the table's full sampling distribution. A scalar target is useful for organising experiments, but cannot guarantee identical difficulty across distinct population tables.

The fourth question concerned sample size and computation. For the tested first-block positive-MI populations with true MI 0.02, both null rates are near 0.05 at  $n = 50,000$ . The corresponding very-small-MI fixed populations remain noticeably conservative at that sample size. Expanded Welch has the same linear-in-cell-count order as Wald but costs about 1.75 times as much per scalar call in the measured implementation. The extra expense is modest compared with a full resampling procedure but not cost-free, particularly for many repeated comparisons.

## 7.2 What the correction can and cannot change

Both methods compute the same  $\widehat{\Delta}_{\text{BC}}$ ,  $\widehat{\text{SE}}$  and  $T$ . With valid positive degrees of freedom, Expanded Welch has a heavier-tailed reference and a two-sided p-value no smaller than Wald's for the same table pair. This gives a direct explanation for many plots: the correction lowers rejection whether the Wald rate is too high or already too low. It is not an adaptive calibration method that can move the threshold in both directions. It also does not alter the leading bias correction or re-estimate the numerator's finite-sample distribution.

The estimated numerator and denominator come from the same count tables. Independence of the two samples allows their variance contributions to be added, but it does not make  $T$ 's numerator independent of its estimated standard error. The representative pilot in Chapter 6 shows that both parts can depart from their first-order approximations. Moment matching the denominator alone therefore cannot guarantee a Student law or correct every finite-sample error.

Near independence, the local calculation gives  $V \approx 2I$ ,  $\tau^2 \approx 4V$ , and hence a population component degree of freedom near  $nI$ . This explains why the reference can become very heavy-tailed at low MI, exactly where the first-order approximation is least reliable. The result identifies a mechanism behind the low effective degrees of freedom.

At exact independence,  $I = 0$  and the population first-order MI variance vanishes. This does not mean finite empirical MI estimates are always zero. It means their leading stochastic variation is quadratic rather than linear in the probability errors. The standard one-table likelihood-ratio statistic uses that second-order behaviour. Neither setting a reference  $Q$  to an independent distribution nor squaring the current

$T$  makes Expanded Welch the classical independence test. Appendix D gives the mathematical distinction.

## 7.3 Interpretation of invalid samples

The validity diagnostics matter most where expected counts are very small. Wald needs a usable total estimated variance. The expanded implementation additionally requires each component sensitivity variance and effective degree of freedom to be positive and finite. An observed table can have a positive total variance while one component is zero. The equivalent combined formula in Equation (4.41) remains defined when one table has  $\hat{V} = \hat{\tau}^2 = 0$ , but the frozen implementation first requires two positive finite component degrees of freedom. Some Expanded-only invalidity therefore reflects the implementation rule rather than the combined formula. The primary unconditional rate records a failed calculation as a non-rejection.

That rate answers an operational question: how often does the implemented procedure produce a valid rejection? It should not be mistaken for the conditional rejection probability among valid tables. Nor does a failure to return a p-value constitute a true negative. The saved validity and paired results allow readers to distinguish fewer rejections from fewer usable tests.

## 7.4 Practical conclusion and limits of the evidence

For the tested task, Normal Wald remains the simpler general baseline. Expanded Welch can be useful as a conservative sensitivity analysis in a regime where its validity is high and Wald's null rejection is demonstrably liberal. It should not be installed as a universal replacement merely because one null table shows a smaller false-positive rate. In applications, the population truth is unknown, so the most relevant calibration study is one matched to the anticipated alphabet, sample size and plausible margins. Selecting between the methods after inspecting a single p-value is not a validated selection rule.

Several limitations bound this conclusion. The population families are deliberately constructed fixed examples, not a probability sample of all joint distributions. Twenty thousand replicates make a rate at a fixed population relatively precise; they do not make the 534 selected population pairs representative of every application. The tables are at most  $8 \times 8$  square and  $3 \times 5$  rectangular, with the stated finite sample ladders and only a two-sided nominal level of 0.05. The main construction emphasises a local four-cell association; spread, rare-cell and log-linear checks

probe but do not exhaust other possibilities. The study does not validate paired observations, continuous estimators, a real-data workflow or growing alphabets.

The protocol was frozen before the final simulation, after exploratory development of the method and design. It is a reproducible final run without external preregistration. Configurations displayed in several sections are the same sampled run. Repeating their figure appearances does not provide independent confirmation. The recorded Monte Carlo intervals also address repeated sampling of each fixed pair, not the broader uncertainty of generalising to an untested population.

The alternative curves use the common nominal 0.05 threshold. They show practical detection under the tested procedures, not power after giving each method a simulation-specific critical value.

Future work should address the joint finite-sample behaviour of the numerator and denominator or use a calibrated reference tailored to sparse tables. The transition near independence is a separate problem because the first-order variance tends to zero. A likelihood-ratio or resampling alternative must be assessed under the equal-MI *weak null*, not imported from a test of full distributional equality or one-table independence without checking its target. These are directions for new validation, not remedies established by the present results.

# Chapter 8

## Conclusion

This thesis examined whether uncertainty in an estimated MI standard error can be represented through an MI-specific Welch–Satterthwaite correction when comparing two independent categorical populations. The null is equality of their population MI values, not independence within either population or equality of the complete joint distributions.

The derivation starts from the pointwise-MI variance governing first-order sampling error. A one-cell perturbation changes the joint probabilities, the margins and the pointwise values. Differentiating the variance functional supplies the sensitivity; the variance of that sensitivity supplies the sampling variability needed for component effective degrees of freedom. The resulting Expanded Welch procedure retains Wald’s bias-corrected MI difference and plug-in standard error but evaluates the shared statistic against a Student reference. This is a principled moment-matching construction, not an exact finite-sample Student law.

The fixed-population experiment found both benefits and costs. Expanded Welch can make a liberal Wald null rate closer to the nominal 5% level. Where Wald is already conservative, the heavier-tailed reference moves rejection farther below that target. In sparse regimes, Expanded Welch also returns fewer valid results under the strict component-validity rule used in the implementation. Some of those failures arise from the rule rather than the combined formula, and a low unconditional rate there is not reliable evidence of good calibration. At ordinary positive MI and sufficiently large samples, both methods approached the nominal null rate in the tested fixed populations, while near-independence settings remained difficult even at large finite sample sizes. The expanded call used about 1.75 times the runtime of Wald in the measured implementation.

Near independence, the component degree of freedom is approximately  $nI$ , so Expanded Welch can impose a very heavy-tailed reference where the first-order

approximation remains weak. A supplementary diagnostic confirms that errors can arise in both the estimated standard error and the numerator. Changing only the reference distribution leaves both approximation errors unchanged and therefore cannot provide a general correction.

The evidence therefore supports Normal Wald as the general baseline. Expanded Welch is only a conditional sensitivity analysis; future corrections must address the numerator, standard error and population structure together rather than changing the reference law alone.



# Appendix A

## Additional Derivation Details

This appendix records algebra that supports Chapters 3 and 4 but is not required to follow the main argument.

### A.1 Gradient of mutual information on the probability simplex

Mutual information can be written as

$$I(P) = \sum_{i,j} p_{ij} \log p_{ij} - \sum_i p_{i+} \log p_{i+} - \sum_j p_{+j} \log p_{+j}. \quad (\text{A.1})$$

Treating the cell probabilities as coordinates gives

$$\frac{\partial I(P)}{\partial p_{ij}} = (\log p_{ij} + 1) - (\log p_{i+} + 1) - (\log p_{+j} + 1) \quad (\text{A.2})$$

$$= \log \frac{p_{ij}}{p_{i+} p_{+j}} - 1 \quad (\text{A.3})$$

$$= \ell_P(i, j) - 1. \quad (\text{A.4})$$

Any valid probability perturbation  $h = (h_{ij})$  satisfies  $\sum_{i,j} h_{ij} = 0$ . Its directional derivative is therefore

$$\left. \frac{d}{d\varepsilon} I(P + \varepsilon h) \right|_{\varepsilon=0} = \sum_{i,j} \{\ell_P(i, j) - 1\} h_{ij} \quad (\text{A.5})$$

$$= \sum_{i,j} \ell_P(i, j) h_{ij}. \quad (\text{A.6})$$

The constant in the unconstrained gradient disappears on the simplex. For the one-cell direction  $h_{ij} = \mathbf{1}\{i = x, j = y\} - p_{ij}$ , Equation (A.6) becomes

$$\ell_P(x, y) - \sum_{i,j} p_{ij} \ell_P(i, j) = \ell_P(x, y) - I(P), \quad (\text{A.7})$$

which is Equation (4.18).

## A.2 Why the variance of pointwise MI appears

For empirical cell probabilities,

$$\hat{p}_{ij} - p_{ij} = \frac{1}{n_P} \sum_{a=1}^{n_P} \{\mathbf{1}(Z_a^{(P)} = (i, j)) - p_{ij}\}. \quad (\text{A.8})$$

Substitution into the first-order Taylor expansion of MI around  $P$  gives

$$\hat{I}(P) - I(P) \approx \sum_{i,j} \ell_P(i, j) (\hat{p}_{ij} - p_{ij}) \quad (\text{A.9})$$

$$= \frac{1}{n_P} \sum_{a=1}^{n_P} \left[ \sum_{i,j} \ell_P(i, j) \{\mathbf{1}(Z_a^{(P)} = (i, j)) - p_{ij}\} \right] \quad (\text{A.10})$$

$$= \frac{1}{n_P} \sum_{a=1}^{n_P} \{\ell_P(Z_a^{(P)}) - I(P)\}. \quad (\text{A.11})$$

The observation-level summand has variance

$$\text{Var}_P\{\ell_P(X, Y) - I(P)\} = \mathbb{E}_P \left[ \{\ell_P(X, Y) - I(P)\}^2 \right] \quad (\text{A.12})$$

$$= \sum_{i,j} p_{ij} \{\ell_P(i, j) - I(P)\}^2 \quad (\text{A.13})$$

$$= V(P). \quad (\text{A.14})$$

This is why the first-order variance of the plug-in MI estimator is  $V(P)/n_P$ , not a variance of the numerical category labels.

## A.3 Mean of the variance influence function

The influence function  $g_P(x, y)$  is centred at the population level. This can be shown without expanding its individual terms. The probability-weighted average of

the one-cell perturbation directions is

$$\sum_{x,y} p_{xy} \{\delta_{(x,y)} - P\} = P - P = 0. \quad (\text{A.15})$$

The first derivative is linear in the direction of change. Therefore, the weighted average of the one-cell derivatives is the derivative in the weighted-average direction:

$$E_P\{g_P(X, Y)\} = \sum_{x,y} p_{xy} \left. \frac{d}{d\varepsilon} V\{(1 - \varepsilon)P + \varepsilon\delta_{(x,y)}\} \right|_0 \quad (\text{A.16})$$

$$= \left. \frac{d}{d\varepsilon} V\left\{P + \varepsilon \sum_{x,y} p_{xy} \{\delta_{(x,y)} - P\}\right\} \right|_0 \quad (\text{A.17})$$

$$= \left. \frac{d}{d\varepsilon} V(P) \right|_0 = 0. \quad (\text{A.18})$$

The same weighted cancellation holds after exact plug-in substitution. The implementation nevertheless subtracts the probability-weighted empirical mean inside Equation (4.38) to protect the variance calculation from floating-point roundoff; this does not change the population definition.

## A.4 Direct table substitutions

For observed counts  $N^P = (N_{ij}^P)$ , the quantities used by expanded Welch are calculated as

$$\hat{p}_{ij} = \frac{N_{ij}^P}{n_P}, \quad \hat{p}_{i+} = \sum_j \hat{p}_{ij}, \quad \hat{p}_{+j} = \sum_i \hat{p}_{ij}, \quad (\text{A.19})$$

$$\hat{\ell}_P(i, j) = \log \frac{\hat{p}_{ij}}{\hat{p}_{i+}\hat{p}_{+j}}, \quad \hat{I}(P) = \sum_{i,j} \hat{p}_{ij} \hat{\ell}_P(i, j), \quad (\text{A.20})$$

$$\hat{V}(P) = \sum_{i,j} \hat{p}_{ij} \{\hat{\ell}_P(i, j) - \hat{I}(P)\}^2. \quad (\text{A.21})$$

All sums involving pointwise MI are restricted to cells with  $\hat{p}_{ij} > 0$ . Direct substitution gives

$$\hat{g}_P(i, j) = \{\hat{\ell}_P(i, j) - \hat{I}(P)\}^2 - \hat{V}(P) \quad (\text{A.22})$$

$$+ 2 \left\{ \hat{\ell}_P(i, j) - \frac{\sum_{j'} \hat{p}_{ij'} \hat{\ell}_P(i, j')}{\hat{p}_{i+}} - \frac{\sum_{i'} \hat{p}_{i'j} \hat{\ell}_P(i', j)}{\hat{p}_{+j}} + \hat{I}(P) \right\}, \quad (\text{A.23})$$

$$\hat{\tau}^2(P) = \sum_{i,j} \hat{p}_{ij} \left\{ \hat{g}_P(i, j) - \sum_{k,l} \hat{p}_{kl} \hat{g}_P(k, l) \right\}^2, \quad (\text{A.24})$$

$$\hat{\nu}_P = \frac{2n_P \hat{V}(P)^2}{\hat{\tau}^2(P)}. \quad (\text{A.25})$$

These expressions are the direct  $O(rc)$  implementation of the population derivation.

## A.5 Numerical derivative checks

The implementation tests the analytic derivatives against centred finite differences on positive probability tables. For a selected cell and a small positive step  $\eta$ , define

$$P_{\pm\eta} = (1 \mp \eta)P \pm \eta\delta_{(x,y)}. \quad (\text{A.26})$$

For sufficiently small  $\eta$ , the checks compare

$$\frac{I(P_\eta) - I(P_{-\eta})}{2\eta} \quad \text{with} \quad \ell_P(x, y) - I(P), \quad (\text{A.27})$$

$$\frac{V(P_\eta) - V(P_{-\eta})}{2\eta} \quad \text{with} \quad g_P(x, y). \quad (\text{A.28})$$

The finite-difference step is selected so that both perturbed tables retain positive probabilities. These checks test the algebra independently of the Monte Carlo calibration experiment.

# Appendix B

## Population Construction and Exact Design Settings

The experiment uses fixed population tables. This appendix states the construction precisely enough to reproduce any plotted point and records the finite parameter grids. Full-precision probabilities for every constructed pair are stored in `population_definitions.csv`; rounded examples in the thesis are for reading, not simulation input.

### B.1 Additive construction

For prescribed positive row vector  $a$  and column vector  $b$ , form  $B = ab^\top$ . The principal first-cell direction  $H$  contains  $\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$  in rows and columns 1 and 2 and zero elsewhere. The rare-cell direction places the same block in the last two rows and columns. The spread direction is

$$H_{ij} = u_i v_j, \quad u_i = -1 + \frac{2(i-1)}{r-1}, \quad v_j = -1 + \frac{2(j-1)}{c-1}. \quad (\text{B.1})$$

Both score vectors sum to zero, so every direction preserves margins.

For  $t \geq 0$ , define  $R(t) = B + tH$ . Calculate the first strength at which a decreasing cell reaches zero,

$$t_{\max} = \min_{H_{ij} < 0} \frac{B_{ij}}{-H_{ij}}. \quad (\text{B.2})$$

Evaluate the path endpoint MI. A target MI must lie strictly below this endpoint, within the stated numerical tolerance. For a positive target, Brent's one-dimensional root search ([Brent, 1973](#)) solves  $I\{R(t)\} = I_{\text{target}}$  on  $[0, t_{\max}]$ ; a zero target uses

$t = 0$ . The final table is accepted only if every cell is strictly positive, the sum is one, both margins agree with their prescribed vectors, and achieved MI agrees with the target. The normalisation and margin tolerances are  $10^{-12}$ ; the MI tolerance is  $10^{-12} + 10^{-8}I_{\text{target}}$ . These numerical checks ensure that a horizontal-axis difference denotes constructed population truth, not an empirical estimate. They do not impose an expected-count floor on the subsequent sample.

The row and column vectors are separately constructed for each population. For shape  $r \times c$ , profile **uniform** uses uniform vectors of lengths  $r$  and  $c$  for both  $P$  and  $Q$ . Profile **same\_skew** uses  $m_r(0.8)$ ,  $m_c(0.8)$  for both. Profile **different\_skew** uses  $m_r(0.7)$ ,  $m_c(0.7)$  for  $P$  and  $m_r(0.8)$ ,  $m_c(0.8)$  for  $Q$ . The extreme regimes instead state the two dominant probabilities directly. The direction of the alternative is a separate parameter and should not be inferred from which population has the more skewed margin.

The log-linear comparison uses the scores  $u_i, v_j$  from Equation (B.1), but starts from positive cell weights

$$K_{ij}(\lambda) = \exp(\lambda u_i v_j) \quad (\text{B.3})$$

instead of adding  $tH$  to the independence table  $B$ . Iterative proportional fitting (Deming and Stephan, 1940) scales rows and columns of these weights until the prescribed margins are restored. At  $\lambda = 0$ , this gives the independence table; a numerical search then chooses  $\lambda$  to reach the target MI. The final table passes the same positivity, normalisation, margin and target-MI checks as the additive table. Unlike the first-block additive construction used in the comparison, it changes cells throughout the table. Matching margins and MI does not imply matching all cell probabilities or expected counts.

## B.2 Frozen statistical grids

Unless a row of Table B.1 says otherwise, both populations use the first-cell additive direction,  $I(P) = I_0$ ,  $I(Q) = I_0 + |\Delta|$ , and the three standard margin profiles. The symbol  $n$  means  $n_P = n_Q = n$ . A requested point was simulated only after the population constructor verified a strictly positive feasible table. The final display contains no infeasible slots, but that is a result of choosing feasible finite grids. Curves should not be extrapolated beyond their last constructed target.

**Table B.1:** Exact statistical experiment grids; MI values are in nats.

| Family           | Settings                                                                                                                                                                                                                                                                        |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Main             | Shapes $2 \times 2, 3 \times 3, 5 \times 5, 8 \times 8$ ; three profiles; $I_0 = 0.02$ ; $ \Delta  \in \{0, 0.001, 0.002, 0.005, 0.01, 0.02\}$ ; $n \in \{2, 5, 10, 20, 50, 100, 250, 500, 1000\}$ .                                                                            |
| Baseline         | Main shapes and profiles; $I_0 \in \{0.0001, 0.001, 0.02\}$ ; main difference grid; $n \in \{20, 100, 1000\}$ .                                                                                                                                                                 |
| Changed cells    | Shapes $3 \times 3, 5 \times 5, 8 \times 8$ ; different-skew margins; $I_0 = 0.0001$ ; $ \Delta  \in \{0, 0.0001, 0.00025, 0.0005, 0.001, 0.002\}$ ; $n \in \{20, 100, 1000\}$ . First and rare blocks for all shapes; spread pattern for $5 \times 5$ and $8 \times 8$ .       |
| Imbalance        | Main shapes, profiles and difference grid; $I_0 = 0.02$ . Sample pairs are $(50, 50), (50, 100), (100, 50), (50, 250), (250, 50), (50, 500), (500, 50), (500, 500)$ . Both $Q$ -higher and $P$ -higher MI directions.                                                           |
| Larger effects   | Shapes $2 \times 2, 3 \times 3$ ; three profiles; $I_0 = 0.02$ ; main difference grid plus $0.05, 0.1, 0.2$ ; $n \in \{20, 100, 1000\}$ .                                                                                                                                       |
| Rectangular      | Shapes $2 \times 3, 3 \times 5$ ; three profiles; $I_0 = 0.02$ ; main difference grid; $n \in \{20, 100, 1000\}$ .                                                                                                                                                              |
| Construction     | Shapes $3 \times 3, 5 \times 5, 8 \times 8$ ; uniform and different-skew profiles; $I_0 \in \{0.0001, 0.02\}$ ; main difference grid; $n \in \{20, 100, 1000\}$ ; additive first block and ordinal log-linear constructors.                                                     |
| Extreme skew     | Shapes $2 \times 2, 3 \times 3, 8 \times 8$ ; dominant margin probability pairs $(\rho_P, \rho_Q) \in \{(0.9, 0.95), (0.99, 0.995), (0.999, 0.9995)\}$ ; $I_0 = 0.00001$ ; $ \Delta  \in \{0, 0.00001, 0.00005, 0.0001, 0.0002\}$ ; $n \in \{1, 2, 5, 20, 100, 1000, 10000\}$ . |
| Rare-cell stress | Shapes $3 \times 3, 8 \times 8$ ; dominant pair $(0.9, 0.95)$ ; first and rare blocks; $I_0 = 0.00001$ ; $ \Delta  \in \{0, 0.00001, 0.000025, 0.00005, 0.0001\}$ ; same $n$ ladder as extreme skew.                                                                            |
| Independence     | Shapes $2 \times 2, 8 \times 8$ ; three profiles; $I_0 = 0$ ; main difference grid; $n \in \{2, 20, 100, 1000\}$ .                                                                                                                                                              |

---

| Family      | Settings                                                                                                                                                                                                                                     |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Convergence | Main square shapes and profiles; equal MI $I(P) = I(Q) \in \{0.0001, 0.02\}$ ; $n \in \{1000, 2500, 10000, 50000\}$ . Additional rare-block controls use $3 \times 3, 5 \times 5, 8 \times 8$ , different-skew margins and common MI 0.0001. |

---

The  $n = 1$  stress setting is retained and not hidden; the statistic is not evaluable there and its result is recorded as invalid. Every unique statistical configuration has 20,000 sampled table pairs. When the same population pair and sample sizes appear in another section, the saved configuration is reused for presentation and not simulated again.



# Appendix C

## Evidence Index and Additional Diagnostics

The main results chapter shows a systematic selection of full-scale and calibration-zoom figures. The full companion atlas remains the source for every sample size and focused regime in the frozen design. The index below specifies how to retrieve exact numbers without averaging regimes into a single summary. Paths are relative to the workspace root `Simulations/`. The final results directory used below is `WelchSatterthwaiteMI/results/thesis_redesign/`. These are workspace paths; Appendix E gives commands for the extracted archive, where `main.tex` and `WelchSatterthwaiteMI/` share one root.

The companion written atlas is at `WelchSatterthwaiteMI/docs/experiments/THESIS_EXPERIMENTS.md`. Its 99 primary figures and seven extra calibration zooms retain the full design. The thesis-specific PDFs in `Thesis Writeup/Welch MI/figures_rewrite/` are generated from the two final CSV files named in the source script. The adjacent `figure_manifest.json` lists, for each thesis PDF, its original section, source files, configuration identifiers and 20,000-replicate denominator. The file `evidence_values.tex` is generated from exact configuration IDs for the numerical examples in Chapter 6.

For each configuration, 20,000 table pairs are sampled. The plotted rate is the number of valid rejections divided by 20,000. The method-valid conditional rate divides valid rejections by that method's valid outputs; it is undefined if there are none. The common-valid rate restricts to outputs where both methods are valid. A paired difference is based on counts of the four joint rejection outcomes for the same table pairs. These distinctions allow a reader to check whether a curve difference is primarily a changed reference or lost validity.

For the paired comparison, let  $D_s$  equal the Wald valid-rejection indicator minus

**Table C.1:** Saved files and the question each answers.

| File in the final results directory                                      | Use                                                                                                                                                                                |
|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>display_manifest.csv</code>                                        | Figure section, row, column, horizontal value and exact configuration identifier; a configuration may recur.                                                                       |
| <code>configuration_manifest.csv</code>                                  | Unique statistical configurations, sample sizes, target populations and simulation identifiers.                                                                                    |
| <code>population_definitions.csv</code>                                  | Full-precision $P$ and $Q$ probability tables, margins, achieved MI, constructor and feasibility diagnostics for every pair.                                                       |
| <code>cell_results.csv</code>                                            | One row per configuration and method: unconditional rejection, method validity, conditional and common-valid rates, Wilson intervals, expected-count and support-loss diagnostics. |
| <code>paired_method_results.csv</code>                                   | Paired rejection counts, rate differences and approximate paired intervals from the same replicates.                                                                               |
| <code>runtime_summary.csv</code>                                         | Per-regime median and interquartile call times in microseconds, validity and Expanded-to-Wald time ratio.                                                                          |
| <code>verification.json</code> and <code>report_verification.json</code> | Run-level population, denominator, pairing, method and report checks.                                                                                                              |

the Expanded Welch valid-rejection indicator for sampled table pair  $s$ , matching the sign used in Table 6.1. Its average is  $\bar{D} = (1/20,000) \sum_{s=1}^{20,000} D_s$ , with approximate 95% interval

$$\bar{D} \pm 1.96 \left\{ \frac{1}{20,000(19,999)} \sum_{s=1}^{20,000} (D_s - \bar{D})^2 \right\}^{1/2}. \quad (\text{C.1})$$

This calculation preserves the within-replicate pairing. In the saved CSV, the generic field is stored as method A minus method B, with Expanded Welch listed as method A; Table 6.1 therefore reports its negative. The individual method intervals use Wilson’s binomial score construction (Wilson, 1927).

For any  $P$  population with sample size  $n_P$ , its minimum expected cell count is  $\min_{ij} n_P p_{ij}$ ; define  $Q$ ’s analogously. No “expected minimum” of random observed counts is calculated. The saved support diagnostics include empty observed rows and columns and the fraction of zero-count cells. Such values describe why a statistic might fail, but do not constitute a universal validity threshold.

The thesis does not treat a descriptive count of regimes closer to 0.05 as a formal paired test over regimes. Wilson intervals quantify binomial Monte Carlo variation at each fixed point, and paired normal intervals quantify a specified within-configuration method difference. Neither interval family is a simultaneous assertion across all configurations. Tables sharing a population but changing  $n$  or  $|\Delta|$  answer different fixed-design questions and should not be collapsed into one mean rate.

# Appendix D

## The Independence Boundary

The main thesis concerns the regular weak null  $I(P) = I(Q) > 0$ . This appendix explains why constructing an independent reference distribution does not turn the Expanded Welch method into a new first-order test of independence.

### D.1 Constructing the independent reference

For a joint distribution  $P = (p_{ij})$ , define

$$q_{ij} = p_{i+}p_{+j}. \quad (\text{D.1})$$

Then  $Q = (q_{ij})$  is a probability distribution because

$$\sum_{i,j} q_{ij} = \left( \sum_i p_{i+} \right) \left( \sum_j p_{+j} \right) = 1. \quad (\text{D.2})$$

It has the same margins as  $P$ , since

$$\sum_j q_{ij} = p_{i+}, \quad \sum_i q_{ij} = p_{+j}. \quad (\text{D.3})$$

The factorisation in Equation (D.1) makes  $X$  and  $Y$  independent under  $Q$ , so  $I(Q) = 0$ .

The divergence from  $P$  to this reference is

$$D_{\text{KL}}(P \| Q) = \sum_{i,j} p_{ij} \log \frac{p_{ij}}{q_{ij}} \quad (\text{D.4})$$

$$= \sum_{i,j} p_{ij} \log \frac{p_{ij}}{p_{i+}p_{+j}} \quad (\text{D.5})$$

$$= I(P). \quad (\text{D.6})$$

Consequently, testing  $I(P) = I(Q)$  for this constructed reference is exactly testing  $I(P) = 0$ , or independence.

## D.2 Why the first-order term vanishes

At independence,  $p_{ij} = p_{i+}p_{+j}$ , and every population pointwise-MI value is zero:

$$\ell_P(i, j) = \log 1 = 0. \quad (\text{D.7})$$

Equation (4.18) therefore gives

$$\left. \frac{dI(P_\varepsilon)}{d\varepsilon} \right|_0 = \ell_P(x, y) - I(P) = 0. \quad (\text{D.8})$$

The first-order variance also vanishes:

$$V(P) = \text{Var}_P\{\ell_P(X, Y)\} = 0. \quad (\text{D.9})$$

The denominator used by Normal Wald and Expanded Welch is thus degenerate under the population independence null. This does not mean that a finite sample must have zero plug-in MI or zero plug-in variance. Sampling noise usually makes the empirical table depart from exact independence. The point is that these random departures do not enter MI through the usual first-order term at the null. They enter through the second-order term derived next, so changing only the degrees of freedom of a first-order statistic cannot supply the missing source of variation.

## D.3 Second-order behaviour

Fix a strictly positive independent table  $P$ , and let  $h = (h_{ij})$  describe a small redistribution of probability, with  $\sum_{i,j} h_{ij} = 0$ . Define the path

$$P_\varepsilon = P + \varepsilon h, \quad f(\varepsilon) = I(P_\varepsilon), \quad (\text{D.10})$$

for values of  $\varepsilon$  small enough that all cells remain positive. Here  $\varepsilon = 0$  is the expansion point and  $f(0) = I(P) = 0$ . The second-order Taylor expansion is

$$f(\varepsilon) \approx \underbrace{f(0)}_{\text{zeroth order}} + \underbrace{f'(0)\varepsilon}_{\text{first order}} + \underbrace{\frac{f''(0)}{2!}\varepsilon^2}_{\text{second order}}. \quad (\text{D.11})$$

The first-order term is

$$f'(0) = \sum_{i,j} \{\ell_P(i,j) - 1\} h_{ij} = 0. \quad (\text{D.12})$$

Both parts vanish:  $\ell_P(i,j) = 0$  at independence, and the probability constraint gives  $\sum_{i,j} h_{ij} = 0$ . Thus, positive and negative cell errors cannot create a linear change in MI around the independence table.

To state the surviving term, define the row and column changes  $h_{i+} = \sum_j h_{ij}$  and  $h_{+j} = \sum_i h_{ij}$ . Differentiating the entropy form of MI twice gives

$$f''(0) = \sum_{i,j} \frac{h_{ij}^2}{p_{ij}} - \sum_i \frac{h_{i+}^2}{p_{i+}} - \sum_j \frac{h_{+j}^2}{p_{+j}}. \quad (\text{D.13})$$

Because  $p_{ij} = p_{i+}p_{+j}$ , define the part of each cell change that cannot be explained by changed margins as

$$a_{ij} = h_{ij} - p_{+j}h_{i+} - p_{i+}h_{+j}. \quad (\text{D.14})$$

Equation (D.13) can then be written as

$$f''(0) = \sum_{i,j} \frac{a_{ij}^2}{p_{i+}p_{+j}}. \quad (\text{D.15})$$

Substituting the zeroth-, first-, and second-order terms into Equation (D.11) gives

$$I(P_\varepsilon) = \underbrace{0}_{\text{zeroth and first order}} + \underbrace{\frac{\varepsilon^2}{2} \sum_{i,j} \frac{a_{ij}^2}{p_{i+}p_{+j}}}_{\text{second order}} + o(\varepsilon^2). \quad (\text{D.16})$$

The remainder  $o(\varepsilon^2)$  is negligible relative to  $\varepsilon^2$  as  $\varepsilon$  approaches zero.

For a multinomial sample, take  $\varepsilon h = \hat{P} - P$ . An empirical cell proportion has standard deviation  $\sqrt{p_{ij}(1-p_{ij})/n}$ , so each  $\varepsilon h_{ij}$  is typically proportional to  $n^{-1/2}$ . The corresponding row- and column-adjusted errors  $\varepsilon a_{ij}$  have the same scale. Equation (D.16) squares these errors, making  $\hat{I}(P)$  proportional to  $n^{-1}$  under independence. Multiplying by  $2n$  therefore produces a statistic with a non-vanishing scale:

$$2n\hat{I}(P) \xrightarrow{d} \chi_{(r-1)(c-1)}^2. \quad (\text{D.17})$$

This is the usual likelihood-ratio, or  $G$ , test of independence under a fixed alphabet and the standard positive-support large-sample conditions (Wilks, 1938; Brillinger, 2004). The chi-squared reference describes the squared cell errors in Equation (D.16).

Expanded Welch instead standardises a signed first-order MI error on the  $n^{-1/2}$  scale. At independence that signed term is absent, so its Student reference is not the chi-squared reference in another form.

Although squaring a regular asymptotic  $N(0, 1)$  statistic gives a  $\chi_1^2$  limit, the present  $T$  is degenerate at independence. Its square is therefore not the likelihood-ratio  $G$  statistic.

The same likelihood-ratio connection appears for conditional information measures. For finite Markov chains, [Barnett and Bossomaier \(2012\)](#) relate transfer entropy to a log-likelihood ratio with an asymptotic chi-squared reference; the wider information-theoretic setting is reviewed by [Bossomaier et al. \(2016\)](#). This is an adjacent application of the same second-order likelihood principle, not a two-sample equal-MI Welch test.

## D.4 What alternative constructions produce

Using the observed margins to set  $\hat{q}_{ij} = \hat{p}_{i+}\hat{p}_{+j}$  gives

$$2nD_{\text{KL}}(\hat{P}||\hat{Q}) = 2n\hat{I}(P), \quad (\text{D.18})$$

which is the likelihood-ratio statistic above. Because  $\hat{Q}$  is constructed from the same sample as  $\hat{P}$ , it is not an independent second table and cannot be inserted into the two-sample Welch variance formula as though it supplied new information.

Repeatedly shuffling one variable preserves its observations while breaking the pairing with the other variable. The resulting empirical distribution is a permutation reference. Repeatedly drawing tables from  $\hat{P}_X\hat{P}_Y$  is a parametric bootstrap reference. A single simulated  $Q$  table adds Monte Carlo variation but does not by itself define a calibrated null distribution.

The independent reference  $Q = P_X P_Y$  is therefore mathematically valid, but it does not create a regular two-sample MI comparison. Analytic improvement of the sparse independence test would require a second-order finite-sample method and is not an immediate extension of the Expanded Welch calculation.

# Appendix E

## Reproducibility and Protocol Record

The final protocol is `WelchSatterthwaiteMI/experiments/THESIS_REDESIGN_PROTOCOL.json`, version 1.0.0, frozen on 7 September 2026 UTC before the final simulation. It specifies the population grids, seeds, methods, 20,000 replicates,  $\alpha = 0.05$ , validation tolerances and timing design. This local freeze followed exploratory development; it was not an external preregistration.

The run contains 3,111 unique configurations, 4,001 display positions and 534 population pairs, corresponding to 62,220,000 sampled table pairs. The saved verification records check the population targets, result denominators, paired method outputs, validity rules and generated atlas. Chapter 6 uses the same frozen tables; no population was resimulated for a thesis figure.

The exact software versions, seeds and source hashes are recorded in `WelchSatterthwaiteMI/results/thesis_redesign/run_metadata.json`. The experiment supplement contains the protocol, implementation, consolidated results, 3,111 configuration checkpoints, complete atlas, mechanism diagnostics and tests. Its README documents minor post-run orchestration changes; the frozen population construction, simulation and method calculations are unchanged.

The evidence audit reconstructs the protocol manifests and checks every figure and Chapter 6 claim against the saved files. Final checksums are in `SHA256SUMS.txt`. After extracting the source and supplement into one archive root, run these commands from the directory containing `main.tex` and `WelchSatterthwaiteMI/`:

```
python figures_rewrite/make_figures.py
python figures_rewrite/audit_evidence.py
latexmk -pdf main.tex
```



# Bibliography

- Lionel Barnett and Terry Bossomaier. Transfer entropy as a log-likelihood ratio. *Physical Review Letters*, 109(13):138105, 2012. doi: 10.1103/PhysRevLett.109.138105.
- Robert M. Bell and Daniel F. McCaffrey. Bias reduction in standard errors for linear regression with multi-stage samples. *Survey Methodology*, 28(2):169–181, 2002. URL <https://www150.statcan.gc.ca/n1/pub/12-001-x/2002002/article/9058-eng.pdf>.
- Thomas B. Berrett and Richard J. Samworth. USP: An independence test that improves on Pearson’s chi-squared and the  $G$ -test. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 477(2256):20210549, 2021. doi: 10.1098/rspa.2021.0549.
- Terry Bossomaier, Lionel Barnett, Michael Harré, and Joseph T. Lizier. *An Introduction to Transfer Entropy: Information Flow in Complex Systems*. Springer, Cham, 2016. doi: 10.1007/978-3-319-43222-9.
- Richard P. Brent. *Algorithms for Minimization without Derivatives*. Prentice-Hall, Englewood Cliffs, New Jersey, 1973.
- David R. Brillinger. Some data analyses using mutual information. *Brazilian Journal of Probability and Statistics*, 18(2):163–183, 2004. URL <https://www.stat.berkeley.edu/~brill/Papers/MIBJPS.pdf>.
- EunYi Chung and Joseph P. Romano. Exact and asymptotically robust permutation tests. *Annals of Statistics*, 41(2):484–507, 2013. doi: 10.1214/13-AOS1090.
- W. Edwards Deming and Frederick F. Stephan. On a least squares adjustment of a sampled frequency table when the expected marginal totals are known. *Annals of Mathematical Statistics*, 11(4):427–444, 1940. doi: 10.1214/aoms/1177731829.
- Robert Drake and Apratim Guha. A mutual information-based  $k$ -sample test for

- discrete distributions. *Journal of Applied Statistics*, 41(9):2011–2027, 2014. doi: 10.1080/02664763.2014.899325.
- Bradley Efron. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1):1–26, 1979. doi: 10.1214/aos/1176344552.
- Michael P. Fay and Barry I. Graubard. Small-sample adjustments for Wald-type tests using sandwich estimators. *Biometrics*, 57(4):1198–1206, 2001. doi: 10.1111/j.0006-341X.2001.01198.x.
- Apratim Guha and Tom Chothia. A two sample test based on mutual information. *Calcutta Statistical Association Bulletin*, 66(1–2):39–54, 2014. doi: 10.1177/0008068320140103.
- Frank R. Hampel. The influence curve and its role in robust estimation. *Journal of the American Statistical Association*, 69(346):383–393, 1974. doi: 10.1080/01621459.1974.10482962.
- Kermit Hutcheson. A test for comparing diversities based on the Shannon formula. *Journal of Theoretical Biology*, 29(1):151–154, 1970. doi: 10.1016/0022-5193(70)90124-4.
- Marcus Hutter. Distribution of mutual information. In *Advances in Neural Information Processing Systems 14*, pages 399–406, 2002. Conference held in 2001; arXiv:cs/0112019.
- Marcus Hutter and Marco Zaffalon. Distribution of mutual information from complete and incomplete data. *Computational Statistics & Data Analysis*, 48(3): 633–657, 2005. doi: 10.1016/j.csda.2004.03.010.
- Guido W. Imbens and Michal Kolesár. Robust standard errors in small samples: Some practical advice. *The Review of Economics and Statistics*, 98(4):701–712, 2016. doi: 10.1162/REST\_a\_00552.
- Arnold Janssen. Studentized permutation tests for non-i.i.d. hypotheses and the generalized Behrens–Fisher problem. *Statistics & Probability Letters*, 36(1):9–21, 1997. doi: 10.1016/S0167-7152(97)00043-6.
- Kirthivasan Kandasamy, Akshay Krishnamurthy, Barnabas Poczos, Larry Wasserman, and James M. Robins. Nonparametric von Mises estimators for entropies, divergences and mutual informations. In *Advances in Neural Information Processing Systems 28*, pages 397–405, 2015.

- Göran Kauermann and Raymond J. Carroll. A note on the efficiency of sandwich covariance matrix estimation. *Journal of the American Statistical Association*, 96(456):1387–1396, 2001. doi: 10.1198/016214501753382309.
- Marius Marinescu and Costel Balcau. On the use of mutual information for testing independence, 2025. arXiv:2502.17636.
- George A. Miller. Note on the bias of information estimates. In Henry Quastler, editor, *Information Theory in Psychology: Problems and Methods*, pages 95–100. Free Press, Glencoe, Illinois, 1955.
- Roelof Moddemeijer. On estimation of entropy and mutual information of continuous distributions. *Signal Processing*, 16(3):233–248, 1989. doi: 10.1016/0165-1684(89)90132-1.
- Roelof Moddemeijer. A statistic to estimate the variance of the histogram-based mutual information estimator based on dependent pairs of observations. *Signal Processing*, 75(1):51–63, 1999. doi: 10.1016/S0165-1684(98)00224-2.
- Ricardo Mora and Javier Ruiz-Castillo. The statistical properties of the mutual information index of multigroup segregation. Working Paper, Economics Series 48 09-84, Universidad Carlos III de Madrid, Departamento de Economía, 2009.
- Ricardo Mora and Javier Ruiz-Castillo. Entropy-based segregation indices. *Sociological Methodology*, 41(1):159–194, 2011. doi: 10.1111/j.1467-9531.2011.01237.x.
- Wei Pan and Melanie M. Wall. Small-sample adjustments in using the sandwich variance estimator in generalized estimating equations. *Statistics in Medicine*, 21(10):1429–1441, 2002. doi: 10.1002/sim.1142.
- Liam Paninski. Estimation of entropy and mutual information. *Neural Computation*, 15(6):1191–1253, 2003. doi: 10.1162/089976603321780272.
- Mark S. Roulston. Estimating the errors on measured entropy and mutual information. *Physica D: Nonlinear Phenomena*, 125(3–4):285–294, 1999. doi: 10.1016/S0167-2789(98)00269-3.
- Franklin E. Satterthwaite. An approximate distribution of estimates of variance components. *Biometrics Bulletin*, 2(6):110–114, 1946. doi: 10.2307/3002019.
- Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27:379–423, 623–656, 1948. Parts I and II.

- Aad W. van der Vaart. *Asymptotic Statistics*. Cambridge University Press, Cambridge, 1998. doi: 10.1017/CBO9780511802256.
- Abraham Wald. Tests of statistical hypotheses concerning several parameters when the number of observations is large. *Transactions of the American Mathematical Society*, 54(3):426–482, 1943. doi: 10.1090/S0002-9947-1943-0012401-3.
- Bernard L. Welch. The generalization of “Student’s” problem when several different population variances are involved. *Biometrika*, 34(1–2):28–35, 1947. doi: 10.1093/biomet/34.1-2.28.
- Samuel S. Wilks. The large-sample distribution of the likelihood ratio for testing composite hypotheses. *The Annals of Mathematical Statistics*, 9(1):60–62, 1938. doi: 10.1214/aoms/1177732360.
- Edwin B. Wilson. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927. doi: 10.1080/01621459.1927.10502953.